Hamiltonian and Hamilton-connected thresholds for planar recolouring graphs

Abstract

For a graph G, let $\mathcal{R}_k(G)$ have the proper colourings of G with a labelled k-colour palette as vertices, with one-vertex recolourings as edges. We prove that $\mathcal{R}_k(G)$ is Hamilton-connected and vertex-pancyclic whenever G is nonempty and d-degenerate and $k \geq d+3$; this uniform palette bound is sharp for both properties and for Hamiltonicity. We determine the common Hamiltonian and Hamilton-connected thresholds as 7 for planar graphs, 5 for triangle-free and bipartite planar graphs, and 4 for planar graphs of girth at least six and for forests. At four colours we also prove Hamiltonicity for planar graphs of girth at least five with at most thirty-three distinct 5-cycles, total degree excess above three at most ten, or at most seven vertices of degree at least four. We also cover those whose blocks all have maximum degree at most three, and two-degenerate planar graphs of girth at least five with total degree excess at most thirteen. The block condition allows unbounded degree at cutvertices. These four-colour proofs combine structural reductions with explicit local constructions given in the appendices. For unrestricted planar graphs of girth at least five, both thresholds remain between 4 and 5.

1 Introduction

A cyclic listing of all proper colourings in which successive colourings differ at one vertex is a colouring Gray code, or equivalently a Hamilton cycle in the recolouring graph. Prescribing the first and last colourings asks for Hamilton-connectedness. Requiring a cycle of every length through each colouring asks for vertex-pancyclicity. We study these stronger spanning and cyclic properties for sparse graphs, with particular emphasis on sharp palette thresholds for planar classes.

All graphs are finite, simple and undirected, and may be disconnected. A graph is nonempty when it has at least one vertex. Put $[k] = \{1, \ldots, k\}$. The vertices of $\mathcal{R}_k(G)$ are the proper maps $\sigma : V(G) \to [k]$; two are adjacent when they differ on exactly one vertex of G. Colours are labelled, and a colouring need not use every colour. A proper colouring is called *frozen* when every vertex sees all the colours other than its own on its neighbourhood; such a colouring is an isolated vertex of the recolouring graph. A cycle has at least three vertices. A graph is Hamilton-connected if every two distinct vertices are joined by a spanning path, and vertex-pancyclic if every vertex lies on a cycle of every length from three to the order of the graph.

For a class $\mathcal{C}$ of nonempty graphs, define

$$h(\mathcal{C}) = \min\{q : \mathcal{R}_k(G) \text{ is Hamiltonian for every } G \in \mathcal{C} \text{ and every } k \geq q\},$$
$$\mathrm{hc}(\mathcal{C}) = \min\{q : \mathcal{R}_k(G) \text{ is Hamilton-connected for every } G \in \mathcal{C} \text{ and every } k \geq q\},$$

when these sets are nonempty. These are uniform thresholds in the palette size; the proofs establish every larger palette separately. A graph is d-degenerate if each nonempty subgraph has a vertex of degree at most d. Write $\mathcal{D}_d$ for this class.

Choo and MacGillivray [3, Theorem 3.4] proved that $\mathcal{R}_k(G)$ is Hamiltonian for $k \geq d+3$ when G is d-degenerate. They also determined the uniform forest threshold and several individual tree and cycle cases [3, Sections 5–6]. Our first result strengthens their general Hamiltonian conclusion while keeping the same palette bound.

Theorem 1.1. *Let G be a nonempty d-degenerate graph. For every $k \geq d+3$, $\mathcal{R}_k(G)$ is Hamilton-connected and vertex-pancyclic. Moreover, $\mathrm{hc}(\mathcal{D}_d) = h(\mathcal{D}_d) = d+3$, and $d+3$ is also the least uniform palette bound for vertex-pancyclicity on $\mathcal{D}_d$.*

Write $\mathcal{P}$ for the planar graphs, $\mathcal{T}$ for the triangle-free planar graphs, $\mathcal{B}$ for the bipartite planar graphs, $\mathcal{P}_g$ for the planar graphs of girth at least g, and $\mathcal{F}$ for the forests. All these classes contain only nonempty graphs; forests have infinite girth.

Theorem 1.2. *The common thresholds $h = \mathrm{hc}$ are 7 for $\mathcal{P}$, 5 for $\mathcal{T}$ and $\mathcal{B}$, and 4 for $\mathcal{P}_6$ and $\mathcal{F}$.*

The large-palette cases follow from degeneracy. The critical palettes require endpoint-preserving restoration of separated low-degree vertices, and, in the triangle-free case, of an adjacent degree-three pair. The lower bounds use frozen colourings and the three-colour recolouring graph of a short path. The equality $h(\mathcal{F}) = 4$ already follows from [3]; here it accompanies the Hamilton-connected forest result.

At four colours the girth-five case requires different local configurations. Write

$$D(G) = \sum_{v \in V(G)} \max\{d_G(v) - 3, 0\}, \qquad \mathcal{H}_4(G) := \{v \in V(G) : d_G(v) \geq 4\}.$$

Let $\mathcal{P}_5^{\leq 33}$ consist of the graphs in $\mathcal{P}_5$ with at most thirty-three distinct 5-cycles. Let $\mathcal{S}_5$ consist of those in $\mathcal{P}_5$ whose every block has maximum degree at most three. Here blocks include bridges and isolated vertices as well as maximal two-connected subgraphs.

Theorem 1.3. *For every $G \in \mathcal{P}_5^{\leq 33} \cup \mathcal{S}_5$ and every $k \geq 4$, $\mathcal{R}_k(G)$ is Hamiltonian. The same holds for $G \in \mathcal{P}_5$ with $D(G) \leq 10$ or $|\mathcal{H}_4(G)| \leq 7$, and for two-degenerate $G \in \mathcal{P}_5$ with $D(G) \leq 13$. Each stated class has uniform Hamiltonian threshold four and Hamilton-connected threshold between four and five.*

In particular, Theorem 1.3 includes all subcubic planar graphs of girth at least five, while allowing vertices of arbitrarily large total degree when that degree is distributed among blocks. Choo and MacGillivray's Gray-code formulation [3] places these questions in the broader recolouring literature. For complete graphs, the recolouring graphs are arrangement graphs, whose Hamiltonicity and fault Hamilton-connectedness have also been studied [6]. General connectivity of colouring graphs is studied by Cereceda, van den Heuvel and Johnson [4]; Bartier et al. [1] proved that $\mathcal{R}_4(G)$ is connected for every $G \in \mathcal{P}_5$. Connectivity supplies paths between states but does not supply a spanning cycle or prescribed endpoints, so these questions remain strictly stronger here. For the full class we obtain

$$4 \leq h(\mathcal{P}_5) \leq 5, \qquad 4 \leq \mathrm{hc}(\mathcal{P}_5) \leq 5.$$

Table 1.1 records the results and these open cases.

The lower-bound witnesses used for the exact entries are collected here. For a frozen colouring, all colour relabellings are distinct isolated states, so the recolouring graph is disconnected.

1.1 Proof strategy and organization

Deleting vertices partitions the larger recolouring graph into fibres above colourings of the smaller graph. At the degenerate bound these fibres are cliques of order at least three. We choose bridge colours between consecutive fibres to prescribe endpoints or cycle lengths. Section 2 proves Theorem 1.1, including the parity obstruction that makes its bound sharp. Section 3 develops the product and independent-set lifts used in the planar proofs.

Table 1.1: Uniform palette thresholds. An entry $\{4,5\}$ means that the integer threshold is known to be either 4 or 5; whether it is 4 or 5 is open.

Graph class	h	hc
Planar	7	7
Triangle-free planar	5	5
Bipartite planar	5	5
Planar, girth at least six	4	4
Forests	4	4
Planar, girth at least five, at most thirty-three 5-cycles	4	$\{4,5\}$
Planar, girth at least five, every block subcubic	4	$\{4,5\}$
Planar, girth at least five, $D(G) \leq 10$	4	$\{4,5\}$
Planar, girth at least five, $\lvert\mathcal{H}_4(G)\rvert \leq 7$	4	$\{4,5\}$
Two-degenerate planar, girth at least five, $D(G) \leq 13$	4	$\{4,5\}$
Planar, girth at least five	$\{4,5\}$	$\{4,5\}$

Table 1.2: Lower-bound witnesses for the sharp palette thresholds and for the sharp $d+3$ bound in Theorem 1.1.

Class	palette	witness	obstruction
Planar	6	icosahedron	frozen six-colouring
Triangle-free and bipartite planar	4	cube Q_3	frozen four-colouring
Planar, girth at least $g \geq 6$	3	C_{3m} with $3m \geq g$	periodic frozen three-colouring
Forests	3	P_3 for h; K_2 for hc	forced cycles; $\mathcal{R}_3(K_2) = C_6$
$\mathcal{D}_d$ (H)	$d+2$	$K_{d+2} - uv$ for $d \geq 1$; K_1 for $d = 0$	forced 4-cycle; $\mathcal{R}_2(K_1) = K_2$
$\mathcal{D}_d$ (HC/VP)	$d+2$	K_{d+1} for $d \geq 1$; $2K_1$ for $d = 0$	permutation-parity obstruction; $\mathcal{R}_2(2K_1) = C_4$

Section 4 proves Theorem 1.2 by combining these lifts with planar degree counts. Section 5 proves Theorem 1.3. The few-cycle argument combines the heavy-vertex structure with restoration of low-degree configurations, including paths whose endpoints share an outside neighbour and cycles with an attached degree-two vertex. A degree-two vertex also supplies a reservoir for an ordinary pentagon whose local fibre can be disconnected. Linked ordinary pentagons and protected products for bowties and hinged pairs supply further restorations. Protected products reduce the degree-four clean-face incidence bound to one. A neighbourhood face count and its equality rigidity finish the minimum-degree-three case at total excess ten. For the subcubic and block-subcubic classes, we restore two separated pentagons. Their product fibres have at most four components. Local cycles retain protected edges at each transition; a bounded-degree spanning tree then joins all component cycles into one spanning cycle, for a base recolouring cycle of arbitrary length.

Appendices A–J give the paths, cycles and auxiliary edge sets used in these local restorations. The finite verification in Section 7 checks the listed colourings, their adjacencies, the symmetry classes, and the common edges required at boundary transitions. The existence of the underlying Hamilton cycles, the extensions to products of arbitrary order, and the spanning-tree splicing arguments are established in the proofs below.

1.2 Notation

For a graph G, write $N_G(v)$ for the open neighbourhood, $N_G[v] = N_G(v) \cup \{v\}$, and $G - S$ for deletion of a vertex set S. The Cartesian product $X \square Y$ joins (x, y) to (x', y') when exactly one coordinate changes along an edge of its factor. A spanning path may be a singleton when its ends coincide in an intermediate construction. In the explicit constructions we use palettes $\{0, \dots, k-1\}$, obtained by relabelling $[k]$. Face lengths count edge-side incidences, so a bridge is counted twice. We distinguish a planar graph, which admits a plane embedding, from a plane graph, whose embedding is fixed. Whenever a facial assertion is used below, fix such an embedding of the planar graph under discussion. Distinct r-cycles are counted as subgraphs (equivalently by their edge sets), with cyclic rotations and reversals identified; a cycle is counted once even if it bounds two faces.

2 Hamilton-connectedness and vertex-pancyclicity

We begin with the general result that supplies all large-palette statements.

Lemma 2.1. *Let $m \geq 3$ and let $L_1, \dots, L_m$ be finite sets of size at least two. Unless m is odd and all L_i are the same two-element set, there are $x_i \in L_i$ with $x_i \neq x_{i+1}$ cyclically.*

Proof. If some L_j has at least three elements, delete position j and colour the resulting path greedily. Its two end values forbid at most two elements of L_j, so choose x_j different from both; this closes the cycle. Now suppose every list is binary and choose consecutive lists $L_j \neq L_{j+1}$. Pick $x_j \in L_j \setminus L_{j+1}$. Both values of L_{j+1} can then be used at the next position, which is the base of the propagation. Inductively, after colouring positions through i, every value of L_i is reachable from the fixed x_j while respecting the path edges: from two reachable values, either value in the next binary list has a different predecessor. Consequently choose at position $j-1$ the value different from x_j. Finally, if all lists are the same binary set, alternate its two values; this works exactly when m is even. □

2.1 Clique fibres

For a vertex v with $H = G - v \neq \varnothing$ and $k \geq d_G(v) + 3$, put

$$L(f) = [k] \setminus f(N_G(v)), \qquad F_f = \{f(c) : c \in L(f)\},$$

where $f(c)$ extends a proper colouring f of H by assigning colour c to v. The sets F_f partition $V(\mathcal{R}_k(G))$ into cliques, and adjacent base colourings f, g satisfy

$$|L(f)| \geq 3, \qquad |L(f) \cap L(g)| \geq |L(f)| - 1 \geq 2. \tag{2.1}$$

The cross edges are exactly $f(c)g(c)$ for $c \in L(f) \cap L(g)$. If the base move changes a nonneighbour of v, then $L(f) = L(g)$. We use these fibres throughout the single-vertex lifts.

Lemma 2.2. *Let $H = G - v \neq \varnothing$ and $k \geq d_G(v) + 3$. If $f_0, f_1, \ldots, f_{s-1}, f_0$ is a cycle of $\mathcal{R}_k(H)$ and $x \in F_{f_0}$, then $\mathcal{R}_k(G)$ has a cycle through x of every length in*

$$[2s + 1,\ \textstyle\sum_i |F_{f_i}|]. \tag{2.2}$$

In particular, if $\mathcal{R}_k(H)$ is Hamiltonian, then so is $\mathcal{R}_k(G)$.

Proof. Put $S_i = L(f_i) \cap L(f_{i+1})$, with cyclic indices. Lemma 2.1 selects bridge colours $y_i \in S_i$ with $y_{i-1} \neq y_i$, unless s is odd and all S_i equal one binary set $\{p, q\}$.

We exclude this exception. By (2.1), each $L(f_i)$ then has size three, and

$$k - 3 = |f_i(N_G(v))| \leq d_G(v) \leq k - 3.$$

Thus the $k-3$ neighbours of v have distinct colours, none uses p or q, and every base move changes a neighbour of v; a nonneighbour move would leave an intersection of size three. Order the neighbours $u_1, \ldots, u_{k-3}$. In the $(k-2)$-element palette $[k] \setminus \{p, q\}$, append the unique unused colour r_i to the tuple $(f_i(u_1), \ldots, f_i(u_{k-3}))$. A base move exchanges one neighbour colour with r_i, reversing permutation parity. An odd number of such exchanges cannot return to the initial tuple. Hence the bridge colours exist.

Choose an m_0-vertex path in F_{f_0} between its bridge states and containing x, with $3 \leq m_0 \leq |F_{f_0}|$. In every other clique choose a bridge-to-bridge path on $2 \leq m_i \leq |F_{f_i}|$ vertices. The sums of these integer intervals are exactly (2.2). The cross edges with v fixed at y_i close these paths into the desired cycle. A Hamilton base cycle with every $m_i = |F_{f_i}|$ gives a Hamilton cycle. □

Lemma 2.3. *Let $t \geq 1$, and let $S_0, \ldots, S_{t-1}$ be finite sets of colours such that $|S_i| \geq 2$ for every i. Suppose that $|S_j| \geq 3$ for some $j \in \{0, \ldots, t-1\}$. For any colours a and b, there exist colours $x_i \in S_i$ satisfying*

$$x_0 \neq a, \qquad x_i \neq x_{i-1} \quad (1 \leq i \leq t-1), \qquad x_{t-1} \neq b.$$

Proof. Set $x_{-1} = a$ and $x_t = b$. Choose $x_0, \ldots, x_{j-1}$ forwards, avoiding the preceding colour, and $x_{t-1}, \ldots, x_{j+1}$ backwards, avoiding the following colour. Each choice excludes only one colour. Finally choose $x_j \in S_j \setminus \{x_{j-1}, x_{j+1}\}$, which is possible since $|S_j| \geq 3$. The boundary assignments also cover $j = 0$, $j = t-1$, and $t = 1$. □

Lemma 2.4. *Let $H = G - v \neq \varnothing$ and $k \geq d_G(v) + 3$. Suppose $Y = \mathcal{R}_k(H)$ is nonempty and Hamilton-connected, and v has a nonneighbour in H. If for every $f \in V(Y)$ and $c \in L(f)$ there is a neighbour g of f with $c \in L(g)$, then $\mathcal{R}_k(G)$ is Hamilton-connected.*

Proof. Fix a nonneighbour w of v. Colour permutations give all k colours at w among the states of Y; deleting at most one state leaves at least $k - 1 \geq 2$ colours. Thus every base path missing at most one state contains a move at w. Its consecutive intersections $S_i = L(f_i) \cap L(f_{i+1})$ have size at least two, and the move at w supplies one of size at least three. Lemma 2.3 now gives bridge colours avoiding any prescribed first and last colours, with distinct entrance and exit colours in every clique.

Hamilton paths inside the cliques concatenate to a spanning path through those fibres with the prescribed endpoints.

Now prescribe distinct endpoints $\alpha(a)$ and $\beta(b)$ in $\mathcal{R}_k(G)$. If $\alpha \neq \beta$, lift a Hamilton path in Y between their restrictions.

If $\alpha = \beta$, choose $c \in L(\alpha) \setminus \{a, b\}$ and a neighbour g preserving c. Since $|V(Y)| \geq k \geq 3$, closing a Hamilton path between the ends of αg gives a Hamilton cycle through that edge. Let h be its other neighbour of α and choose $q \in (L(\alpha) \cap L(h)) \setminus \{c\}$. Since q cannot equal both a and b, and a spanning path may be reversed, we may interchange a and b so that $q \neq a$.

Partition F_α into paths $P : \alpha(a) \rightsquigarrow \alpha(c)$ and $P' : \alpha(q) \rightsquigarrow \alpha(b)$. Use a singleton for P' if $q = b$, and its endpoint edge otherwise; put all remaining vertices in P. Deleting α from the chosen base cycle leaves a path missing only that state. Lift it from $g(c)$ to $h(q)$ through all other fibres. The cross edges at c, q join this lifted path to P, P', giving the prescribed spanning path. □

Theorem 2.5. *Let G be a nonempty d-degenerate graph and $k \geq d + 3$. Then $\mathcal{R}_k(G)$ is Hamilton-connected.*

Proof. Induct on $n = |V(G)|$, starting with $\mathcal{R}_k(K_1) = K_k$. If $G = K_n$, then $\mathcal{R}_k(G)$ is the arrangement graph $A_{k,n}$. Here $A_{k,n}$ is the graph of ordered n-tuples of distinct elements of $[k]$, with one-coordinate changes. In [6] the number of symbols is k and the tuple length is n, and Theorem 1 there assumes that their difference is at least two. Since K_n has minimum degree $n - 1$, we have $d \geq n - 1$ and $k \geq d + 3 \geq n + 2$, so $A_{k,n}$ is Hamilton-connected by [6, Theorem 1].

Otherwise a minimum-degree vertex v has degree at most d and a nonneighbour. Put $H = G - v$; induction makes $\mathcal{R}_k(H)$ Hamilton-connected. For each base colouring f and $c \in L(f)$, choose a vertex of degree at most d in H. It has at least three available colours, so can be recoloured avoiding its present colour and c. This base move preserves c at v. Lemma 2.4 applies. □

Theorem 2.6. *Let G be a nonempty d-degenerate graph and $k \geq d + 3$. Then $\mathcal{R}_k(G)$ is vertex-pancyclic.*

Proof. Induct on $|V(G)|$, starting with $\mathcal{R}_k(K_1) = K_k$. For larger G, choose v with $d_G(v) \leq d$, put $H = G - v$, use the clique fibres above, and write $Y = \mathcal{R}_k(H)$, $M = |V(Y)|$, and $N = |V(\mathcal{R}_k(G))|$. Colour permutations give $M \geq k \geq 3$. Fix a state $x = \alpha(a)$ of $\mathcal{R}_k(G)$.

The clique F_α already supplies a triangle through x. For lengths four, five and six, recolour a vertex of degree at most d in H to a colour avoiding its present colour, its neighbours, and a. At most $d + 2$ colours are excluded, so this gives a neighbour β of α with $a \in L(\beta)$. Choose $p \in (L(\alpha) \cap L(\beta)) \setminus \{a\}$, $q \in L(\alpha) \setminus \{a, p\}$, and $r \in L(\beta) \setminus \{a, p\}$. The following cyclic sequences have the required lengths:

$$\begin{aligned} 4 &: \ \alpha(a), \alpha(p), \beta(p), \beta(a), \\ 5 &: \ \alpha(a), \alpha(q), \alpha(p), \beta(p), \beta(a), \\ 6 &: \ \alpha(a), \alpha(q), \alpha(p), \beta(p), \beta(r), \beta(a). \end{aligned}$$

For each $3 \leq s \leq M$, induction supplies a base s-cycle C_s through α. Put $T_s = \sum_{f \in V(C_s)} |F_f|$. Lemma 2.2 gives every length in $[2s+1, T_s]$. These intervals overlap successively, since $T_s \geq 3s \geq 2s+3$ for $s \geq 3$. Also $T_M = N$. They therefore cover every length from seven to N, completing the proof. □

Remark 2.7. Vertex-pancyclicity here cannot be strengthened to edge-pancyclicity (cycles of every length through each edge). For any $k \geq 4$, colour the leaves of the 1-degenerate star $G = K_{1,k-2}$ distinctly with $3, \ldots, k$. The extensions with centre colours 1 and 2 form an edge of $\mathcal{R}_k(G)$ in no triangle: such a triangle would recolour only the centre, which has no third available colour.

Proposition 2.8. *For every $d \geq 0$, $h(\mathcal{D}_d) = \mathrm{hc}(\mathcal{D}_d) = d + 3$. The same uniform palette bound for vertex-pancyclicity cannot be reduced.*

Proof. The upper bounds are the preceding theorems. For $d \geq 1$, first take $G = K_{d+1}$ and $k = d+2$. Append the unique unused colour to each ordered colouring of G. The resulting permutation of $d + 2$ colours changes by a transposition at every recolouring. Thus $\mathcal{R}_{d+2}(G)$ is bipartite by permutation parity, with equal parts and even order $(d + 2)! \geq 6$. A spanning path must have endpoints in opposite parts, so the graph is not Hamilton-connected. It has no triangle, so it is not vertex-pancyclic.

For the Hamiltonian lower bound, let $G = K_{d+2} - uv$ and again put $k = d + 2$ (for $d = 1$ this is P_3). This graph is d-degenerate. The d vertices of the core clique use d distinct colours, leaving two colours p, q for u, v. If u and v receive different colours, every core vertex is frozen and the state has exactly two neighbours, obtained by changing u or v to the other colour. Hence, for each fixed core colouring, the four states with colour pairs

$$(p, p),\ (p, q),\ (q, q),\ (q, p)$$

at (u, v) form a forced 4-cycle in every Hamilton cycle. There are $2(d + 2)! > 4$ states in total, so $\mathcal{R}_{d+2}(G)$ is not Hamiltonian. For $d = 0$, $\mathcal{R}_2(K_1) = K_2$ is not Hamiltonian, while $\mathcal{R}_2(2K_1) = C_4$ is not Hamilton-connected or vertex-pancyclic. □

3 Endpoint lifting and restoration

A colouring of a deleted graph determines the available colours at the vertices to be restored. The resulting extension graph is a fibre. In all the constructions below, assignments retain their actual coordinate values: equal assignments in two different fibres are distinct vertices, and a cross edge joins their two copies. In particular, a move of one vertex of the deleted graph may change several available lists at once.

Lemma 3.1. *Let $s \geq 1$ and $F = \square_{j=1}^{s} K_{A_j}$, where every A_j is finite and $|A_j| \geq 2$. If all lists have size two, F has a Hamilton path between two prescribed vertices if and only if those vertices lie in opposite bipartition classes. If at least one list has size at least three, F is Hamilton-connected.*

Proof. In the binary case F is the hypercube Q_s, whose bipartition classes have equal size. The endpoint condition is therefore necessary. For sufficiency, induct on s, starting with the single edge Q_1. Choose a coordinate on which the prescribed endpoints differ and split along it. Their projections into Q_{s-1} have the same parity. Choose a projected vertex of the opposite parity; use the two induction paths to and from its copies, joined by the edge between those copies.

For the second assertion, we first show that $Y \square K_r$ is Hamilton-connected whenever Y is Hamilton-connected and $|V(Y)| \geq 3$. Induct on r, starting with $r = 1$. If the two endpoints lie in different copies of Y, order the copies starting and ending at those copies. Choose successive bridge vertices in $V(Y)$, with the first different from the first endpoint projection, consecutive bridges different, and the last different from the last endpoint projection. Such choices exist because $|V(Y)| \geq 3$: choose greedily, leaving the last bridge until the end, when it has at most two forbidden values. Prescribed-end Hamilton paths inside the copies then join into a spanning path. If both endpoints lie in one copy, take a Hamilton path between them there and choose an edge xy of it. By induction the other $r - 1$ copies have a spanning path from a copy of x to a copy of y in the same one of those copies. Insert that path in place of xy, using the two cross edges. Start with a complete-graph factor of order at least three and add the remaining factors to obtain the claim. □

Lemma 3.2. *Let $m \geq 3$, $s \geq 1$, and let*

$$F_i = \square_{j=1}^{s} K_{A_{i,j}}, \qquad |A_{i,j}| \geq 2, \qquad 0 \leq i < m,$$

be disjoint fibres arranged along a path. Join equal assignments in consecutive fibres. At every transition assume that at least one list is unchanged and that, for every coordinate,

$$|A_{i,j} \setminus A_{i+1,j}| \leq 1, \qquad |A_{i+1,j} \setminus A_{i,j}| \leq 1.$$

If an internal fibre F_r, $1 \leq r \leq m-2$, has a factor of order at least three, then every prescribed $p \in F_0$ and $q \in F_{m-1}$ are joined by a path covering all vertices of all fibres exactly once.

Proof. Write

$$J_i = \square_{j=1}^{s} K_{A_{i,j} \cap A_{i+1,j}}$$

for the graph of common assignments. The intersections are nonempty, and an unchanged coordinate supplies an edge of J_i.

For each i, write $J_i^{[i]} \subseteq V(F_i)$ and $J_i^{[i+1]} \subseteq V(F_{i+1})$ for the two copies of the same assignment set. Thus every $z \in V(J_i)$ gives the cross edge $z^{[i]}z^{[i+1]}$. Call an endpoint $z \in V(J_i)$ reachable if a partial path covers $F_0, \ldots, F_i$ exactly once and ends at $z^{[i]}$. An edge of J_i is one-sided, respectively fully, reachable if at least one, respectively both, of its endpoints are reachable. This distinction is needed because a binary first fibre can have only one endpoint of a given edge of J_i reachable from a prescribed p.

Suppose an incoming edge of J_{i-1} is one-sided reachable, and let $x^{[i]}$ be a reachable endpoint after its cross edge into F_i. For any outgoing edge $e \in E(J_i)$, if F_i is binary, its two endpoints have opposite parity, so one of them, say y, is at the opposite parity from x. If F_i is nonbinary, choose an endpoint $y \in e$ different from x. The product lemma gives a Hamilton path of F_i from x to y in either case. Together with the two cross edges this makes e one-sided reachable. Starting from p, the same argument applies in F_0 (opposite parity in the binary case and a different endpoint in the nonbinary case), so a one-sided reachable edge exists in every graph J_i before F_r.

At the internal fibre F_r, choose an outgoing edge of J_r avoiding the incoming endpoint x. Such an edge exists because the unchanged large coordinate gives a triangle after deleting x, while if that coordinate changes, its intersection has at least two values and an unchanged coordinate gives a $K_2 \square K_2$, which also has an edge after deleting x. Since F_r is nonbinary and Hamilton-connected, each endpoint of this outgoing edge is reachable from x; hence the edge is fully reachable.

Full reachability propagates: if both endpoints of the incoming edge are reachable, then for each endpoint y of an outgoing edge choose an incoming endpoint of opposite parity when F_i is binary, and choose an incoming endpoint different from y when F_i is nonbinary. The product lemma again supplies the required Hamilton path, so the outgoing edge is fully reachable. At the last fibre, choose from the fully reachable incoming edge an endpoint of opposite parity to q in the binary case, or an endpoint different from q in the nonbinary case, and join it to q by a Hamilton path.

Backtracking these finite choices gives the desired path. Each selected fibre path uses all vertices of its own disjoint fibre exactly once, and the cross edges use the displayed legal copies, so no vertex is repeated and every fibre is covered. □

Lemma 3.3. *Let S be a nonempty independent set, $H = G - S \neq \varnothing$, and $k \geq 4$. Suppose*

$$d_G(u) \leq k-2 \quad (u \in S), \qquad |N_G(x) \cap S| \leq |S| - 1 \quad (x \in V(H)).$$

Assume that $\mathcal{R}_k(H)$ is Hamilton-connected and that some proper k-colouring of H leaves at least three colours available at a vertex of S. If α, β are proper k-colourings of G with $\alpha|_H \neq \beta|_H$, then $\mathcal{R}_k(G)$ has an α–β Hamilton path.

Proof. Take a Hamilton path of $\mathcal{R}_k(H)$ between the two restriction colourings. For a base colouring f, the fibre is exactly

$$\square_{u\in S} K_{A_f(u)}, \qquad A_f(u) = [k] \setminus f(N_G(u)).$$

Every list is at least binary. A move at an actual vertex x of H leaves the list of at least one non-neighbour in S unchanged; every other list loses and gains at most one colour. These are the transition conditions of Lemma 3.2, even when several lists change together.

A colouring with a nonbinary fibre has at least k distinct colour permutations: fix a vertex of the nonempty H and send its colour to each of the k labels. Every resulting fibre is nonbinary. The base Hamilton path has only two endpoints, so one of these colourings is internal. Lemma 3.2 now lifts the path with the specified, actual colours of α and β on S. □

For use of this lemma, first choose a vertex w on which two prescribed colourings differ, and then choose S with $w \notin S$. This ensures different restriction colourings; the lemma does not claim to handle arbitrary endpoints in the same fibre.

Lemma 3.4. *Let $m \geq 3$ disjoint fibres $F_i = \square_{j=1}^{s} K_{A_{i,j}}$, with $s \geq 1$ and $|A_{i,j}| \geq 2$, be cyclically ordered. Join equal assignments in consecutive fibres and define $J_i = \square_j K_{A_{i,j}\cap A_{i+1,j}}$, with indices modulo m. Suppose every J_i contains an edge, F_0 has a factor of order at least three, and there are distinct edges $e_0 \in E(J_0)$ and $e_{m-1} \in E(J_{m-1})$ (compared as actual assignment pairs). Then the layered graph has a Hamilton cycle.*

Proof. Choose an edge e_i in every remaining graph J_i, and choose $z_0 \in V(e_0) \setminus V(e_{m-1})$. Every endpoint of e_{i-1} and e_i is an assignment in the middle fibre F_i, since $J_{i-1}, J_i \subseteq F_i$. If F_i is binary, each edge of J_i has one endpoint in each hypercube parity class, so match opposite parities. If F_i has a factor of order at least three, the product lemma permits any two distinct endpoints; choose a bijection with no fixed assignment between the two endpoint sets. Such a bijection is obtained by swapping equal sets, mapping the common member to the other target member when the sets meet in one point, and using any bijection when they are disjoint. Thus, for $1 \leq i < m$, there is a bijection $\phi_i : V(e_{i-1}) \to V(e_i)$ such that each $z, \phi_i(z)$ is an admissible pair of Hamilton-path endpoints in F_i.

Put $z_i = \phi_i(z_{i-1})$. Traverse F_i from z_{i-1} to z_i for $1 \leq i < m$. Since $z_{m-1} \in V(e_{m-1})$, $z_{m-1} \neq z_0$, so the nonbinary F_0 has a Hamilton path from z_{m-1} to z_0. Join the fibre paths by the cross edges at the assignments z_i. They cover every fibre once and form one cycle. □

Lemma 3.5. *Let S be a nonempty independent set, $H = G - S \neq \varnothing$, and $k \geq 4$. Suppose*

$$d_G(u) \leq k-2 \quad (u \in S), \qquad |N_G(x) \cap S| \leq |S| - 1 \quad (x \in V(H)).$$

If $\mathcal{R}_k(H)$ is Hamiltonian and some proper k-colouring of H leaves at least three colours available at a vertex of S, then $\mathcal{R}_k(G)$ is Hamiltonian.

Proof. Take a Hamilton cycle of the base and use the same product fibres as in Lemma 3.3. Every graph J_i contains an edge. Index a nonbinary fibre as F_0 and choose a large coordinate in it. In either adjacent graph J_i, that coordinate supplies a triangle if unchanged, or a $K_2\square K_2$ together with a different unchanged coordinate if it changes. Thus both adjacent graphs have at least two edges, and distinct edges e_0, e_{m-1} can be chosen. Apply Lemma 3.4. □

Lemma 3.6. *Let $H = G - v \neq \varnothing$ and $k \geq d_G(v) + 3$. If v has a nonneighbour, $\mathcal{R}_k(H)$ is Hamilton-connected, and $\delta(\mathcal{R}_k(H)) > d_G(v)$, then $\mathcal{R}_k(G)$ is Hamilton-connected.*

Proof. For each base colouring f and $c \in L(f)$, at most $d_G(v)$ moves make c unavailable at v: only a neighbour of v recoloured to c can do so. The state-degree bound therefore supplies a neighbour preserving c. Apply Lemma 2.4. □

A Hamilton-connected graph of order at least four has minimum degree at least three. A vertex of degree two would, with its neighbours chosen as the two endpoints, force a spanning path to have only three vertices; degrees zero and one are excluded as well. Every nonempty colourable H has at least k states in $\mathcal{R}_k(H)$, by colour permutation. These observations will supply the state-degree condition in the low-degree inductions.

4 Hamilton-connected planar classes

4.1 All planar graphs

We first establish the marked structural statement needed to retain a coordinate on which the prescribed endpoint colourings differ.

Lemma 4.1. *If G is planar and $\delta(G) \geq 5$, then, for any $w \in V(G)$, there are two degree-five vertices different from w whose distance is at least three. Vertices in different components have infinite distance.*

Proof. After fixing an embedding, use the standard maximal-plane augmentation on the same vertex set: join the components by noncrossing edges in the outer face, and then add noncrossing diagonals until every face is a triangle. Here every component has at least six vertices, because $\delta(G) \geq 5$, so this construction introduces no small exceptional case and yields a simple plane triangulation T. Since the added edges only increase degrees, a degree-five vertex of T has degree five in G; and since G is a subgraph of T, a pair at distance at least three in T has that property in G. Moreover $\delta(T) \geq 5$. It suffices to prove the assertion in T. For a plane triangulation,

$$\sum_{z \in V(T)} (6 - d_T(z)) = 12,$$

so T has at least twelve degree-five vertices.

For a degree-five vertex x, put $W = N_T[x]$, $U = V(T) \setminus W$, $r = |U|$, and let B be the vertices at distance exactly two from x; write $b = |B|$. The five neighbours of x form a cycle in their cyclic order, so $e_T(W) \geq 10$. Contract the connected W and simplify parallel edges. Every vertex of B has a neighbour in W, and after simplification at most one edge from the contracted vertex to each such vertex remains. Thus the resulting plane graph has exactly $e_T(U) + b$ edges. As $|V(T)| \geq 12$, the planar edge bound gives $e_T(U) + b \leq 3r - 3$. Since $e(T) = 3r + 12$,

$$\sum_{z \in U} d_T(z) = e_T(U) + e(T) - e_T(W) \leq 6r - b - 1.$$

Let F_x be the degree-five vertices at distance at least three from x. The same sum is exactly

$$6r - b - |F_x| + \sum_{z \in B} (d_T(z) - 5) + \sum_{\substack{z \in U \setminus B \\ d_T(z) \geq 6}} (d_T(z) - 6).$$

It follows that $|F_x| \geq 1$, and if $|F_x| = 1$, every vertex of B has degree five.

Suppose every pair of degree-five vertices at distance at least three contains w, and let L be all degree-five vertices except w. For every $x \in L$, the nonempty F_x must be $\{w\}$. Hence w is at

distance at least three from L, and every vertex at distance exactly two from x belongs to L. Take a shortest path from w to L, ending at x. The vertex two steps before x is also in L and is closer to w, a contradiction. □

Theorem 4.2. *For every nonempty planar graph G and every $k \geq 7$, $\mathcal{R}_k(G)$ is Hamilton-connected. Consequently $h(\mathcal{P}) = \mathrm{hc}(\mathcal{P}) = 7$.*

Proof. Prove the seven-colour assertion by induction on $|V(G)|$, allowing disconnected graphs throughout. If $|V(G)| \leq 5$, the graph is four-degenerate and Theorem 2.5 applies. For $|V(G)| \geq 6$, suppose first that some vertex v has degree at most four; it has a nonneighbour. The smaller $H = G - v$ is Hamilton-connected by induction. For every seven-colouring f of H, which has at least five vertices,

$$d_{\mathcal{R}_7(H)}(f) = \sum_{x \in V(H)} (6 - |f(N_H(x))|) \geq 6|V(H)| - 2|E(H)| \geq 12 > d_G(v).$$

Lemma 3.6 restores v.

Otherwise $\delta(G) \geq 5$. For prescribed distinct colourings α, β, choose an actual vertex w with $\alpha(w) \neq \beta(w)$. Lemma 4.1 supplies $S = \{u, v\}$, avoiding w, with $d(u) = d(v) = 5$ and $d_G(u, v) \geq 3$. Thus S is independent, no outside vertex is adjacent to both members, and $H = G - S$ is smaller and nonempty. Choose two nonadjacent neighbours x, y of u; they exist since its five neighbours cannot induce a planar K_5. Properly six-colour the planar $H - \{x, y\}$, and give both x, y colour seven. This is a proper colouring of H in which the neighbours of u use at most four colours, leaving at least three available. The restriction colourings differ at w, so Lemma 3.3 gives an α–β Hamilton path.

For $k \geq 8$, planar five-degeneracy and Theorem 2.5 suffice. For sharpness, use the following explicit model of the icosahedron. Put $\varphi = (1 + \sqrt{5})/2$ and take the twelve points

$$\{(0, \pm 1, \pm\varphi), (\pm 1, \pm\varphi, 0), (\pm\varphi, 0, \pm 1)\},$$

joining two points at distance 2. The six antipodal pairs are independent, and a direct check of the five neighbours of any point shows that they contain exactly one point from each of the other five pairs. Colour each antipodal pair with its own colour. Every vertex then sees all other five colours, so the colouring is frozen: no one-vertex recolouring is possible. Relabelling the six colours gives distinct frozen states, so the recolouring graph is disconnected. This is the standard icosahedral obstruction (see also [2]), and proves that both thresholds are seven. □

Planar five-degeneracy and Theorem 2.6 also give vertex-pancyclicity for $k \geq 8$.

4.2 Triangle-free and bipartite planar graphs

The critical five-colour step needs an adjacent-pair fibre, which is not a product of complete graphs. We give its local construction separately from the arbitrary-size lifting argument.

For $A, B \subseteq \{0, 1, 2, 3, 4\}$, put

$$F(A, B) = \{(a, b) \in A \times B : a \neq b\},$$

with adjacency given by changing one coordinate. When $|A|, |B|$ are three or four, exchange the coordinates if necessary so that $|A| \leq |B|$. There are exactly seven types $(|A|, |B|, |A \cap B|)$.

Lemma 4.3. *For each of the seven types*

$$331, 332, 333, 342, 343, 443, 444,$$

every edge of $F(A, B)$ lies on a Hamilton cycle. The type 333 is itself a 6-cycle. The types $342, 343, 443, 444$ are Hamilton-connected. If J is the common state set of one of the transitions allowed in Lemma 4.4, then every $p \in F(A, B)$ has a neighbour $q \in J$, and hence a p–q Hamilton path in $F(A, B)$.

Proof. Appendix A gives one short Hamilton cycle for each of the seven representatives and checks its orbit under the independent row and column permutations that agree on $A \cap B$ (with the stated side exchange for type 332). The resulting edge orbits are the full edge sets. The 53 endpoint words in Appendix A prove the four Hamilton-connected types. Their symmetry coverage and an exact subset dynamic program verify the complete endpoint classification.

For the transition exit, exchange coordinates if needed so that B is unchanged, and put $A_0 = A \cap A'$. All allowed transitions, including the unchanged case, give $|A_0| \geq 2$ and $|B| \geq 3$. For $p = (a, b)$ with $a \in A_0$, choose (a, d) with $d \in B \setminus \{a, b\}$; otherwise choose (c, b) with $c \in A_0 \setminus \{b\}$. Either choice is a neighbour in J. Deleting this edge from a Hamilton cycle gives the required p–q Hamilton path. □

Lemma 4.4. *Arrange $m \geq 3$ full fibres $F(A_i, B_i)$, $0 \leq i < m$, along a base path. Each list has size three or four in a fixed five-colour palette. At each step either both lists stay unchanged, or exactly one changes. A changed list is either replaced by another three-element list differing in one colour, or changes between a three-element list and a containing four-element list. Join the copies of equal assignments at consecutive layers. If an internal layer has $|A_r| = |B_r| = 4$, there is a spanning path between any prescribed first-layer and last-layer states.*

Proof. For each transition put

$$J_i = F(A_i \cap A_{i+1}, B_i \cap B_{i+1}).$$

Write $J_i^{[i]}$ and $J_i^{[i+1]}$ for its copies in the two adjacent fibres; for $z \in J_i$, the edge $z^{[i]}z^{[i+1]}$ is the corresponding cross edge. The endpoint lemma gives a forward path for every $p \in F(A_i, B_i)$ from p to some $z^{[i]} \in J_i^{[i]}$. It also gives the reverse path needed below: interchange the two endpoint pairs (A_i, B_i) and (A_{i+1}, B_{i+1}) (and exchange the two coordinate names if the convention $|A| \leq |B|$ requires it). The common state set is still J_i, so every $q \in F(A_{i+1}, B_{i+1})$ has a Hamilton path to some $z^{[i+1]} \in J_i^{[i+1]}$. Reversing this path and using $z^{[i]}z^{[i+1]}$ propagates an entrance state across the transition in either direction; the cross edge is an actual one-coordinate recolouring.

There are $m \geq 3$ layers; choose an internal r with $1 \leq r \leq m-2$ and $|A_r| = |B_r| = 4$. Propagate one reachable entrance from the prescribed first state through layers $0, \ldots, r-1$. The outgoing graph J_r has at least nine states: it has at least twelve when both lists are unchanged, and otherwise the changed list contributes three common values while the other list has four, giving at least $3 \cdot 4 - 3 = 9$ unequal assignments. The fibre $F(A_r, B_r)$ is one of $443, 444$ and is Hamilton-connected. From a reachable entrance p, all states of J_r except possibly p are therefore reachable; in particular two distinct common states are reachable.

If $r < m-2$, the next layer has one four-element list and is one of $342, 343, 443, 444$, hence Hamilton-connected. Given two distinct reachable entrances and any prescribed state q in that layer, choose an entrance different from q and use Hamilton-connectedness. Thus all states of the next layer are reachable. If $r = m-2$, the same argument reaches the prescribed final state directly.

Once a whole layer is reachable, every state in the following layer is reachable. For a target $q \in F(A_{i+1}, B_{i+1})$, use the reverse endpoint statement above to choose $z \in J_i$ and a Hamilton path from $z^{[i+1]}$ to q; the fully reachable incoming graph supplies the path to $z^{[i]}$, and the cross edge joins the two. This propagates to the final layer, where it includes the prescribed endpoint. Backtracking the finitely many choices gives one Hamilton path in every fibre; the fibres are disjoint and the selected cross edges use only legal common assignments, so no state is repeated or omitted. □

Theorem 4.5. *If G is a nonempty triangle-free planar graph, then $\mathcal{R}_k(G)$ is Hamilton-connected for every $k \geq 5$. For the class of nonempty triangle-free planar graphs, $h = \mathrm{hc} = 5$.*

Proof. For $k \geq 6$, use three-degeneracy and Theorem 2.5. For $k = 5$, induct on the order of G, permitting disconnected graphs. Orders at most three are handled by two-degeneracy and the same theorem. At larger orders, suppose first that some vertex v has degree at most two; it has a nonneighbour. The induction hypothesis gives a Hamilton-connected base of at least five states, thus of minimum degree at least three; Lemma 3.6 applies.

Assume now $\delta(G) \geq 3$. Let L be its degree-three vertices and $s = |L|$. The triangle-free planar edge bound gives

$$s \geq 8 + \sum_{d_G(z) \geq 4} (d_G(z) - 4).$$

Given distinct α, β, choose w on which they differ. There are two exhaustive cases.

If $L \setminus \{w\}$ contains adjacent u, v, denote their other neighbours by u_1, u_2 and v_1, v_2. The four actual outside vertices are distinct: a common neighbour of u and v would form a triangle. In $H = G - \{u, v\}$ the restriction endpoints still differ at w, so induction gives a base Hamilton path between them. Its fibre over f is

$$F(A_f, B_f), \qquad A_f = [5] \setminus \{f(u_1), f(u_2)\}, \quad B_f = [5] \setminus \{f(v_1), f(v_2)\}.$$

The lists have size three or four. One actual base move affects at most one of them and gives precisely a transition allowed in Lemma 4.4. Edges among the four outside vertices restrict the base colourings, but do not alter this statement.

There is a base colouring with both lists four-element: use Grötzsch's theorem [5] to three-colour $H - \{u_1, u_2, v_1, v_2\}$, then give u_1, u_2 colour four and v_1, v_2 colour five. Each pair is independent by triangle-freeness, and edges across the pairs have distinct colours. At least five global colour permutations give distinct double-four-list base states. Since the Hamilton path has only two endpoints, one is internal. Apply Lemma 4.4 with the specified local endpoint states to complete the restoration.

Otherwise $S = L \setminus \{w\}$ is independent and has at least $s - 1 \geq 7$ vertices; put $H = G - S$. For any outside vertex z of degree at least four, the displayed Euler inequality implies $d_G(z) \leq s - 4 < |S|$. The only possible degree-three outside vertex is w, whose degree is also less than $|S|$. Thus no outside vertex sees all of S. Choose $u \in S$ and two of its neighbours x, y. They are nonadjacent. Three-colour $H - \{x, y\}$ by Grötzsch's theorem and give both x, y colour five. At most two colours then occur on the three neighbours of u, so u has at least three available colours. All conditions of Lemma 3.3 hold, and w is retained; it gives the desired endpoint path.

For sharpness, on the cube Q_3 view the colours as $\mathbb{Z}_2^2$ and set

$$f(x, y, z) = x(1, 0) + y(0, 1) + z(1, 1).$$

The three neighbours of every vertex have all three other colours, so the colouring is frozen and the complete four-colour recolouring graph is disconnected. The upper and lower bounds give both thresholds. □

Corollary 4.6. *The class of nonempty bipartite planar graphs has* $h = \mathrm{hc} = 5$. *For every graph in this class,* $\mathcal{R}_k(G)$ *is vertex-pancyclic for* $k \geq 6$.

Proof. Bipartite planar graphs are triangle-free, and the cube belongs to this class. The vertex-pancyclic assertion follows from three-degeneracy and Theorem 2.6. □

The same vertex-pancyclic bound $k \geq 6$ holds for all triangle-free planar graphs. This conclusion uses the degenerate theorem separately from Hamilton-connectedness.

4.3 Girth at least six and forests

Lemma 4.7. *If* G *is planar,* $\mathrm{g}(G) \geq 6$, *and* $\delta(G) \geq 2$, *then for any* $w \in V(G)$ *there are two degree-two vertices different from* w *whose distance is at least three.*

Proof. Every component has a cycle. For a connected component C, every facial walk in a fixed plane embedding contains a simple cycle and therefore has length at least six; a bridge, when present, is counted twice. To see the first assertion, remove the immediate backtracking pairs contributed by bridge excursions. The remaining closed walk cannot lie in a forest: a leaf of that forest would force an immediate reversal. Thus it contains a simple cycle. Consequently

$$2e(C) \geq 6f(C), \qquad 2e(C) \leq 3v(C) - 6$$

by Euler's formula. Writing $n_2(C)$ for the number of degree-two vertices of C, the degree sum is

$$2e(C) = 3v(C) - n_2(C) + \sum_{d_C(z) \geq 3} (d_C(z) - 3),$$

and hence

$$n_2(C) \geq 6 + \sum_{d_C(z) \geq 3} (d_C(z) - 3) \geq 6.$$

Thus every component has at least six degree-two vertices. If there are at least two components, choose such a vertex different from w in each of two components. Suppose G is connected and let n_2 be its number of degree-two vertices. The displayed component inequality gives $n_2 \geq 6 + \sum_{d_G(z) \geq 3}(d_G(z) - 3)$. Let L consist of all degree-two vertices except w; then $|L| \geq n_2 - 1 \geq 5$. Suppose all pairs in L have distance at most two.

If $u, v \in L$ are adjacent, let a, b be their respective other neighbours. A further common vertex in the two closed distance-two balls centred at u and v would give a cycle of length at most five, and the same girth condition gives $d_G(a, b) = 3$. Thus their intersection is exactly $\{u, v, a, b\}$. Under the assumption that all pairs in L have distance at most two, a and b cannot both lie in L; hence at most three of these four vertices lie in L, a contradiction.

Otherwise L is independent. Its two-element neighbourhoods are pairwise intersecting; they are distinct because equal neighbourhoods would give a 4-cycle. At least four distinct pairwise intersecting two-element sets have a common element: if $\{a, b\}, \{a, c\}$ occur and a third set omits a, it must be $\{b, c\}$, after which no fourth distinct set is possible. Hence some vertex z is adjacent to all of L. Then $d_G(z) \geq n_2 - 1 \geq 5$, whereas the Euler inequality gives

$$n_2 \geq 6 + d_G(z) - 3 \geq n_2 + 2,$$

a contradiction. □

Theorem 4.8. *For every finite integer $g \geq 6$, the class of nonempty planar graphs of girth at least g has $h = \mathrm{hc} = 4$. Every graph G in this class has Hamilton-connected $\mathcal{R}_k(G)$ for $k \geq 4$, and vertex-pancyclic $\mathcal{R}_k(G)$ for $k \geq 5$.*

Proof. For $k \geq 5$, use two-degeneracy and Theorems 2.5 and 2.6. At $k = 4$, induct on the order, starting with orders at most two, which are forests and covered by the one-degenerate theorem. If $|V(G)| \geq 3$ and $d_G(v) \leq 1$, then v has a non-neighbour. The smaller nonempty base is Hamilton-connected by induction and has at least four states, hence minimum degree at least three. Restore v using Lemma 3.6.

Otherwise apply Lemma 4.7 to a vertex w at which the prescribed endpoint colourings differ. It gives separated degree-two vertices u, v, both different from w. Set $S = \{u, v\}$ and $H = G - S$. The set S is independent, no outside vertex sees both its members, and H is nonempty. The neighbours x, y of u are nonadjacent. The two-degenerate $H - \{x, y\}$ can be three-coloured; giving x, y colour four leaves three colours available at u. Lemma 3.3 now restores a Hamilton path between the actual prescribed states.

For the lower bound, take a cycle of length at least g divisible by three and repeat the colours $0, 1, 2$ around it. This is a frozen proper three-colouring, so its complete three-colour recolouring graph is disconnected. Both class thresholds are therefore four. □

Theorem 4.9. *The class of nonempty forests has $h = \mathrm{hc} = 4$. Every such forest F has Hamilton-connected and vertex-pancyclic $\mathcal{R}_k(F)$ for $k \geq 4$.*

Proof. The upper bounds follow from one-degeneracy and Theorems 2.5 and 2.6. For the Hamiltonian lower bound, $\mathcal{R}_3(P_3)$ has twelve states. Its six states using three distinct colours have degree two, so every edge incident with them is forced in a spanning cycle. For each of the three middle-vertex colours b, these forced edges form a 4-cycle

$$(a, b, a), (c, b, a), (c, b, c), (a, b, c), \qquad \{a, b, c\} = \{0, 1, 2\}.$$

The three disjoint forced cycles prevent a Hamilton cycle. For the Hamilton-connected lower bound, $\mathcal{R}_3(K_2) = C_6$ is already not Hamilton-connected. □

5 Hamiltonian four-colour subclasses at girth five

The reductions below splice Hamilton cycles and establish Hamiltonicity at four colours. They do not prescribe arbitrary endpoints, so the Hamilton-connected thresholds of these subclasses remain in $\{4, 5\}$. Appendices B–J supply their local constructions.

All facial statements in this section refer to the fixed plane embedding specified in the notation. A hypothesis such as "G is planar" is an abstract-graph hypothesis; the embedding is chosen when a face count or a facial cycle is used.

A pentagon is *ordinary* if all five of its vertices have degree three. These degrees are measured in the graph before deletion. Vertices of degree three are called *light*, and vertices of degree at least four are called *heavy*; thus the heavy vertices are precisely $\mathcal{H}_4(G)$.

5.1 Graphs with few five-cycles

Lemma 5.1. *Let $S = \{u_1, \ldots, u_s\}$ be independent, $s \geq 3$, and suppose $N(u_i) = \{w, x_i\}$, where $N(w) = S$, $w \notin S$, and the x_i are distinct and belong to $J = G - (S \cup \{w\})$. If $\mathcal{R}_4(J)$ is Hamiltonian and some proper colouring of J gives the same colour to two distinct x_i, then $\mathcal{R}_4(G)$ is Hamiltonian.*

Proof. Take a Hamilton cycle C of $\mathcal{R}_4(J)$. Write three distinct chosen edges as $e_r = f_r g_r$ $(r = 1, 2, 3)$. Since one edge of C changes only one vertex of J and the x_i are distinct, choose $j(r)$ so that $x_{j(r)}$ is unchanged on e_r, and put $a_r = f_r(x_{j(r)}) = g_r(x_{j(r)})$.

There is a tree T on the four colours whose edge t_r contains a_r. Indeed, if the three labels are distinct, use the star centred at the fourth colour; if exactly two labels are equal, say a, a, b, use the edges ab, ac, bd, where c, d are the two remaining colours; and if all labels are equal, use the star centred at that common label. Make four copies of C, indexed by the colour of w. For each r, remove the copy of e_r from the two copies indexed by the ends of t_r, and add the two cross edges joining corresponding J-colourings while changing the colour of w. These are legal edges of $\mathcal{R}_4(G-S)$ because w is isolated in $G-S$. The three two-edge switches join the four cycles according to the tree T, so the resulting 2-regular connected graph is a Hamilton cycle B of $\mathcal{R}_4(G-S)$.

The only edges of B that change the colour of w are these cross edges. At the one indexed by r, the unchanged vertex $x_{j(r)}$ has colour a_r, and a_r is one of the two endpoint colours of t_r. Thus every such transition has an unchanged x_j whose colour is one of the two colours of w; this is the required control property.

Over a base colouring f the u_i-lists are

$$A_i(f) = [4] \setminus \{f(w), f(x_i)\}.$$

They have size at least two. At a move in J, one list changes and another coordinate is unchanged. At a move of w, all intersections are $[4] \setminus \{a, b, f(x_i)\}$, and the control property leaves a coordinate with a two-element intersection. Thus the graph of common assignments contains an edge at every transition, and the cyclic product-fibre lift applies once we exhibit its nonbinary fibre and the two distinct edges in the adjacent graphs of common assignments. Let h be the assumed colouring of J with $h(x_p) = h(x_q) = c$, and take the base state with restriction h and w coloured c. The two lists at u_p, u_q then both have size three. If an incident edge of B changes a vertex of J, at most one of these two lists changes, so the other unchanged list supplies a K_3 in the graph of common assignments. If it changes w from c to b, both common lists contain $[4] \setminus \{c, b\}$ and hence supply the two distinct edges of $K_2 \square K_2$. Consequently the two adjacent graphs of common assignments contain at least two distinct edges, from which distinct edges can be chosen for the cyclic lift. Its fibre over the displayed base state has a factor K_3; the lift therefore gives a Hamilton cycle through every extension of every base state, proving the lemma. □

Lemma 5.2. *Let C be a connected plane graph of girth at least five and minimum degree at least two. Let p be the number of 5-faces, $Q = \sum_{|F|\geq 7}(|F| - 6)$, $s = |\{v : d(v) = 2\}|$, and $D = \sum_v (d(v) - 3)$ over vertices of degree at least three. Then*

$$p = 12 + Q + 2D - 2s.$$

Every 5-face is a simple 5-cycle, and distinct faces give distinct cycles unless $C = C_5$.

Proof. A facial walk has no immediate reversal and contains a cycle, hence has length at least five. Counting face lengths gives $2e = 6f - p + Q$, while the degree sum is $2e = 3n - s + D$; with $n - e + f = 2$, subtraction yields the identity. A repeated vertex on a facial walk of length five would give a shorter cycle. If one 5-cycle bounds two faces, all its vertices have degree two and connectedness gives $C = C_5$. □

Corollary 5.3. *A planar graph of girth at least five with at most eleven distinct 5-cycles is two-degenerate.*

Proof. Otherwise a connected subgraph has minimum degree at least three. In its plane embedding the identity has $s = 0$, and therefore at least twelve 5-faces, which are distinct 5-cycles. This contradicts the hypothesis. □

5.2 Local degree-two reductions

Lemma 5.4. *Let u, v be adjacent degree-two vertices, with $N(u) = \{v, x\}$ and $N(v) = \{u, y\}$, where $x \neq y$. If $\mathcal{R}_4(G - \{u, v\})$ is Hamiltonian, then $\mathcal{R}_4(G)$ is Hamiltonian.*

Proof. For a base colouring f, the extension fibre is $F(A, B)$, where $A = [4] \setminus \{f(x)\}$ and $B = [4] \setminus \{f(y)\}$. For $A = B = \{a, b, c\}$ use the cycle (ab, ac, bc, ba, ca, cb); for $A = \{a, b, c\}$, $B = \{a, b, d\}$ use $(ab, ad, bd, ba, ca, cd, cb)$.

In fact, any two fibre Hamilton cycles have at least two common edges when at most one list changes. For an unchanged fibre, equal lists give C_6, while unequal lists give four distinct forced edges at the two degree-two states using the common colours. By exchanging coordinates suppose only A changes. If one value equals $B = \{a, b, c\}$, the common fibre is the path (ab, ac, bc, ba); its two end edges are forced in both cycles. Otherwise let a, b be the old and new excluded colours, c the fixed excluded colour and d the fourth colour. The common fibre has five vertices ca, cb, cd, da, db and six edges. The old outside states ba, bd occur consecutively because bd has degree two with neighbours ba, cd. Thus the old cycle uses exactly four common-fibre edges; the new cycle does likewise with ab, ad, cd. The cycles share at least $4 + 4 - 6 = 2$ edges.

Since $x \neq y$, a base step changes at most one list. Cut an edge of a base Hamilton cycle and process its fibres in path order. At each splice the current fibre has lost at most one edge and the next is fresh, so a common edge remains. Cutting its two copies and adding the two actual base-step edges merges the circles. The fibres partition all states, giving a spanning cycle. □

Lemma 5.5. *Let G be planar of girth at least five and let u, v be nonadjacent vertices of degree two with $N(u) \cap N(v) = \varnothing$. If $H = G - \{u, v\}$ has Hamiltonian $\mathcal{R}_4(H)$, then $\mathcal{R}_4(G)$ is Hamiltonian.*

Proof. Set $S = \{u, v\}$. Every vertex of H has at most one neighbour in S. Let x, y be the neighbours of u; they are distinct and nonadjacent. The planar triangle-free graph $H - \{x, y\}$ admits a proper three-colouring by Grötzsch's theorem [5] (the empty graph has its empty colouring). Use colours $1, 2, 3$ there and give both x, y colour 4. This is a proper colouring of H leaving three colours available at u. The hypotheses of Lemma 3.5 hold with $k = 4$ and $|S| = 2$, so that lemma supplies the required Hamilton cycle. □

5.3 Restoring a $(2, 3, 2)$ path

The following reduction uses only a Hamilton cycle of the deleted graph. It processes an entire constant-boundary run together with its two neighbouring nonconstant fibres, retaining the edges needed for the next splice. The three external neighbours remain distinct actual vertices.

Proposition 5.6. *If $F = \mathcal{R}_3(P_3)$, then $F \,\square\, P_\ell$ is Hamiltonian for every $\ell \geq 2$.*

Proof. Partition F into its three centre-colour squares Q_b. Order each square as (q_0, q_1, q_2, q_3), where q_0, q_2 are its monochromatic-endpoint vertices. On $Q_b \square P_2$ use

$$(0, q_0), (1, q_0), (1, q_1), (0, q_1), (0, q_2), (1, q_2), (1, q_3), (0, q_3).$$

This cube cycle contains all four edges between layers 0 and 1. For larger ℓ, append each subsequent square by an ordinary edge splice using an internal edge of the newest layer. The induction invariant

is that the strip is one cycle, contains every vertex in the first ℓ layers, retains all four edges between layers 0 and 1, and has at least two internal cycle edges reserved for later splices in its newest layer. The initial strip has two such edges; after one splice the new layer has three, so a further splice can avoid any prescribed reserved edge. Thus the invariant holds for every $\ell \geq 2$, and the four original interlayer edges are never cut.

The three centre-colour squares of F are connected by three edges between monochromatic-endpoint vertices, one for each pair of centre colours. Choose a tree of two of these edges. Each chosen edge xy produces a square with vertices $(0,x),(1,x),(1,y),(0,y)$. Delete its two interlayer edges and add its two copies of xy, joining the corresponding strip cycles. The two selected interlayer edges at the middle strip are distinct. Two such splices give a single spanning cycle, for arbitrary $\ell \geq 2$. □

Let x, y, z be a path of degrees $2, 3, 2$. Its external neighbours, in that order, are p, q, r. Assume they are distinct and outside the path. For a base colouring with boundary $t = (f(p), f(q), f(r))$, write

$$B_t = \{(u,v,w) : u \neq v,\ v \neq w,\ u \neq t_0,\ v \neq t_1,\ w \neq t_2\}, \qquad u,v,w \in \{0,1,2,3\}.$$

A boundary is *bad* if all three entries agree, and *good* otherwise. Up to colour permutations and path reversal the good types are $001, 010, 012$. Their orders are $6+4+4=14$, $9+4+4=17$, and $6+6+4=16$, respectively. The bad type 000 has order 12.

Good fibres and safe edges. The cycles D_E, D_M, D_A for the normalized boundaries $001, 010, 012$ are given explicitly in Appendix B, together with the required edge counts; the coordinates are uvw. Allow all their colour images and path reversals. Legality, cyclic adjacency, distinctness, and the counts above verify that each is a Hamilton cycle of its fibre.

For a bad boundary aaa, partition B_{aaa} into squares Q_c indexed by $c \neq a$, with fixed middle coordinate c. A good neighbour of aaa is obtained by changing position j from a to $b \neq a$. If j is an end position, then for each $c \notin \{a,b\}$ there is exactly one edge common to the good fibre and Q_c. Let P_t consist of these two edges. For middle-change or all-distinct boundaries set $P_t = \varnothing$. A good-cycle edge outside P_t is called *safe*. Directly from Appendix B, every allowed cycle has the following properties:

1. For an end change, it contains the unique edge in each Q_c, $c \notin \{a,b\}$, and two opposite edges of Q_b.
2. For a middle change it has no vertex in Q_b and contains three edges of each of the other two squares.

For $t = 001$, the two special edges are 123–323 and 132–232; the two edges in Q_1 are 313–213 and 212–312. For $t = 010$, the retained three-edge paths are $121, 321, 323, 123$ and $132, 232, 231, 131$. These descriptions also prove the assertions for every symmetry image.

Lemma 5.7. *For adjacent good boundaries t, s, every allowed cycle at t and every allowed cycle at s have at least two common edges outside $P_t \cup P_s$.*

Proof. Partition the common fibre graph into its fixed-middle layers H_v and the edge set H_* changing the middle coordinate. In any block, if m edges remain outside $P_t \cup P_s$ and the two cycles retain x, y of them, they share at least $\max(0, x+y-m)$ safe edges. These bounds add over disjoint blocks. Appendix B classifies all allowed transitions and gives sufficient block counts for seven families, each yielding at least two.

Let a, b, c, d be distinct and use the selected cycles defined there. For $aab \to bab$, the common middle-b prism has no special edges. The E cycle retains one edge in each row triangle and two vertical edges. The cycle M_0 contains both these triangle edges, and M_1 both these vertical edges, giving two safe common edges in either case.

For $aba \to cba$, H_a has nine edges. Both cycles omit *bab–bac* and *bab–bad*: the M layer isolates *bab*, while the A path starts *bab*, *dab*. They retain three and five edges in this seven-edge universe, giving one common edge. In H_c they retain three and two of four edges, giving another. Both special sets are empty. Colour permutations and reversal transport all arguments, proving the assertion for every allowed labelled pair. □

Bad blocks.

Lemma 5.8. *Consider consecutive fibres with boundaries*

$$t, \underbrace{aaa, \ldots, aaa}_{\ell \text{ times}}, s, \qquad \ell \geq 1,$$

where t, s are good and each differs from aaa in one coordinate. When $\ell = 1$ suppose $t \neq s$. Choose any allowed good cycles at the two ends, and reserve at most one safe edge in each. The layered graph on these fibres has a Hamilton cycle retaining both reserved edges. Only consecutive-layer matching edges are used.

Proof. Call a common edge with Q_c *available* if it belongs to the chosen good cycle and is not its reserved edge. An end-change good cycle has available edges in all three squares: the two singleton edges are special, hence cannot have been reserved, and at least one of its two Q_b edges remains. A middle-change good cycle has at least two available edges in each of its two supported squares.

First suppose $\ell = 1$. Let S_L, S_R be the sets of square colours supported by available edges of the two chosen good cycles. Then $|S_L|, |S_R| \geq 2$ and $S_L \cup S_R = [4] \setminus \{a\}$: the only exception could be two middle changes to the same colour, which would give $t = s$.

Choose a common square with different available edges on the two sides. If either change is at the middle, any $c \in S_L \cap S_R$ works, since that side offers at least two edges. If both changes are at ends, choose $c \notin \{a, b_L, b_R\}$, where b_L, b_R are their changed colours. Its two singleton edges are different: different ends give different coordinate directions, while changes at the same end have $b_L \neq b_R$ and give opposite square edges.

Start with the two good cycles and three bad squares. Attach the chosen square to both good cycles using these different edges, and attach each other square to a supporting good cycle. These four incidences form a tree. Cuts at a good cycle lie in different middle-colour squares, and the two bridge-square cuts are distinct. Along each tree edge delete the two selected internal edges and add the two matching edges. Each splice joins disjoint cycles into one; all reserved edges survive.

Now let $\ell \geq 2$. Each good end supports at least two squares, so choose an available left edge in Q_c and an available right edge in Q_d with $c \neq d$. Build the three square-strip cycles as in Proposition 5.6. The cube formula there permits either perfect matching of a square to be used in the first layer, by rotating its cyclic order; the last layer of a two-layer strip uses the other matching. Consequently its first layer can retain any specified square edge. For a two-layer strip the same is true at its last layer; our two requirements are on different strips and are independent.

If $\ell \geq 3$, a last-layer requirement is met by appending each new square with a splice edge different from the required edge. There are initially two choices and subsequently three in the current last layer, so this is always possible. The new last square retains all edges except the splice edge. First-layer horizontal edges and all four original vertical edges survive these extensions.

Join the three strips using the two vertical-edge switches in Proposition 5.6; these do not affect either required horizontal edge. Finally join the left and right good cycles to their selected strip edges by two ordinary edge splices. The result is one Hamilton cycle. The reserved edges were never cut. Thus the construction works for every $\ell \geq 2$. □

Restoration over a base cycle.

Lemma 5.9. *Suppose the three external neighbours of a degree-$(2,3,2)$ path are distinct and outside it. If the recolouring graph of the deleted graph is Hamiltonian at four colours, then so is $\mathcal{R}_4(G)$.*

Proof. Fix an arbitrary base Hamilton cycle. There must be a good vertex on it. Otherwise every boundary is constant; consecutive constant triples can only be equal, as at most one coordinate changes. The external colour would therefore be fixed throughout the cycle. A global colour permutation of a proper colouring supplies another proper colouring with a different external colour, a contradiction to Hamilton coverage, so the cycle contains a good vertex.

Every good base vertex has at most one bad neighbour in the entire base recolouring graph: a two-colour boundary has exactly one possible coordinate change making it constant, and an all-distinct boundary has none. The full neighbouring colouring, if proper, is uniquely determined by that change. Hence no good vertex on the Hamilton cycle can lie between two bad vertices. In particular there is an edge joining two good vertices.

Cut such an edge. In the resulting base Hamilton path, group each maximal bad run together with its two neighbouring good vertices; make each remaining good vertex a singleton block. These blocks are disjoint and partition the path. A bad run has constant boundary, and its two neighbours are distinct. If the run has length one their boundaries are different: otherwise both neighbours are obtained from the intervening colouring by the same one-vertex recolouring and would be the same base vertex.

For each labelled good boundary fix one allowed cycle once and for all. Consecutive blocks meet at a good–good boundary transition. If the boundary changes, Lemma 5.7 provides two common safe edges. If it stays unchanged, the fixed cycles coincide and have at least $14 - 2$ safe edges. Select one connecting edge for each interblock transition in path order. If a singleton block already has an incoming selected edge, choose a different outgoing one, which is possible from the two candidates. In a nonsingleton block the two selected edges are in its distinct end fibres, so cannot conflict.

Use Lemma 5.8 to construct each nonsingleton block cycle preserving its selected end edges; singleton blocks already have their fixed cycles. Splice the resulting disjoint block cycles along the selected transitions. Each step joins two cycles into one, and every later selected edge is still present. The one-block case needs no final splice. All transitions are legal: a local step changes one path vertex, and a matching cross-edge performs its actual base recolouring while holding the local colouring fixed. Finally the fibres are disjoint and exhaust all proper colourings. Thus the resulting single cycle is Hamiltonian, including its closing edge. □

5.4 Paths with a common outside neighbour

For $r \in \{4,5\}$ and $t = (a, b_1, \ldots, b_{r-2}) \in Q^{r-1}$, where $Q = \{0,1,2,3\}$, let $F_t^{(r)}$ consist of the proper colourings of the path $v_0 \cdots v_{r-1}$ that avoid the respective colours $(a, b_1, \ldots, b_{r-2}, a)$. Two words are adjacent when they differ in one coordinate. The palette Q is a relabelling of $[4]$; the first boundary coordinate is the colour of one outside neighbour shared by the endpoints.

Lemma 5.10. *There are Hamilton cycles $C_t^{(r)}$ in all $F_t^{(r)}$ such that whenever t, t' differ in one coordinate, the chosen cycles have at least two common edges.*

Proof. Appendix E specifies five and fifteen explicit cycle representatives for $r = 4$ and $r = 5$, respectively. Normalize colours in first-appearance order, append unused colours in increasing order, and restore each word by the inverse palette map. The computation checks coverage and cyclic adjacency for all 64 and 256 labelled boundaries, then intersects the chosen cycle edge sets at all 576 and 3,072 directed one-coordinate transitions. The minimum intersections have three and five edges, respectively. In particular, changing a changes both endpoint exclusions in one step. □

Lemma 5.11. *Let G have girth at least five. Suppose $v_0 \cdots v_{r-1}$ is a path, where $r \in \{4, 5\}$, its endpoints have degree two and a common neighbour w outside the path, and its internal vertices have degree three. Put $H = G - \{v_0, \ldots, v_{r-1}\}$. If $\mathcal{R}_4(H)$ is Hamiltonian, then $\mathcal{R}_4(G)$ is Hamiltonian.*

Proof. There are no additional internal edges. The third neighbours of the internal vertices are distinct and different from w: a prohibited coincidence would create a triangle or a 4-cycle. The extension fibres are therefore exactly the graphs in Lemma 5.10.

Cut an edge of an arbitrary base Hamilton cycle and process its fibres in path order, giving each its specified cycle. An outside move either leaves the boundary word unchanged or changes one boundary coordinate; changing the shared coordinate a changes both endpoint exclusions. The chosen cycles share at least two internal edges. Both endpoints of a common edge are legal in both fibres, so the two matching cross edges perform the actual outside recolouring.

The current last fibre has lost at most its one incoming splice edge. Choose a remaining common edge, remove it from the current cycle and the fresh fibre cycle, and add the two matching cross edges. This joins the cycles. Iteration gives one cycle through every extension of every base colouring, and hence a Hamilton cycle of $\mathcal{R}_4(G)$. □

5.5 Further degree-two reductions

A shared-hub path has degree-two endpoints with the same neighbour outside the path. For $m \in \{5, 6\}$, a tailed m-cycle consists of an induced cycle $v_0 \cdots v_{m-1}v_0$ and an additional vertex u whose only neighbour on the cycle is v_0, where

$$d(v_0) = 4, \qquad d(v_1) = d(u) = 2, \qquad d(v_i) = 3 \quad (2 \le i < m).$$

The other neighbour of u is outside this configuration. All degrees in the following lemma are degrees in G.

Lemma 5.12. *Let G have girth at least five, and let X be one of the following configurations:*

1. *a degree-four vertex and three of its degree-two neighbours;*
2. *an induced shared-hub path with degree sequence $(2, 4, 3, 2)$, $(2, 4, 3, 3, 2)$, $(2, 3, 4, 3, 2)$, or $(2, 3, 3, 3, 3, 2)$, allowing reversal;*
3. *a tailed 5-cycle or tailed 6-cycle.*

If $\mathcal{R}_4(G - X)$ is Hamiltonian, then $\mathcal{R}_4(G)$ is Hamiltonian.

Proof. Appendix F specifies seven explicit cycle families for these configurations. Each boundary position records the colour of an outside neighbour; for a shared-hub path, one position determines the exclusions at both endpoints. Colours are normalized by first appearance, with unused colours appended in increasing order. The inverse palette map selects one fixed cycle at every labelled boundary.

Two distinct boundary positions can denote the same outside vertex only if all pairs of their local neighbours have internal distance at least three; otherwise a triangle or a 4-cycle results. Appendix F

lists all allowed pairs. Their compatibility graphs have no triangle, so no outside vertex can occupy three distinct boundary positions. Several disjoint pairs of positions may be identified, but an actual outside move changes one singleton boundary coordinate or one listed pair of boundary coordinates; a listed pair may represent one outside vertex. Both entries of a pair change together from the same old colour to the same new colour.

The computation reconstructs the seven fixed internal graphs and all legal states, checks the 421 normalized cycles, and verifies all 8,704 labelled boundaries and 158,976 directed transitions, including the paired changes. The chosen cycles share at least two edges at every transition. A separate computation restores every cycle to the original colour labels and directly checks the same transition inequalities. Table F.1 gives the individual counts.

Cut an edge of a base Hamilton cycle and process its fibres in path order. Consecutive fibres have at least two common chosen-cycle edges; an unchanged boundary has identical chosen cycles. The current last fibre has lost at most its incoming splice edge. Choose another common edge, remove its two copies, and add the two matching cross edges, which perform the actual outside recolouring with the local state fixed. This joins two cycles and preserves the invariant. Iteration gives one cycle through all extensions of all base colourings, as required. □

5.6 Linked pentagons and three hub contacts

Two ordinary pentagons sharing exactly one edge have the model

$$(0, 1, 4, 3, 2, 0), \qquad (0, 1, 7, 6, 5, 0),$$

possibly with the extra edge 36. The shared endpoints already have degree three; among the remaining vertices, only 36 is compatible with girth five. The exterior boundary positions are $(2, 3, 4, 5, 6, 7)$ in the bare model and $(2, 4, 5, 7)$ in the chorded model. Each boundary position excludes one outside colour; every other list is full.

We also use two disjoint ordinary pentagons joined by one edge, or by two edges whose endpoints are nonadjacent on each pentagon. Their models are the cycles 012340 and 567895, with edge 05 and, in the second case, edge 27. Finally, the three-contact path has vertices $v_0, \ldots, v_6$ and exclusion pattern

$$(a, b, c, a, d, e, a).$$

Here a is one outside boundary position, with contacts at v_0, v_3, v_6.

Lemma 5.13. *The five models above have simultaneously selected Hamilton cycles at every labelled boundary, sharing at least two edges across every admissible outside recolouring.*

Proof. Appendix H supplies the explicit cycles. Its two computations reconstruct the legal states, check exact cyclic coverage, and verify every boundary transition. The numbers of representatives for the shared-edge bare and chorded models, the three-contact path, and the one-link and two-link models are respectively $187, 15, 51, 451, 67$. Their observed minimum common-edge counts are $56, 94, 10, 219, 390$.

The first three use first-appearance palette normalization, with unused colours appended increasingly. The linked models also minimize over their fixed graph automorphisms. The inverse maps select one cycle at each labelled boundary. An independent computation in the original colour labels reproduces the distributions of common-edge counts.

Two distinct exterior boundary positions may represent the same outside vertex only when all pairs of their contacts have internal distance at least three. The programs reconstruct these compatibility graphs and check their complete matchings of permissible identifications. None has a

compatible triple, so a move changes one boundary position or one equal pair simultaneously. This includes the single three-list hub move on the path. Static identifications of other disjoint pairs do not add another move type. All 75,008 labelled boundaries and 2,577,024 directed transitions are covered; Appendix H gives the individual counts. □

Lemma 5.14. *Let G have girth at least five, and let X be one of the following induced configurations:*

1. *the bare or chorded shared-edge model, with all eight vertices of degree three in G;*
2. *two disjoint ordinary pentagons with exactly one joining edge, or exactly two joining edges with nonadjacent endpoints on each cycle;*
3. *a path $v_0 \cdots v_6$ with degree-two endpoints, all other vertices of degree three, and a vertex w outside it adjacent to v_0, v_3, v_6.*

If $\mathcal{R}_4(G - X)$ has a Hamilton path, then $\mathcal{R}_4(G)$ is Hamiltonian.

Proof. The actual extension fibres have exactly the prescribed models and outside boundary positions. In the last case girth excludes any further contact of w with the path; its other neighbours are unrestricted. Give each fibre its selected cycle and process the base Hamilton path in order. An unchanged boundary has identical cycles. Otherwise Lemma 5.13 supplies at least two common edges. The current last fibre has lost at most its incoming splice edge, so another common edge remains. Remove its two copies and insert the two matching cross edges, which perform the actual outside recolouring with a local assignment legal on both sides. Each switch joins two cycles. Iteration covers all extensions of all base colourings. □

Lemma 5.15. *Let C, D be distinct ordinary facial pentagons in a plane graph G of girth at least five. There is a set $X \subseteq V(C) \cup V(D)$ such that Hamiltonicity of $\mathcal{R}_4(G - X)$ implies Hamiltonicity of $\mathcal{R}_4(G)$.*

Proof. If the two faces intersect, every shared vertex lies on a shared edge, since it has degree three. Shared edges cannot be consecutive, since only one facial angle uses a given pair of edges at such a vertex. If there are r common edges, their union has $10 - 2r$ vertices and $10 - r$ edges. The planar girth-five edge bound gives

$$10 - r \le \frac{5}{3}(8 - 2r), \qquad 7r \le 10,$$

so $r = 1$. Its induced union is the bare or chorded shared-edge model, and Lemma 5.14 applies.

Otherwise the cycles are disjoint, and their joining edges form a matching. If two matching endpoints are adjacent on one cycle, their mates are nonadjacent on the other, by girth. The adjacent edge, the two joining edges and a two-edge arc between those mates form an ordinary pentagon sharing exactly one edge with the first cycle. Its induced shared-edge union is again one of the two checked models; the constructed pentagon need not be facial.

In the remaining case the matching endpoints are independent on both cycles, so there are at most two joining edges. With no joining edge, Lemma 5.28 applies. With one or two, use Lemma 5.14. □

5.7 Pentagonal reservoirs

Lemma 5.16. *With palette $Q = \{0, 1, 2, 3\}$, the following models admit fixed Hamilton cycles in every component of every boundary fibre. The last column lists the boundary positions whose colours*

each local vertex must avoid; cycle vertices are ordered $v_0, \dots, v_4$, followed by z when present.

model	*internal graph*	*excluded boundary positions*
R	$C_5 \sqcup \{z\}$	$\{0\}, \{1\}, \{2\}, \{3\}, \{4\}, \{5, 6\}$
F	C_5	$\varnothing, \{0\}, \{1\}, \{2\}, \{3\}$
P	$C_5 + v_0 z$	$\varnothing, \{0\}, \{1\}, \{2\}, \{3\}, \{4\}$

Every fibre of F and P is connected. A fibre of R has two components exactly when its first five boundary entries are equal, and is otherwise connected. Across every admissible boundary move, each component on either side overlaps a component on the other side. For every overlapping pair, the selected component cycles have at least four common edges in R, and at least two in F and P.

Proof. Appendix G specifies the three explicit cycle families. Normalize colours by first appearance, appending unused colours in increasing order, and restore words by the inverse palette map. This selects the cycles simultaneously at all labelled boundaries. The computations reconstruct the legal states and components, check cyclic coverage and adjacency, and test all labelled transitions. A separate computation in the original colour labels reproduces the distributions of common-edge counts.

The three families have respectively $715, 15, 51$ normalized boundaries, $720, 15, 51$ component cycles, and 16,384, 256, 1,024 labelled boundaries. The numbers of directed boundary transitions are 466,944, 3,072, 16,896, with minimum common-edge counts $4, 11, 8$. For R, all 470,208 overlapping component transitions are checked.

In R, the outside vertices recorded in the first five positions are distinct, as are those recorded in the last two. The possible extra identifications are the pairs (i, j) with $0 \le i < 5$ and $j \in \{5, 6\}$. Their simultaneous identifications form the 31 matchings of this bipartite graph. In F all recorded outside vertices are distinct. In P the only possible pairs are $(1, 4)$ and $(2, 4)$, and they cannot both coincide. Girth excludes every other identification. Checking all labelled moves at one position and simultaneous moves at equal pairs covers these possibilities, including other static identifications in the same boundary. □

Lemma 5.17. *Let G have girth at least five, and let X be one of the following:*

1. *an ordinary pentagon and a degree-two vertex outside it with no neighbour on the pentagon;*
2. *a pentagon with one degree-two vertex and all other vertices of degree three;*
3. *an ordinary pentagon and an adjacent degree-two vertex outside it.*

If $\mathcal{R}_4(G - X)$ is Hamiltonian, then $\mathcal{R}_4(G)$ is Hamiltonian.

Proof. Girth excludes chords and gives exactly the three models of Lemma 5.16. In the last case the outside degree-two vertex meets the pentagon only once; deleting it leaves a full list at that cycle vertex. Every matching cross edge at a common local state performs the actual base recolouring, since that state is legal in both fibres.

Cut a base Hamilton cycle into a path and give each component of each fibre its fixed selected cycle. For the two connected-fibre models, splice successive cycles in path order. The current last fibre has lost at most one incoming splice edge. Of at least two common cycle edges with the next fresh fibre, one remains; deleting its two copies and adding the matching cross edges joins the cycles. An unchanged boundary uses identical cycles.

For R, form a graph whose nodes are the components of the layers, joining overlapping components in consecutive layers. It has maximum degree at most four. Some layer is connected. Otherwise the five cycle-neighbour colours form a constant word at every base state. A recolouring move changes

at most one of these five entries, so that constant word is unchanged along the base path. A palette permutation changing its colour produces another proper base colouring, contradicting Hamilton coverage.

Every component can step to a component in either adjacent layer. Walking towards a connected layer therefore proves that the component graph is connected. Choose a rooted spanning tree and attach each fresh component cycle to its already attached parent. Before an attachment the parent has lost at most three internal edges, since its tree degree is at most four and one attachment remains. At least four common selected-cycle edges are available, so an uncut edge remains for the switch. An unchanged boundary uses identical component cycles, which also have at least four edges. Each switch joins two cycles. The final cycle covers every component of every fibre, hence all proper colourings of G. □

5.8 Degree-two vertices in a minimum counterexample

In the next two lemmas, a minimum counterexample means a graph of minimum order with non-Hamiltonian $\mathcal{R}_4(G)$ in a fixed class of nonempty planar graphs of girth at least five closed under nonempty vertex deletion. Thus every smaller nonempty induced graph has a base Hamilton cycle.

Lemma 5.18. *A minimum counterexample is connected and has minimum degree at least two. Let S be its set of degree-two vertices. If $|S| \geq 2$, its members have a common heavy neighbour. If $S \neq \varnothing$, every 5-cycle meets a heavy vertex.*

Proof. The singleton case gives K_4. Products of Hamiltonian graphs are Hamiltonian, by alternating two distinct splice edges along a factor Hamilton path, so G is connected. Lemma 2.2 excludes degree at most one, and Lemma 5.4 makes S independent.

Suppose $|S| \geq 2$. If no outside vertex is adjacent to all of S, every outside vertex has at most $|S| - 1$ neighbours in S. Put $H = G - S$ and choose $u \in S$ with neighbours x, y. They are nonadjacent by girth. Three-colour $H - \{x, y\}$ by Grötzsch's theorem [5], and give both x, y colour four. This leaves three colours available at u, so Lemma 3.5 contradicts minimality. There is consequently a common neighbour w outside S. It cannot have degree two, since S is independent, and degree three would give Lemma 5.9; the three external neighbours of that path are distinct by girth. Thus w is heavy. Write $N(u_i) = \{w, x_i\}$; the x_i are distinct because a repetition creates a 4-cycle.

If $S = \{z\}$, a pentagon avoiding heavy vertices either contains z and has all other degrees three, or is ordinary. Pair an ordinary pentagon with z, using its adjacency or nonadjacency to the cycle. All three possibilities are excluded by Lemma 5.17. If $|S| \geq 2$, a cycle containing a member of S must contain w. A pentagon avoiding heavy vertices is therefore ordinary; pairing it with any member of S gives the same contradiction. □

For a plane minimum counterexample write

$$s = |S|, \qquad h = |\{v : d(v) \geq 4\}|, \qquad D = \sum_{d(v)\geq 4} (d(v) - 3), \qquad Q = \sum_{|F|\geq 7} (|F| - 6),$$

and let p count pentagonal faces. Lemma 5.2 gives

$$p = 12 + Q + 2D - 2s. \tag{5.1}$$

Pentagonal faces give distinct 5-cycles unless $G = C_5$, already covered by Theorem 5.21.

Lemma 5.19. *If a minimum counterexample has $s = 1$, then*

$$10 + D + Q \le 3h, \qquad h \ge 5, \qquad p \ge 20.$$

If $s \ge 2$, write

$$N(u_i) = \{w, x_i\}, \qquad d(w) = s + t,$$

and let h' count the heavy vertices other than w, with $D' = \sum_{v \ne w,\ d(v) \ge 4} (d(v) - 3)$. Then $t \ge 1$ and

$$p = 6 + Q + 2t + 2D', \tag{5.2}$$

$$D' + 6 + Q \le 3h', \qquad D' \ge h' \ge 3. \tag{5.3}$$

Proof. If $s = 1$, every pentagonal face meets a heavy vertex by Lemma 5.18. Counting its incidences gives $p \le 3h + D$. Combine this with (5.1) and $D \ge h$ to obtain all three assertions.

For $s \ge 2$, substituting $D = s + t - 3 + D'$ gives (5.2). A pentagonal face having no heavy vertex other than possibly w must contain w. It cannot use two edges from w to S: its other two cycle vertices would give the regular shared-hub path of order four in Lemma 5.11. Thus it uses an edge from w to $Z = N(w) \setminus S$, so there are at most $2t$ such faces. Counting all remaining pentagonal faces at other heavy vertices gives $p \le 2t + 3h' + D'$. This proves (5.3), since $D' \ge h'$.

If $t = 0$, then $s = d(w) \ge 4$. Among the distinct x_i, two are nonadjacent, since otherwise three form a triangle. Put $J = G - (S \cup \{w\})$. Three-colour J with these two vertices removed by Grötzsch's theorem, then give both colour four. Induction on J and Lemma 5.1 contradict minimality. Therefore $t \ge 1$. □

For $s \ge 2$, retain the nonnegative deficit

$$R = 2t + 3h' + D' - p = 3h' - D' - 6 - Q \ge 0. \tag{5.4}$$

Let p_0 count pentagonal faces with no heavy vertex other than possibly w, and let $a(F)$ count incidences of heavy vertices other than w on a face F. Summing these incidences gives the exact decomposition

$$R = (2t - p_0) + \sum_{|F| \ne 5} a(F) + \sum_{|F| = 5,\ a(F) > 0} (a(F) - 1). \tag{5.5}$$

Every term is nonnegative. Incidences count repeated occurrences on a nonpentagonal facial walk.

Lemma 5.20. *In a connected plane graph of girth at least five and minimum degree at least two, a nonsimple facial walk has length at least ten. Put $Q = \sum_{|F| \ge 7} (|F| - 6)$. If $Q \le 1$, every face has a simple boundary of length at most seven, and distinct angles at any vertex belong to distinct faces.*

Proof. A facial walk is cyclically nonbacktracking. Splitting it at a repeated vertex gives two nonempty closed subwalks without internal immediate reversal. Each contains a simple cycle: a nontrivial closed walk in a tree reverses at an internal leaf. Both subwalks therefore have length at least five. If $Q \le 1$, no face has length at least eight, so every boundary is simple. A simple boundary visits a vertex at most once, proving the last assertion. □

5.9 Protected pentagon restoration

Theorem 5.21. *Let G be a nonempty finite simple planar graph of girth at least five and maximum degree at most three. Then $\mathcal{R}_k(G)$ is Hamiltonian for every $k \ge 4$. Disconnected graphs are allowed. The uniform Hamiltonian threshold of this class is exactly four, and its Hamilton-connected threshold lies between four and five.*

We prove a local restoration lemma using only Hamiltonicity of the smaller recolouring graph, and then supply the planar structure. Appendices C–D specify the component and product cycles, auxiliary edge sets and transition records for 2,304 labelled pentagon boundaries and 313 one-entry transitions from standard boundaries. We first prove the protected-edge invariant, then use it in an arbitrary-order restoration.

Local constructions. Put $Q = \{0, 1, 2, 3\}$. A boundary t has five entries in Q, with at most one entry $*$. Its local graph F_t consists of the proper C_5 colourings x satisfying $x_i \neq t_i$; $*$ excludes no colour. In the constructions $*$ is encoded by 4. There are eleven ordinary boundary orbits, with 590 legal states in total. The constant ordinary boundary 00000 has two components, each a 15-cycle; every other ordinary boundary orbit has one component. Eight of those ten nonconstant ordinary components have the printed Hamilton cycles, while 00001 and 00112 are the two exceptional components handled by path-block and path constructions. There are also eleven starred boundary orbits, with 811 legal states in total; each starred fibre is connected and has a printed Hamilton cycle. Thus the catalogue has twelve ordinary component records (the constant boundary is split) and eleven starred records, for 23 component types in all. The complete boundary-orbit and component index is Table 5.1.

The eleven ordinary boundary representatives, after rotation, reflection and colour permutation, are

$$00000,\ 00001,\ 00011,\ 00012,\ 00101,\ 00102,\ 00112,\ 00121,\ 00123,\ 01012,\ 01023,$$

and the eleven starred representatives are

$$0000*,\ 0001*,\ 0010*,\ 0011*,\ 0012*,\ 0101*,\ 0102*,\ 0110*,\ 0112*,\ 0120*,\ 0123*.$$

We record the four local facts used in the restoration as four lemmas.

Lemma 5.22. *Every one of the 23 component types B has a partition into paths $P_1, \dots, P_r$ of order two or three, with an actual B edge joining P_i to P_{i+1} for each $i < r$.*

Proof. Twenty-one component records have the printed Hamilton cycles. The two exceptional ordinary records use the 44-state ordered matching-chain blocks for 00001 and the 51-state Hamilton path for 00112 in Appendix C. Delete a cycle edge where necessary, and divide the resulting path into consecutive two-vertex blocks, with one three-vertex block if its order is odd. For 00001, Appendix C supplies the ordered path blocks and their actual connectors. The program checks coverage and every one-coordinate difference. □

For a Hamilton component A with specified cycle C_A, designate an ordered auxiliary edge set P_A, of size five for either constant component and size eight otherwise. Put $K_A = E(C_A) \setminus P_A$. The first two auxiliary edges join copies of C_A in a two- or three-row path product; the remaining auxiliary edges supply at least two splice reserves, all outside K_A. For an exceptional component, specify cycles of $A \square P_2$ and $A \square P_3$, and a set D_A of internal edges occurring in *every* row of both product cycles. For each of 00001 and 00112, Appendices C–D specify a two-row cycle and derive its three-row product cycle by the same three-path insertion rule. Both use $|D_A| = 13$ and an auxiliary edge set P_A of size two. Put $K_A = D_A \setminus P_A$; the two distinct auxiliary edges supply the splice reserves.

Here is the exact normalization convention. Enumerate $s = 0, \dots, 4$, then $\epsilon = +1, -1$, then the permutations $p \in S_4$ in lexicographic order. Take the first standard type $n_i = p(t_{s+\epsilon i})$, retaining $*$. Restoration puts the standard entry x_i at position $s + \epsilon i$ with colour $p^{-1}(x_i)$. In particular a standard boundary first normalizes by the identity.

Lemma 5.23. *Let the source be a standard component A of F_t, and change one non-star boundary entry. Normalize the target by the preceding rule. Every target component B whose restored state set meets A has an edge in common with K_A after restoration of K_B. For an unchanged boundary use the same component and the same map.*

Proof. The ordinary and starred orbit representatives cover $4^5 + 5 \cdot 4^4 = 2304$ labelled boundaries. The displayed cyclic words give their specified component cycles and the two-row exceptional product cycles: their entries are precisely the legal words, without repetition, and successive entries, including last and first, differ in one coordinate. Both three-row product cycles follow from the three-path insertion rule in Appendix C. For 00001, the 40-state core path together with the four displayed endpoint states covers the 44-state fibre; for 00112, the 47-state core path together with its four displayed endpoint states covers the 51-state fibre. These are the explicit `COREPATH-00001` and `COREPATH-00112` blocks. In both cases the resulting three-row product cycle retains all thirteen common edges in every row. Every listed common edge occurs in every row of both corresponding product cycles. Removing the listed auxiliary edge sets therefore defines the asserted protected sets. For the 313 standard-source transitions, Appendix D gives 47 individual common edges, twenty-one rows of selected vertices covering 236 cases, and six constant-source families covering 30 cases. Endpoint membership in the specified representatives and outside their auxiliary edge sets proves each individual record. The selector rows use fixed protected incident edges against the opposite component's full degree, with the prescribed first normalizer and auxiliary edge sets. The constant-source families use two normalizing directions and four protected target edges. This is a complete finite case proof for standard sources; labelled sources are obtained by transport, with the target normalized relative to the source as described after Lemma 5.25. □

Products and overlaps.

Lemma 5.24. *For any two of the 23 component types A, B, the graph $A \square B$ has a Hamilton cycle retaining a copy of every edge of K_A in every B row.*

Proof. Use the ordered path blocks $P_1, \dots, P_r$ of Lemma 5.22 for B. For a two-row block with rows b_0, b_1, remove the same auxiliary edge xy from the two copies of C_A and add

$$(x, b_0)(x, b_1), \qquad (y, b_0)(y, b_1).$$

This is one cycle on $A \square P_2$. For a three-row block use two distinct auxiliary edges x_1y_1, x_2y_2 and perform this operation between rows b_0, b_1 and b_1, b_2, respectively. The two cuts in the middle row are distinct, so the result is again one cycle. For either exceptional component, use the explicit two- or three-row product cycle; the specified cycle contains every edge of D_A in every row. In all cases every edge of K_A survives in every row. For an ordinary component retain two of the unused auxiliary edges as reserves; for an exceptional component the two auxiliary edges in its product cycle are the reserves. Thus every reserve is outside K_A.

Fix two such reserves e and f. The construction above puts both edges in every B-row of every block product cycle, and neither belongs to K_A, so either edge is available for a later switch. Between P_i and P_{i+1}, let $u_i \in P_i$ and $v_i \in P_{i+1}$ be the specified adjacent states. Cut the copies of e in the two rows u_i, v_i when i is even, and the copies of f when i is odd; add the two vertical edges obtained by holding the A-coordinate fixed and changing u_i to v_i. If two consecutive transitions use the same row, their opposite parity choices cut different reserve edges. Thus no block is cut at the same edge on both sides, no edge of K_A is cut, and every displayed cut lies in the two cycles being joined. The two-edge switch therefore joins two distinct cycles into one. Induction on the ordered

blocks proves that the partial product is one cycle after each switch, including the one-block case. All added edges are actual one-coordinate changes in B. □

Lemma 5.25. *For a fixed star position, or no star, let two boundaries be equal or differ in one non-star position. Any overlapping component pair has at least ten common vertices.*

Proof. An unchanged constant component has 15 vertices. Between a constant and nonconstant boundary, one additional colour is excluded at one coordinate of that 15-cycle. A cyclic permutation of its three colours preserves each component: an order-three action on a two-element component set is trivial. Hence each colour appears five times at each coordinate and precisely ten states remain. In every other case, both components are their entire local state sets. The intersection lists have size at least three except possibly one list of size two. Start at that position, proceed around the cycle, and finally avoid the two already coloured neighbours. This gives at least $2 \cdot 2^3 \cdot 1 = 16$ colourings. With no changed position the bound only improves. Full lists cause no difficulty. □

Relative restoration maps. Call a coordinate and colour map Ψ a restoration map for a boundary t when $\Psi^{-1}(t)$ is standard; it restores the specified cycles, auxiliary edge sets and protected sets of that standard type. Let t' equal t or differ from it in one non-star entry. Put $\Psi' = \Psi$ if $t' = t$, and otherwise $\Psi' = \Psi \circ N_u$, where $u = \Psi^{-1}(t')$ and N_u is the canonical restoration map of u. Transporting Lemma 5.23 by Ψ shows that overlapping components at t and t', restored by Ψ and Ψ', have a common protected edge; the program also verifies this rule over all labelled boundaries and admissible maps Ψ. The dependence of Ψ' on Ψ is necessary: for the labelled boundaries 00002 and 00003, both of type 00001, the two canonical maps restore protected sets with no common edge, although the components overlap.

Now let two consecutive layers carry maps related in this way in each factor; the two factors use independent maps and may change at the same base edge. If xx' is a common protected edge of overlapping first-factor components and $y_1, \dots, y_{10}$ are common states of overlapping second-factor components, then every $\{(x, y_i), (x', y_i)\}$ is an internal edge of both product cycles of Lemma 5.24 with the first factor as cycle factor, since these retain every protected edge in every row. Thus the two product cycles share at least ten internal edges.

Lemma 5.26. *Let $h_0 h_1 \cdots h_{m-1}$ be a Hamilton path in the recolouring graph of the graph outside two pentagons. For $j \in \{1, 2\}$, let $T_j(i)$ be the boundary list seen by the jth pentagon at h_i, and let $c_j(i) = 1$ when $F_{T_j(i)}$ is a constant two-component fibre, and $c_j(i) = 0$ otherwise. Form the graph whose vertices are the product components of $F_{T_1(i)} \square F_{T_2(i)}$, $0 \le i < m$, and whose edges record a common local state across $h_i h_{i+1}$, $0 \le i < m - 1$. This component graph is connected.*

Proof. Give the two components of a constant fibre labels $0, 1$ and give a nonconstant fibre the single label $*$. A base edge changes at most one entry in either boundary list. If both consecutive fibres of a factor are constant, their boundary lists are equal and the label is preserved; otherwise, by Lemma 5.25, every label on one side meets every label on the other. The two factors follow these rules independently, and this describes the product component graph exactly.

For each j, some r_j has $c_j(r_j) = 0$. Otherwise every $T_j(i)$ is a constant word. Two constant words differing in at most one position are equal, so the whole Hamilton path would have one fixed boundary word. A global permutation of the four colours changing that word would give another proper colouring of the outside graph, which the Hamilton path must contain, a contradiction.

From any product component, walk layer by layer to r_1 and then to r_2, choosing label 0 whenever a constant fibre is entered from a nonconstant one. The first label is $*$ at r_1 and then evolves by this fixed rule, while the second label is $*$ at r_2. Hence every walk ends at the same component of layer r_2, and the graph is connected. □

Lemma 5.27. *Let Q be a connected graph of maximum degree at most Δ. For each $v \in V(Q)$ let X_v be a vertex-disjoint fibre carrying a Hamilton cycle C_v. Suppose that, for every $uv \in E(Q)$, the two reference cycles share at least $r \geq \Delta$ internal edges, and each shared edge has the two corresponding cross edges in the ambient graph. Then the union of the fibres has a Hamilton cycle.*

Proof. Root a spanning tree of Q and process its vertices from the root outwards. Maintain one cycle on the fibres already processed; the only edges removed from a processed fibre are the internal edges used for its earlier tree attachments. When a child v is attached to its parent u, at most $\deg_Q(u) - 1 \leq \Delta - 1$ shared edges at u have been removed. Since $r \geq \Delta$, a shared edge xy remains at u and is still present in the fresh cycle C_v. Delete xy from both cycles and add the two cross edges at x and y. This joins the two cycles into one and preserves the invariant. Each tree edge is processed once, so the final cycle covers every fibre. □

Restoration over a base cycle.

Lemma 5.28. *Suppose G has girth at least five and contains two vertex-disjoint 5-cycles with no edge between them. Each cycle has vertices of degree three except for at most one vertex of degree two. Let H be obtained by deleting both cycles. If $\mathcal{R}_4(H)$ is Hamiltonian, so is $\mathcal{R}_4(G)$.*

Proof. Each degree-three cycle vertex has one exterior neighbour; a degree-two vertex gives a full list. An exterior vertex meets a given 5-cycle at most once, since two attachments would create a 3- or 4-cycle. Thus a base recolouring changes at most one list in each factor; the two factors may change simultaneously. All exterior neighbours remain in H, which is nonempty.

Delete one edge of a base Hamilton cycle to obtain a Hamilton path $h_0, \ldots, h_{m-1}$. In each factor give $T_j(0)$ its canonical restoration map, and propagate maps along the path by the relative rule above. For every product component at h_i, take the product cycle of Lemma 5.24 restored by the two maps at h_i, with the first pentagon as cycle factor. The component graph over the path is connected by Lemma 5.26. It has at most four vertices per layer, hence maximum degree at most eight. Along each of its edges the two product cycles share at least ten internal edges. A cross edge at either endpoint of such an edge is the actual base recolouring, because the complete local colouring is held fixed while one exterior vertex changes colour. Lemma 5.27 with $\Delta = 8$ and $r = 10$ gives one cycle through all extensions of all base colourings. □

5.10 Protected bowtie products

A bowtie consists of two pentagons meeting only at a vertex of original degree four, with all eight other vertices of original degree three. All degrees in these definitions are measured before deletion. Label its cycles $(0, 1, 2, 3, 4, 0)$ and $(0, 5, 6, 7, 8, 0)$. Girth excludes chords within either cycle. An extra edge can only join a vertex of $\{2, 3\}$ to one of $\{6, 7\}$. Two such edges either exceed a prescribed degree or form a 4-cycle with edges 23 and 67. Up to reflecting the arms, the induced union is therefore the bare bowtie or the same graph with edge 26 added. In the bare model the exterior boundary positions are $1, \ldots, 8$; in the chorded model they are $1, 3, 4, 5, 7, 8$. Each boundary position excludes its outside colour. The other local lists are full, including those at vertices $2, 6$ of the chorded model.

Lemma 5.29. *Let A be a graph, let $K \subseteq E(A)$, and let $e, f \in E(A) \setminus K$ be distinct. Suppose $A \square K_2$ and $A \square K_3$ have specified Hamilton cycles retaining $K \cup \{e, f\}$ in every row. Suppose B has a partition into cliques of orders two and three, with actual B-edges joining the blocks as a tree and at most two connector incidences at each B-vertex. Then $A \square B$ has a Hamilton cycle retaining every edge of K in every B-row.*

Proof. The connector graph on the actual vertices of B is a forest: a cycle would project to a closed trail with distinct edges in the quotient tree. Its maximum degree is two, so label the edges of each nontrivial component alternately e, f. Put the appropriate two- or three-row product cycle on each clique block. For a connector xy labelled $e = ab$, remove the copies $(a, x)(b, x)$ and $(a, y)(b, y)$, and add $(a, x)(a, y)$ and $(b, x)(b, y)$. Use f for the other label. No row uses the same reserve twice, so each cut remains present. The quotient tree ensures that each switch joins two distinct current cycles. No edge of K is removed, and the final cycle covers $A \square B$. Neither K nor the two reserves need form a matching. □

Lemma 5.30. *For every boundary t of either bowtie model, its local recolouring graph B_t has the following simultaneous finite data.*

1. *Hamilton cycles of $B_t \square K_2$ and $B_t \square K_3$ retain a matching M_t in every row. Two specified edges $e_t, f_t \in M_t$ are reserves, and $K_t = M_t \setminus \{e_t, f_t\}$ is the protected core.*
2. *The vertices of B_t have a partition into edges and triangles. Joining edges form a tree on these blocks, with at most two joining edges incident with each vertex of B_t.*
3. *Across every actual outside move, the selected protected edge sets in the original coordinates have at least nine common edges in the bare model and twenty-four in the chorded model. An unchanged boundary uses the same core.*

In particular every B_t is connected, and consecutive bowtie fibres have at least eighteen common states.

Proof. Appendix I supplies the two product cycle families and their trees of clique blocks for all 451 bare and 107 chorded canonical boundaries. Normalize colours by first appearance, append unused colours increasingly, and choose the first minimizing vertex permutation among the eight bare or two chorded symmetries. Restoring by this fixed map selects the product cycles in the original coordinates, matching and tree at every labelled boundary.

The two computations independently enumerate all legal states, verify each product cycle's complete coverage and closing edge, and check the matching in every row of both product cycles. The tree computation checks a partition without repetitions, every pairwise clique edge, all connectors, the quotient tree and the connector degree at each vertex. In particular, a three-vertex path is insufficient for a triangle block.

An outside vertex meets each pentagon at most once. Its possible boundary positions within a bowtie are thus singletons or pairs at internal distance at least three; no triple is possible. The programs check every such move on all 4^8 bare and 4^6 chorded labelled boundaries, including the simultaneous change of equal-coloured paired boundary positions. There are 2162688 and 89088 directed transitions. After the two reserves are removed, primary and independent original-label checks give identical transition histograms and the stated core margins. Unchanged boundaries retain the same core, of at least the corresponding minimum size. Finally, nine common matching edges have eighteen distinct endpoints. □

The ordinary-pentagon component product cycles already used in Lemma 5.23 retain K_A and two distinct reserves outside K_A in every row of $A \square P_2$ and $A \square P_3$. They are also valid product cycles for the corresponding clique products, since $P_3 \subseteq K_3$. The ordinary product cycles are checked again in Appendix I. Its protected sets are arbitrary edge sets, as permitted by Lemma 5.29.

Theorem 5.31. *Let G have girth at least five. Suppose an ordinary pentagon C and a bowtie X are vertex-disjoint, with no edge between them. Put $H = G - (V(C) \cup V(X))$. If $\mathcal{R}_4(H)$ is Hamiltonian, then $\mathcal{R}_4(G)$ is Hamiltonian.*

Proof. Cut a base Hamilton cycle into a path. The fibre over each base colouring is $A_t \square B_s$, where A_t is the ordinary-pentagon fibre and B_s is the appropriate bowtie fibre. Propagate the ordinary restoration maps along this path by the relative rule following Lemma 5.25; retain the map at an unchanged boundary. For each component A of A_t, apply Lemma 5.29 with its ordinary product cycles and the specified tree of clique blocks of B_s. This gives a cycle in $A \square B_s$ retaining every K_A-edge in every bowtie row.

Consecutive overlapping A-components have a common protected edge by Lemma 5.23. The bowtie fibres have at least eighteen common states by Lemma 5.30. Their selected product cycles therefore share at least eighteen internal edges. Each has its two actual matching cross edges: the whole local assignment is legal on both sides and is held fixed during the base recolouring. This also covers an outside vertex meeting one vertex of C and two vertices of X. The two deleted factors have no joining edge, so both fibres and transition intersections factorize despite shared outside neighbours.

Form the component graph over this base path. Each layer has at most two components, since B_s is connected, so its maximum degree is four. Every component crosses each adjacent transition, by the ordinary-pentagon overlap property and the bowtie state overlap. Some ordinary fibre is connected. Otherwise all five exterior colours of C would be constant at every base state. A move changes at most one of those entries, so the constant word could never change. A global palette permutation changing its colour produces another proper base colouring, contradicting coverage by the Hamilton path. These five distinct exterior neighbours belong to H, since C and X are nonadjacent, so H is nonempty.

Walking toward that connected layer proves the component graph connected. Lemma 5.27, with $\Delta = 4$ and $r = 18$, joins all its reference cycles. They cover precisely all proper colourings of G. □

Theorem 5.32. *Let G have girth at least five and contain two vertex-disjoint bowties with no edge between them. Delete their eighteen vertices to obtain H. If $\mathcal{R}_4(H)$ has a Hamilton path, then $\mathcal{R}_4(G)$ is Hamiltonian.*

Proof. Use the first bowtie as factor A and the second as factor B. Lemmas 5.29 and 5.30 give a Hamilton cycle in every fibre $A_t \square B_s$, retaining the first factor's core in every second-factor row. Across a base move, at least nine first-factor core edges and eighteen second-factor states are common. The two reference cycles therefore share at least $9 \cdot 18 = 162$ internal edges. This remains true if one outside vertex changes two boundary positions in each bowtie: local assignments and their intersections still factorize.

Process the base Hamilton path in order. The most recently attached fibre has lost only its incoming splice edge. Choose another common internal edge, remove its two copies and insert the two matching cross edges. Both cross edges are genuine outside recolourings. Each switch joins two cycles, and the final cycle covers all extensions of all base colourings. The outside graph is nonempty: each bowtie has exterior contacts and the other bowtie is nonadjacent. □

Remark 5.33. The bare boundary $(3, 0, 0, 2, 3, 0, 0, 1)$ has 1432 states and three leaves, so its fibre has no Hamilton path. The restoration uses trees of clique blocks and protected products; neither restoration assumes a spanning path in a bowtie fibre.

Lemma 5.34. *Let G have girth at least five. Suppose a bowtie X and a degree-two vertex z outside it are disjoint and have no edge between them. If $\mathcal{R}_4(G - (V(X) \cup \{z\}))$ has a Hamilton path, then $\mathcal{R}_4(G)$ is Hamiltonian.*

Proof. A fibre is $B_t \square K_A$, where A is the available set at z, of size two or three. Use the specified clique product cycle, identifying its rows with the colours in A. It retains every edge of M_t in every

row. Across an outside move, $A \cap A' \neq \varnothing$: the two neighbours of z are distinct, and only one can change. The bowtie matchings have at least nine common edges by Lemma 5.30, even after its reserves are removed. In any common colour row these give at least nine common product cycle edges. This remains valid when the move changes two bowtie boundary positions and one neighbour colour of z simultaneously. If the bowtie boundary is unchanged, the same matching is used.

Process the base path as in Lemma 5.14. Only one incoming edge has been cut at the newest fibre, so a common edge remains for each next switch. The result is one cycle through all extensions. No claim that an isolated bowtie fibre has a Hamilton path is needed. □

5.11 Pentagons sharing an edge at a heavy vertex

A *hinged pair* consists of the pentagons

$$(0, 2, 3, 4, 1, 0), \qquad (0, 1, 7, 6, 5, 0),$$

with vertex 0 of original degree four and all seven other vertices of original degree three. Its induced graph may also have edge 36. The exterior boundary positions are $(0, 2, 3, 4, 5, 6, 7)$ in the bare model and $(0, 2, 4, 5, 7)$ in the chorded model. Each boundary position excludes one outside colour. Vertex 1, and the chord endpoints when present, have full lists.

Two facial pentagons with the same sole degree-four heavy vertex form a hinged pair when their angles there are adjacent. At a common light vertex the faces share an edge, and common edges cannot be consecutive. If they share r edges, their union has $10 - 2r$ vertices and $10 - r$ edges. The planar girth-five bound gives

$$10 - r \leq \frac{5}{3}(8 - 2r), \qquad 7r \leq 10,$$

so $r = 1$. Among unsaturated local vertices only 36 can be added without creating a triangle or a 4-cycle. Opposite facial angles instead give the bare or chorded bowtie defined above: another common vertex would give two distinct paths of length at most two from the centre, and hence a triangle or 4-cycle.

Lemma 5.35. *Every labelled boundary t of either hinged-pair model has a selected matching M_t and Hamilton cycles of $A_t \square K_2$ and $A_t \square K_3$ retaining M_t in every row, where A_t is the local recolouring fibre. Two specified edges of M_t are reserves; the remaining core K_t has at least six common edges across every bare transition and twenty across every chorded transition. Every A_t also has a partition into edges and triangles, with joining edges that form a tree on the blocks and meet each fibre vertex at most twice. Thus the fibres are connected, and consecutive fibres have at least twelve common states.*

Proof. Appendix J supplies unchanged explicit product cycles and trees. The bare and chorded models have respectively 379 and 31 canonical boundaries, with 241774 and 26051 states in total. Normalize by first appearance after taking the minimum over the identity and the arm exchange $(0, 1, 5, 6, 7, 2, 3, 4)$; append unused colours increasingly. Minimizing the full normalization tuple fixes all ties. Its inverse selects data in the original coordinates at every labelled boundary.

The two computations independently reconstruct all legal states and check complete product cycle coverage, legality, closure and matching retention in every row. The first computation also checks every clique partition, connector, quotient tree and connector degree bound. An ordered three-vertex path does not suffice for a triangle block.

An outside vertex can occupy only one boundary position or two whose contacts have internal distance at least three. The complete bare pairs of boundary positions are

$$(1,5),(1,6),(2,4),(2,5),(2,6),(3,4),(3,5),$$

and the chorded pairs are $(1,4),(2,3)$. The compatibility graphs have no triangle; their complete matchings of permissible identifications number 22 and 4. Both computations check all 16384 and 1024 labelled boundaries and all 430080 and 16896 directed transitions, including simultaneous changes at paired boundary positions. The independent computation restores explicit colour words in the original coordinates at every boundary and reproduces the distributions of matching and protected-edge intersections. Their core minima are six and twenty; unchanged boundaries use identical cores of sufficient size. Six common matching edges give twelve distinct common states. □

Lemma 5.36. *Let G have girth at least five and contain a hinged pair X. Suppose X is separated from a degree-two vertex, another hinged pair, a bowtie, or an ordinary pentagon, where separated means vertex-disjoint with no edge between the sets. Delete both displayed configurations to obtain H. If $\mathcal{R}_4(H)$ has a Hamilton path, then $\mathcal{R}_4(G)$ is Hamiltonian. Permitted chords and shared outside neighbours are allowed.*

Proof. For a degree-two vertex z, the fibre is $A_t \square K_L$, where L is its available set of two or three colours. Use the corresponding clique product cycle. An outside move leaves $L \cap L' \neq \varnothing$, since the neighbours of z are distinct and only one can change. In a common row the product cycles retain at least six common matching edges, even if the same move changes two boundary positions of X. The base contains the two neighbours of z, so is nonempty.

For another hinged pair, apply Lemma 5.29 with X as factor supplying the product cycles and the second factor's specified tree of clique blocks. The resulting cycles retain the first core in every second-factor row. Consecutive cycles share at least $6 \cdot 12 = 72$ internal edges. For a bowtie use its tree of clique blocks and its eighteen common states from Lemma 5.30, giving at least $6 \cdot 18 = 108$ common internal edges. Fibre intersections factorize because the two deleted factors have no joining edge. These counts therefore include one outside move affecting two boundary positions in each factor simultaneously.

In these three cases process the base Hamilton path in order. The newest fibre has lost at most its incoming splice edge. Choose another common internal edge, remove its copies and add the two matching cross edges. Each cross edge performs the actual outside recolouring with a fixed local assignment legal on both sides. Each switch joins two cycles, and iteration covers all extensions.

For an ordinary pentagon C, use each ordinary component B as the factor supplying the product cycles and A_t as the factor partitioned into clique blocks. The ordinary P_2/P_3 product cycles are valid K_2/K_3 product cycles. Propagate the ordinary restoration maps along the base path by the relative rule following Lemma 5.25, retaining the map at an unchanged boundary. Lemma 5.29 gives a product cycle retaining every ordinary protected edge in every hinged row. Overlapping ordinary components have a common protected edge, and consecutive hinged fibres have twelve common states. Their product cycles thus share at least twelve internal edges, including combined two-boundary position/one-boundary position outside moves.

The component graph over the base path has at most two nodes per layer and maximum degree at most four. Every component crosses each adjacent transition. Some ordinary fibre is connected: otherwise its five exterior colours would be a constant word at every base colouring. A move changes at most one entry, so the word could not change, contrary to a global palette permutation and Hamilton coverage. Those five exterior neighbours belong to H. Walking to a connected layer proves the component graph connected. Apply Lemma 5.27 with $r = 12$ and $\Delta = 4$. The resulting cycle covers all proper colourings of G. □

Remark 5.37. The bare boundary $(2,2,1,1,3,1,1)$ has 526 states and three leaves, so its fibre has no Hamilton path. Appendix J checks this obstruction as well as the protected product cycles; the restoration uses the products.

5.12 Sharper bounds for degree-two vertices

Retain the minimum-counterexample convention and the notation of Lemma 5.19. All degrees and facial incidences below are measured in the original plane graph, before any deletion.

Lemma 5.38. *If a minimum counterexample has exactly one degree-two vertex, then it has* $|\mathcal{H}_4(G)| \geq 8$, $D \geq 16$, *and at least forty-two distinct* 5*-cycles.*

Proof. Let z have neighbours x, y. Every pentagon meets a heavy vertex by Lemma 5.18, so

$$p = 10 + 2D + Q, \qquad \Delta = 3h + D - p = 3h - D - 10 - Q \geq 0.$$

The deficit Δ counts nonpentagonal heavy incidences and excess heavy incidences on pentagons. Thus at most Δ pentagons have two or more heavy vertices.

Call a face near if it meets $N[z]$, and far otherwise. At most $d(x) + d(y) - 2$ pentagonal faces are near. If zx, zy are not bridges, their two distinct incident faces both contain x, y. If they are bridges, their common bridge face repeats both x and y, giving the stronger bound $d(x) + d(y) - 3$. Every face meeting z meets x, y. Put

$$B = \{x, y\} \cap \{v : d(v) \geq 4\}, \qquad D_B = \sum_{v \in B} (d(v) - 3),$$

and let b_B count the vertices of B of degree at least five. The near-face bound is $4 + D_B$.

Two sole-heavy far pentagons at a degree-four vertex form a hinged pair or a bare/chorded bowtie, according as their angles are adjacent or opposite. They are separated from z, contradicting Lemma 5.36 or Lemma 5.34. Hence a degree-four vertex is sole heavy on at most one far pentagon, and a vertex of higher degree is sole heavy on at most $d(v)$ of them. If a counts degree-four vertices, the sum of these bounds is

$$C = a + \sum_{d(v) \geq 5} d(v) = 3h + D - 3a.$$

Let C_B be its contribution at B. Since $D_B - C_B = -3b_B$,

$$p \leq 4 + D_B + C - C_B + \Delta \quad \Longrightarrow \quad 2D + 3a + 16 + 2Q + 3b_B \leq 6h.$$

Write $e = \sum_{d(v) \geq 6} (d(v) - 5)$, so $D = 2h - a + e$. The last inequality gives

$$D \geq 16 + 2Q + 3e + 3b_B, \qquad 2h \geq 16 + a + 2e + 2Q + 3b_B.$$

Thus $D \geq 16$, $h \geq 8$, and $p \geq 42$. These pentagonal faces give distinct 5-cycles, as noted after (5.1). □

Suppose now that $s \geq 2$. Write $N(u_i) = \{w, x_i\}$ and $Z = N(w) \setminus S$, where $|Z| = t \geq 1$. Recall that p_0 counts pentagonal faces having no heavy vertex other than possibly w. Such a face contains w and cannot use an S–S angle there. Set $b = 2t - p_0$.

Lemma 5.39. *Let the cyclic order of the neighbours of* w *be written as an* S/Z *word. If a* Z*-vertex is flanked by* S*-vertices, its two mixed faces cannot both be counted by* p_0*. If the lengths of the maximal* Z*-runs are* $\ell_1, \ldots, \ell_r$*, then*

$$b \geq \sum_{j=1}^{r} \max\{1, \ell_j - 1\} \geq 1. \tag{5.6}$$

Proof. Suppose the two faces beside wz, with $z \in Z$ isolated in the word, are pentagons

$$(w, u, x, a, z, w), \qquad (w, z, b, y, v, w),$$

where $u, v \in S$ are distinct. Every vertex other than w, u, v displayed here has degree three. A further degree-two vertex on either pentagon would have to use its edge to w on that cycle, contrary to the mixed angle.

The seven vertices u, x, a, z, b, y, v are distinct. An internal vertex cannot coincide with u, z, v without creating a triangle. The equalities $x = y$, $x = b$, and $a = y$ create 4-cycles through w. Finally $a = b$ is impossible, since the two faces beside wz use different other neighbours of the degree-three vertex z. The only possible chord of this seven-vertex path is xy: the degrees at u, v, z are exhausted, and any other edge among x, a, b, y creates a triangle or a 4-cycle. If xy exists, u, x, y, v is the regular shared-hub path of order four. Otherwise the induced path has the three-contact outside neighbour w at its first, fourth, and seventh vertices. Lemma 5.11 or Lemma 5.14, respectively, contradicts minimality.

There are $2t$ edge-side incidences at the Z-edges of w. Each p_0-face uses one at a mixed angle or two at a Z–Z angle. Thus each Z–Z angle costs at least one in b, and an isolated Z-vertex costs at least one by the preceding argument. These costs use disjoint incidences. A run of length $\ell \geq 2$ contains $\ell - 1$ Z–Z angles, proving the assertion. Pentagonal facial boundaries are simple by Lemma 5.20; repeated occurrences on longer faces do not affect this incidence count. □

Let T' be the heavy vertices other than w, and let a count its degree-four vertices. A clean face at such a vertex q is a pentagon whose sole heavy vertex is q and which avoids w.

Lemma 5.40. *Every degree-four vertex in T' has at most one clean face, with at most one exception. An exceptional vertex has exactly two clean faces, can occur only if $s = 2$, and requires both x_1, x_2 to have degree three. If κ counts exceptional vertices and C counts all clean incidences at degree-four vertices of T', then*

$$\kappa = 0 \ (s \geq 3), \qquad 0 \leq \kappa \leq \min\{1, a\} \ (s = 2), \qquad \delta = a + \kappa - C \geq 0. \tag{5.7}$$

Proof. Every noncentral vertex of a clean pentagon has degree three: any cycle through a member of S contains w. A clean pentagon contains at most one attachment x_i. Otherwise two attachments at cycle distance one or two, together with their degree-two neighbours, give a regular or one-degree-four shared-hub path of order four or five. The paths are induced by girth and the endpoint degrees. This also covers the case that one of the attachments is the centre. Apply Lemmas 5.11 and 5.12.

Two clean faces at q form a hinged pair or a bare/chorded bowtie and contain at most two attachments. If an attachment x_i is absent, then u_i is separated from this pair, and Lemma 5.36 or Lemma 5.34 applies; its base contains w. Thus $s \geq 3$ excludes two clean faces. If $s = 2$, both attachments must lie on their union. They must be light: a heavy attachment can only be the centre, in which case each clean face already contains it and cannot contain the other attachment.

A light x_i has just one facial angle that can avoid w, since its other two angles use $x_i u_i$ and continue through w. Its clean face is therefore fixed. The two faces containing x_1, x_2 fix their unique heavy centre, so at most one centre has two clean faces. Three clean faces there would include a pair missing an attachment, contradicting the same restoration. Summing the capacities proves the stated slack identity and inequalities. □

For a pentagonal face F, let $a(F)$ count its heavy vertices other than w, and let $a_4(F)$ count those of degree four. Let N_4, N_+ count the nonpentagonal incidences of vertices in T' of degree four

and at least five, respectively, with multiplicity on facial walks. Put

$$e = \sum_{v \in T',\ d(v) \geq 6} (d(v) - 5), \qquad D' = 2h' - a + e, \tag{5.8}$$

$$W = N_4 + 2N_+ + \sum_{|F|=5,\ a(F) \geq 2} \big(2(a(F) - 1) - a_4(F)\big). \tag{5.9}$$

Every summand in W is nonnegative, since $a_4(F) \leq a(F) \leq 2(a(F) - 1)$ when $a(F) \geq 2$.

Lemma 5.41. *For $s \geq 2$, the quantities above satisfy*

$$a + b + \delta + W + 2Q + 2e \leq 2h' - 12 + \kappa, \tag{5.10}$$

$$D' \geq 12 + b - \kappa + 2Q + 3e + \delta + W, \tag{5.11}$$

$$p \geq 30 + 2t + 2b - 2\kappa + 5Q + 6e + 2\delta + 2W. \tag{5.12}$$

Proof. Let A_4 count pentagons containing w and exactly one other heavy vertex, of degree four. Such a pentagon cannot use an S–S angle: its complementary path would be the one-degree-four order-four reduction. It uses a Z-edge incidence not used by a p_0-face, so $A_4 \leq b$. Partitioning the $4a$ degree-four angles gives

$$4a = C + A_4 + N_4 + \sum_{|F|=5,\ a(F) \geq 2} a_4(F).$$

Combine this identity with (5.5) and (5.9) to obtain

$$3a - \kappa + \delta + W = 2R - 2b + A_4 \leq 2R - b.$$

Substitute $R = 3h' - D' - 6 - Q$ and $D' = 2h' - a + e$ to get the first two inequalities. Substitution in $p = 6 + 2t + 2D' + Q$ gives the third. All identities count repeated nonpentagonal incidences with multiplicity. □

Lemma 5.42. *A minimum counterexample containing a degree-two vertex has $|\mathcal{H}_4(G)| \geq 8$, $D \geq 14$, and at least thirty-four distinct 5-cycles.*

Proof. The unique case is Lemma 5.38. Suppose $s \geq 2$. Since $a \geq \kappa$ and $b \geq 1$, (5.10) forces $h' \geq 7$, giving eight heavy vertices including w. For $s \geq 3$, use $t \geq 1$ and $\kappa = 0$ in (5.12); for $s = 2$, the heavy hub gives $t \geq 2$ and $\kappa \leq 1$. In both cases $p \geq 34$.

Equation (5.11) gives $D' \geq 13$ unless $s = 2$ and $\kappa = 1$. Since $D = D' + s + t - 3$ and w is heavy, this already gives $D \geq 14$ outside that exceptional case. Suppose then that $s = 2$, $\kappa = 1$ and $D \leq 13$. The same equation and $t \geq 2$ force

$$t = 2, \quad D' = 12, \quad b = 1, \qquad Q = e = \delta = W = 0.$$

Both attachments have degree three by Lemma 5.40. Since $Q = 0$, every face has a simple boundary of length five or six. An S–S pentagon therefore gives the regular shared-hub path of order four. On an S–S hexagon, $W = 0$ excludes any heavy vertex other than w, so the complementary path is the regular shared-hub path of order five. Both contradict Lemma 5.11. There is consequently no S–S angle. The cyclic word at w must be S, Z, S, Z, and its two isolated Z-runs give $b \geq 2$ by (5.6), a contradiction. Thus $D \geq 14$. Finally the pentagonal faces give distinct 5-cycles, as noted after (5.1). □

5.13 Two separated five-faces

Lemma 5.43. *Let X be a connected plane graph that contains a cycle, has girth at least five, and has maximum degree at most three. If p and q are the numbers of vertices of degrees one and two, respectively, then*

$$f_5(X) = 12 - 4p - 2q + \sum_{\substack{F \text{ a face of } X \\ |F|\geq 7}} (|F| - 6). \tag{FC}$$

Here $f_5(X)$ counts facial walks of length five. Every facial walk has length at least five, and a facial walk of length five is a simple 5-cycle.

Proof. The boundary walk of a face is a closed walk. Since X is connected and contains a cycle, each such walk contains a simple cycle; a bridge, when it occurs on a boundary walk, is traversed twice on the way to and from the cyclic block. The girth assumption therefore gives length at least five. A non-simple closed walk contains a simple cycle and at least two further edge traversals, so in this graph it has length at least seven. Thus a length-five facial walk is a simple 5-cycle.

Put

$$L_X = \sum_{|F|\geq 7} (|F| - 6).$$

The face-length sum is $2e = 6f - f_5 + L_X$, while the degree sum is $2e = 3n - 2p - q$. Combining these equalities with Euler's identity $n - e + f = 2$ gives

$$f_5 = 6f + L_X - 2e = 12 - 6n + 4e + L_X = 12 - 4p - 2q + L_X,$$

as claimed. □

Lemma 5.44. *Every connected plane graph of girth at least five with $2 \leq d(v) \leq 3$ and at most one degree-two vertex contains two vertex-disjoint 5-cycles with no edge between them.*

Proof. We use Lemma 5.43 throughout. Set

$$q := |\{v \in V(G) : d_G(v) = 2\}| \in \{0, 1\}, \qquad p := |\{v \in V(G) : d_G(v) = 1\}| = 0.$$

If $q = 1$, let z denote the unique degree-two vertex; when $q = 0$ there is no vertex z in use. Thus G has at least $12 - 2q$ facial 5-cycles. If there is a bridge ab, take the side C not containing the unique degree-two vertex, if it exists, and relabel the endpoints so that $a \in C$. (If there is no degree-two vertex, choose either side.) In the graph induced by this side, deleting ab leaves a of degree two and every other vertex of degree three. It is cyclic: otherwise, being connected, it would be a tree, which is impossible when every vertex has degree at least two. Lemma 5.43 gives at least ten facial 5-cycles. Since a has two face angles, at most two of these cycles contain a; choose one, S, that avoids a.

Let D be the other side. If the original degree-two vertex is in $D \setminus \{b\}$, then D has two degree-two vertices; if it is b, then b is its only degree-one vertex; and if there is no such vertex, D has one degree-two vertex. Thus $4p_D + 2q_D \leq 4$ in all cases. The side D is cyclic, since a tree would have at least two degree-one vertices. The face-count lemma therefore gives at least eight facial 5-cycles in D. Any one of them is disjoint from S; it has no edge to S because ab is the only edge between the two sides and $a \notin S$.

Assume now there is no bridge. The subcubic graph is 2-connected: at a cutvertex every branch would require at least two incident edges to avoid a bridge, forcing degree at least four. Take a facial 5-cycle S and let B be its exterior neighbours. Girth makes them distinct; $b = |B|$ is five, or

four when $q = 1$ and S contains the unique degree-two vertex z. Set $H = G - S$, embedded with the deleted region outside. Every member of B is on that outside face. A vertex outside S has at most one neighbour on S, by the girth assumption, and has degree at least two in G; hence H has no isolated vertex. Each member of B has degree two in H, except that, if $q = 1$ and $z \in B$, the vertex z becomes a leaf. There is at most one leaf, so every component of H is cyclic (a finite tree has at least two leaves).

In one such component, let r count its members of B, let q' count the degree-two vertices of G in the component that are not in B, and let ℓ be its outside facial length. Define

$$\varepsilon := \begin{cases} 1, & q = 1 \text{ and } z \in B \text{ lies in this component,} \\ 0, & \text{otherwise,} \end{cases} \qquad 0 \le q' \le 1.$$

When $q = 0$, these definitions give $\varepsilon = q' = 0$. The degree-one and degree-two counts in this component are

$$p_H = \varepsilon, \qquad q_H = r - \varepsilon + q'.$$

Applying Lemma 5.43 and removing the outside face gives

$$f_5^{\text{in}} = \ell + 6 - 2r - 2\varepsilon - 2q' + L, \qquad L = \sum_{\substack{F \text{ internal} \\ |F| \ge 7}} (|F| - 6) \ge 0. \tag{**}$$

Indeed,

$$4p_H + 2q_H = 4\varepsilon + 2(r - \varepsilon + q') = 2r + 2\varepsilon + 2q'.$$

The outside boundary is a closed walk containing a cycle, so $\ell \ge 5$. If $\ell = 5$ it contributes one facial 5-cycle, while if $\ell \ge 6$ it contributes $\ell - 6$ to the face-length excess; in either case the subtraction gives the displayed formula. A member of B belongs to at most one internal 5-face: its degree is at most two, with an outside face angle, and a leaf belongs to none. If H is disconnected, some component has $r \le 2$ since $|B| \le 5$. Since $\varepsilon + q' \le 1$ and $\ell \ge 5$, (**) gives

$$f_5^{\text{in}} \ge 5 + 6 - 2r - 2(\varepsilon + q') \ge 9 - 2r > r.$$

It therefore has more internal 5-faces than attachment vertices. One avoids B, so it is disjoint from S and has no edge to S.

Suppose H is connected. Let $F_1, \dots, F_t$ be the distinct faces on the exterior side of the edges from S to B. Deleting the five vertices of S and the $5 + b$ incident edges gives $n(H) = n(G) - 5$ and $e(H) = e(G) - 5 - b$; since H is connected, Euler's formula gives $f(H) = f(G) - b$. In the deletion, the 5-face bounded by S and the t exterior faces merge into the new outside face, so $t = b$. Thus the exterior faces are indeed distinct; call them $F_1, \dots, F_b$. Their merger has outside length

$$\ell = \sum_{i=1}^{b} |F_i| - (2b + 5), \tag{***}$$

because every edge of S occurs once among the F_i, while every edge from S to B occurs twice (once for each incident face).

If $q = 0$, any S has $b = 5$, $\ell \ge 10$, and $\varepsilon = q' = 0$. Formula (**) gives

$$f_5^{\text{in}} \ge 10 + 6 - 2 \cdot 5 = 6 > |B|.$$

If $q = 1$ and z lies on a 5-face, choose that face as S. Now $b = 4$, $\ell \ge 7$, and $\varepsilon = q' = 0$ (the unique degree-two vertex is on S, not in B). Hence

$$f_5^{\text{in}} \ge 7 + 6 - 2 \cdot 4 = 5 > |B|.$$

It remains that $q = 1$ and z lies on no 5-face. There is a 5-face adjacent to two distinct non-5-faces. To see this, consider the plane dual. It is loopless because G is bridgeless. Since G is 2-connected, its facial boundaries are cycles. Thus the faces of the dual are triangles, one for each degree-three vertex of G, except for the unique digon corresponding to z. The digon's endpoints are distinct (there is no cutvertex) and represent the two faces incident with z; both are non-5-faces because z lies on no 5-face. Let W be the set of dual vertices representing non-5-faces. Suppose, for a contradiction, that every dual vertex outside W has at most one distinct neighbour in W. Then in every triangular dual face whose two consecutive vertices lie in W, the third vertex also lies in W; otherwise it would have two distinct neighbours in W.

Start with the digon edge. Its endpoints lie in W. Walk around either endpoint in the fixed plane embedding. Each intervening face is triangular, except possibly the unique digon, so the preceding paragraph shows inductively that every successive neighbour is in W; the digon only repeats the same neighbour. The same argument applies at every newly reached vertex of W. Hence W is closed under adjacency. The dual is connected, so this would give $V(G^*) = W$, contradicting the at least ten 5-faces of G, whose dual vertices are not in W. Therefore some 5-face is adjacent to two distinct non-5-faces, as asserted. Choose the resulting 5-face S. If its deletion disconnects the graph, use the case already proved; otherwise $b = 5$ and two distinct neighbours in (***) have length at least six. Thus $\ell \geq 12$. If $q = 1$ and $z \in B$, then $\varepsilon = 1, q' = 0$; if $q = 1$ and $z \notin B$, then $\varepsilon = 0, q' \leq 1$. In both cases $\varepsilon + q' \leq 1$, so (**) gives

$$f_5^{\text{in}} \geq 12 + 6 - 2 \cdot 5 - 2 = 6 > |B|.$$

In every case one avoids B, making it disjoint and nonadjacent to S. □

Completing the subcubic induction.

Proof of Theorem 5.21. Induct on the number of vertices, allowing disconnected graphs. The one-vertex case is $\mathcal{R}_4(G) = K_4$. A vertex of degree at most one is handled by Lemma 2.2. If there are two degree-two vertices, adjacent ones use Lemma 5.4. Ones at distance two use Lemma 5.9, unless their middle vertex has degree two, when an adjacent pair already applies. Ones at distance at least three use Lemma 5.5, whose spare-colour witness uses Grötzsch's theorem [5]. Pairs in different components have disjoint neighbourhoods and are covered by the same separated-pair argument.

The remaining graph has minimum degree at least two and at most one degree-two vertex. Apply Lemma 5.44 inside any component. Delete its two separated pentagons. They are vertex-disjoint and nonadjacent. If they lie in one connected component, a path between those pentagons contains a vertex outside their ten vertices; if they lie in different components, another component remains. Thus the remaining graph is nonempty and belongs to the same class, so induction gives its four-colour Hamilton cycle. Lemma 5.28 restores both pentagons. This proves the assertion at four colours. For every $k \geq 5$ use Theorem 4.5, with larger colours separately supplied there by degeneracy. The frozen 012012 colouring of C_6 gives the Hamiltonian threshold lower bound four. □

5.14 Graphs with subcubic blocks

Theorem 5.45. *Let G be a nonempty finite simple planar graph of girth at least five. If every block B satisfies $\Delta(B) \leq 3$, then $\mathcal{R}_k(G)$ is Hamiltonian for every $k \geq 4$. Cutvertices may have arbitrarily large degree in G. The uniform Hamiltonian threshold of this class is exactly four. Its Hamilton-connected threshold lies between four and five.*

Proof. Treat bridges as two-vertex blocks and isolated vertices as singleton blocks. The class is closed under vertex deletion: a nontrivial two-connected block of a subgraph is a two-connected subgraph of G, hence is contained in one original block, with no larger internal degrees. Planarity and girth are also preserved.

Products of Hamiltonian graphs are Hamiltonian: copy one factor cycle along a spanning path of another factor cycle, and merge successive copies using two alternating, distinct internal edges. This also handles disconnected original graphs. Choose a smallest four-colour counterexample, which may therefore be assumed connected. The singleton base is K_4, and the single-vertex lift excludes vertices of degree at most one. If there is only one block, the previous subcubic girth-five theorem applies. Thus assume minimum degree at least two and at least one cutvertex.

Lemmas 5.4, 5.9, and 5.5 exclude adjacent degree-two vertices, a $(2,3,2)$ path, and two degree-two vertices at distance at least three. Consequently any two degree-two vertices not already giving a reduction must have a common neighbour of degree at least four in G.

The block-cut tree has at least two endblocks. No endblock is a bridge, since its noncut endpoint would be a leaf. Let B be an endblock and x its unique cutvertex of G. Every vertex of $B - x$ has its full G-degree in B, hence degree two or three; also $d_B(x) \in \{2,3\}$. Let q count the degree-two vertices of $B - x$ and put $\varepsilon = [d_B(x) = 2]$. If $q \geq 2$, any pair of these vertices has a common neighbour of G-degree at least four by the excluded reductions. Their neighbours all lie in B, so only x can be this common neighbour. Every one of the q vertices is therefore adjacent to x, giving $q \leq d_B(x) = 3 - \varepsilon$. Together with the immediate $q \leq 1$ case, this proves $q + \varepsilon \leq 3$.

Euler's identity in the two-connected block B now gives

$$f_5(B) = 12 - 2(q + \varepsilon) + \sum_{|F| \geq 7} (|F| - 6) \geq 6.$$

Each five-face is a simple pentagon, while at most $d_B(x) \leq 3$ faces meet x, so choose one avoiding x. If $q \leq 1$, it contains at most one degree-two vertex of G. If $q \geq 2$, a cycle through any of the q vertices must use its edge to x, so the chosen pentagon contains none. All its other vertices have G-degree three.

Choose these pentagons in two distinct endblocks. Distinct endblocks intersect only in cutvertices, and the selected pentagons avoid their endblock cutvertices. Every edge joining two blocks belongs to a common block, so no edge joins the two selected pentagons. Their vertices are therefore noncutvertices of G, belong only to their respective endblocks, and the pentagons are disjoint. Lemma 5.28 applies, including its allowance of one degree-two vertex per pentagon and simultaneous exterior list changes. The smaller graph remains in the class and has a base Hamilton cycle by induction. Restoration contradicts minimality.

For every $k \geq 5$ separately use Theorem 4.5. Finally C_6 belongs to the class and its periodic 012012 colouring is frozen in $\mathcal{R}_3(C_6)$, giving the lower bound four. □

In particular, any remaining minimum counterexample for general planar girth-five four-colour Hamiltonicity contains a block with a vertex of degree at least four inside that block.

5.15 Small total degree excess

Recall that $D(G)$ is the total degree excess above three. It cannot increase under vertex deletion. All degree conditions in the following structural arguments refer to the original plane graph.

Lemma 5.46. *Let G be a connected plane graph of minimum degree at least three and girth at least five. Suppose a hinged pair X consists of two facial pentagons. Put $B = N(X) \setminus X$ and $D_B = \sum_{v \in B}(d(v) - 3)$. At most $15 + D_B$ pentagonal faces meet $X \cup B$.*

Proof. Use the local labels $(0,2,3,4,1,0)$ and $(0,1,7,6,5,0)$ for the pentagons, with sole heavy vertex 0. Their union is a disk. In the bare model its exterior boundary has seven outside incidences, at seven different local vertices. The sectors between them account for at most seven further faces meeting X, hence at most nine pentagonal faces in total. Bridges may merge sectors and only reduce this count. In the chorded model the chord 36 divides the exterior into two regions and leaves five outside incidences. A region with contacts has at most as many sector faces as contacts; a contact-free region has at most one. At most one region is contact-free, giving at most eight pentagonal faces meeting X.

Let $c \in \{0,1\}$ indicate the chord, put $r = 7 - 2c$, and let $j = r - |B|$. At $v \in B$ write $r_v = |N(v) \cap X|$. The permitted outside-boundary position identifications give $r_v \le 2 < d(v)$. At least $r_v + 1$ angles in its circular rotation use an edge into X. Thus at most $d(v) - r_v - 1$ angles can belong to a pentagon avoiding X. Summing over B bounds the additional pentagonal faces by $2|B| - r + D_B$. Consequently

$$p_{\text{near}} \le 16 - 3c - 2j + D_B.$$

This proves the desired bound unless $c = j = 0$.

In that remaining case all seven outside neighbours are distinct. The endpoint with no outside neighbour 1 gives the exterior sector $4,1,7$. If its face is not pentagonal, the bound of nine loses one. Otherwise that face is $(4,1,7,b_7,b_4,4)$, where $b_4, b_7 \in B$ are distinct, so b_4b_7 is an edge on a cycle. Its other incident face is different. At both endpoints its angle on that side uses no edge into X: the endpoint has degree at least three and just one neighbour in X. These are two angles included in the preceding candidate sum. If the other face is a pentagon avoiding X, they count the same face twice. If it meets X, it was already counted there; if it is nonpentagonal, neither angle contributes a new pentagon. In every case at least one unit is saved, proving $p_{\text{near}} \le 15 + D_B$. □

Lemma 5.47. *Let G be a connected plane graph of minimum degree at least three, girth at least five, and $D(G) \le 11$. If it has exactly one ordinary facial pentagon, that pentagon is separated from a hinged pair or bowtie.*

Proof. Let a, b count vertices of degrees four and at least five, respectively, and put $e = \sum_{d(v)\ge 6}(d(v)-5)$. Thus $h = a + b$ and $D = a + 2b + e$. If S is the unique ordinary facial pentagon, the face identity and heavy-incidence count give

$$p = 12 + 2D + Q, \qquad \Delta = 3h + D - (p-1) = 2a + b - e - 11 - Q \ge 0.$$

The deficit Δ counts heavy incidences on nonpentagonal faces and excess heavy incidences on pentagonal faces.

Suppose there is no stated partner. Delete S and let B be its five distinct exterior neighbours. Every residual component C has minimum degree at least two. The connected set consisting of S and the other components lies in one face of C; choose it as outside. All attachments lie on that face. Let $r = |B \cap V(C)|$, let ℓ be its outside facial length, and let L be its internal face excess above six. Use a_C, b_C, e_C, D_C for its original degree counts and excess, and let β_+ count its attachments of original degree at least five. Its degree and face sums give the internal pentagon count

$$I_C = \ell + 6 - 2r + 2D_C + L.$$

Its internal faces are unchanged faces of G. Let Δ_C be their contribution to the heavy-incidence deficit, so $\sum_C \Delta_C \le \Delta$.

An attachment of original degree d has at most $d-2$ internal angles. Thus at most $r + D_{B\cap V(C)}$ internal pentagons meet attachments. None of the other internal pentagons is ordinary.

A nonattached degree-four vertex is sole heavy on at most one of them: two facial pentagons at that vertex form a hinged pair or bowtie separated from S. At a vertex of higher degree, the corresponding bound is d. Multi-heavy pentagons consume at least one unit of Δ_C each. The sum of the sole-heavy incidence bounds is $a_C + 5b_C + e_C$. Subtracting the contributions of the attached vertices cancels $D_{B\cap V(C)}$ at degree four and leaves -3 per higher-degree attachment. Hence

$$I_C \le r + a_C + 5b_C + e_C - 3\beta_+ + \Delta_C,$$

or

$$\ell + 6 - 3r + a_C - b_C + e_C + L + 3\beta_+ \le \Delta_C. \tag{5.13}$$

Every residual component is cyclic, so $\ell \ge 5$. If there are $c \ge 2$ components, their attachment counts sum to five. Summing (5.13) and using the displayed value of Δ gives

$$D \ge 11c - 4 + Q + 3e \ge 18,$$

a contradiction. Thus $G - V(S)$ is connected. Deleting five vertices and ten edges reduces the face count by five; the incident faces merge into one, so the five exterior faces beside S are distinct. If ϵ is their total excess above five, their merger has outside length $10 + \epsilon$. Substitution in (5.13), now with $r = 5$, gives

$$D \ge 12 + Q + 3e + \epsilon + L + 3\beta_+ \ge 12,$$

again contradicting $D \le 11$. □

Lemma 5.48. *Let G be a connected plane graph of minimum degree at least three, girth at least five, and $D(G) \le 9$. It has two ordinary facial pentagons, an ordinary facial pentagon separated from a hinged pair or bowtie, or a hinged pair separated from a hinged pair or bowtie.*

Proof. Two ordinary faces give the first alternative; a unique one is handled by Lemma 5.47. Suppose there are none. Retain a, b, e, h, D, Q from the preceding proof. Then

$$p = 12 + 2D + Q, \qquad \Delta = 3h - D - 12 - Q \ge 0.$$

The nonpentagonal and multi-heavy pentagonal incidences at degree-four vertices total at most 2Δ: a pentagon with $k \ge 2$ heavy vertices has at most $k \le 2(k-1)$ such incidences. The number C_4 of sole-heavy pentagonal incidences at degree-four vertices therefore satisfies $C_4 \ge 4a - 2\Delta$. Since

$$a - \Delta = 12 - h + e + Q > 0$$

by $h \le D \le 9$, we have $C_4 > 2a$. Some degree-four vertex has at least three sole-heavy pentagonal faces, including two adjacent in its rotation. Their union is a hinged pair X.

Put $B = N(X) \setminus X$ and call a pentagon far if it avoids $X \cup B$. Suppose X has no separated hinged or bowtie partner. A degree-four vertex outside $X \cup B$ is sole heavy on at most one far pentagon, since two would form such a partner. A vertex of higher degree is sole heavy on at most d far pentagons. The sum of these incidence bounds is

$$C = a + \sum_{d(v)\ge 5} d(v) = 3h + D - 3a.$$

Subtract the contribution one of the heavy vertex of X, and the contribution C_B from the heavy vertices of B. At most Δ far pentagons are multi-heavy. Lemma 5.46 consequently gives

$$p \le 15 + D_B + C - 1 - C_B + \Delta.$$

If β_+ counts vertices of B of degree at least five, then $D_B - C_B = -3\beta_+$. Subtracting the asserted upper bound from the actual value of p yields

$$p - (15 + D_B + C - 1 - C_B + \Delta) = 10 - D + 3e + 2Q + 3\beta_+ > 0,$$

a contradiction. Thus the last alternative holds. □

Lemma 5.49. *Let G be a connected plane graph of minimum degree at least three and girth at least five, with no ordinary facial pentagon. Let X be a hinged pair of facial pentagons, called a facial hinge, and put $B = N(X) \setminus X$, $D_B = \sum_{v\in B}(d(v) - 3)$. Write a, h for the numbers of degree-four and heavy vertices, and put*

$$e = \sum_{d(v)\geq 6} (d(v) - 5), \qquad Q = \sum_{|F|\geq 7} (|F| - 6), \qquad \Delta = 3h - D - 12 - Q.$$

Assign reference value one to a degree-four vertex and $d(v)$ to a higher-degree vertex. Their sum is $C = 3h + D - 3a$; let C_B be their sum at heavy vertices of B. Let n_X count pentagonal faces meeting $X \cup B$, and let f_1, f_2 count pentagonal faces avoiding that set with exactly one and at least two heavy vertices, respectively. Set

$$\nu_X = 15 + D_B - n_X, \qquad \sigma_X = C - 1 - C_B - f_1, \qquad \omega_X = \Delta - f_2.$$

Then, with $\beta_+ = |\{v \in B : d(v) \geq 5\}|$,

$$\nu_X + \sigma_X + \omega_X = D - 10 - 3e - 2Q - 3\beta_+. \tag{5.14}$$

The quantities ν_X, ω_X are nonnegative. If X has no separated hinged or bowtie partner, then $\sigma_X \geq 0$ as well.

Proof. Lemma 5.46 gives $\nu_X \geq 0$. If no stated partner exists, a degree-four vertex outside $X \cup B$ is sole heavy on at most one far pentagon; at a higher-degree vertex use its degree. This gives $\sigma_X \geq 0$.

Let $k(F)$ count heavy vertices on a pentagonal face and put

$$E = N_{\text{nonpent}} + \sum_{|F|=5,\ k(F)\geq 2} (k(F) - 2),$$

where N_{nonpent} counts heavy incidences on nonpentagonal facial walks, with multiplicity. The heavy-incidence deficit gives

$$\omega_X = E + |\{F : |F| = 5,\ k(F) \geq 2,\ F \cap (X \cup B) \neq \varnothing\}| \geq 0. \tag{5.15}$$

Finally $p = n_X + f_1 + f_2 = 12 + 2D + Q$, $D_B - C_B = -3\beta_+$, and $D = 2h - a + e$. Substitution gives the exact identity (5.14); it remains valid when σ_X is negative. □

Lemma 5.50. *Under the hypotheses of Lemma 5.46, suppose $\nu_X := 15 + D_B - n_X = 0$, where n_X counts pentagons meeting $X \cup B$. The hinge is bare and its seven outside neighbours are distinct. Label its cycles $(0, 2, 3, 4, 1, 0)$ and $(0, 1, 7, 6, 5, 0)$, and call the face through $4, 1, 7$ special. If the special face is pentagonal, every face meeting $X \cup B$ is pentagonal. Otherwise it is the only nonpentagonal face meeting X.*

Proof. The coarse bound in Lemma 5.46 is $n_X \le 16 - 3c - 2j + D_B$, where c indicates the chord and j counts repeated outside contacts. Equality $n_X = 15 + D_B$ forces $c = j = 0$.

There are at most nine distinct faces meeting X; let p_X count their pentagons. At each $v \in B$, exactly $d(v) - 2$ angles avoid its single edge into X, giving $7 + D_B$ candidate angles. Let U count candidates whose face is nonpentagonal or already meets X. For each new pentagon count its remaining candidate incidences, and let V be their total excess over one per face. Then $U, V \ge 0$ and

$$\nu_X = (9 - p_X) + U + V - 1. \tag{5.16}$$

If the special face is pentagonal, it is $(4, 1, 7, b_7, b_4, 4)$. The other side of b_4b_7 supplies two candidate angles. If that face is a new pentagon, then $V \ge 1$; otherwise both angles are unusable, so $U \ge 2$. The latter is incompatible with (5.16). The former forces $p_X = 9$, $U = 0$, and $V = 1$. All faces meeting X and all remaining candidate angles are therefore pentagonal, proving the first assertion. If the special face is nonpentagonal, then $9 - p_X \ge 1$; equality forces $p_X = 8$ and $U = V = 0$. Since at most nine faces meet X, it is the only nonpentagonal one among them. This counts actual angles and distinct faces, so bridges or merged exterior sectors cause no exception. □

Lemma 5.51. *Let G be a connected plane graph of minimum degree at least three and girth at least five, with every face of length five or six. Let $\mathcal{C}$ consist of its degree-four vertices whose four incident faces are single-heavy pentagons. Suppose $\mathcal{C} \neq \varnothing$ and every facial hinge X centred in $\mathcal{C}$ satisfies $\nu_X = 0$, has no vertex of degree greater than four in $B = N(X) \setminus X$, and has every pentagon meeting $X \cup B$ single-heavy. Then every face is a single-heavy pentagon, all heavy vertices belong to $\mathcal{C}$, and $D(G) = 6$.*

Proof. All facial boundaries are simple by Lemma 5.20. Fix $q \in \mathcal{C}$. Its four adjacent face pairs are four hinges. Each neighbour z of q occurs in three of their unions, whose shared light endpoints are three distinct neighbours of q. If the third face at z were hexagonal, it would have to be the special face of all three hinges by Lemma 5.50. It would contain those three neighbours but not q, whose faces are pentagonal. Two of three vertices on a hexagon have cycle distance at most two; their edges to q would create a triangle or a 4-cycle. Thus every special face at q is pentagonal.

Lemma 5.50 now makes every face meeting $X \cup B$ pentagonal for each hinge at q. By hypothesis all these faces are single-heavy. Every heavy vertex in such a B has degree four and four single-heavy incident faces, hence belongs to $\mathcal{C}$.

Write a q-face as $F = (q, a, b, c, d, q)$. Its neighbours across qa, qd contain q. Its neighbours across ab, cd are special faces of the hinges sharing qa, qd; their sole heavy vertices lie in the respective B-sets and thus in $\mathcal{C}$. Let p, r be the third neighbours of b, c. They are outside either hinge containing F, since those hinges are bare. The exterior face across bc is a single-heavy pentagon (b, c, r, z, p, b). If p or r is heavy, it belongs to the corresponding B and hence to $\mathcal{C}$.

Otherwise both are light and z is heavy. Use the hinge sharing qa, whose other face contains a light vertex b' beside a. Its special face is (b, a, b', h, p, b), with h heavy. These two faces lie on opposite sides of bp. They are distinct, since their angles at b differ and facial boundaries are simple. Original degree three at p therefore makes its next neighbours h, z distinct. The third face at p contains both heavies. But p is in the hinge's B, and every such face is single-heavy, a contradiction. The neighbour across bc also contains a vertex of $\mathcal{C}$.

The nonempty set of faces containing $\mathcal{C}$ is thus closed under dual adjacency. The dual is connected, so these are all faces. Every heavy vertex is the unique $\mathcal{C}$-vertex on an incident face, hence belongs to $\mathcal{C}$. If $m = |\mathcal{C}|$, heavy incidence counting gives $p = 4m$, while Euler gives $p = 12 + 2m$. Consequently $m = D(G) = 6$. □

Theorem 5.52. *For every nonempty planar graph G of girth at least five with $D(G) \leq 10$, the graph $\mathcal{R}_k(G)$ is Hamiltonian for every $k \geq 4$. This class has uniform Hamiltonian threshold four.*

Proof. The class is hereditary. A minimum four-colour counterexample is connected with minimum degree at least two by Lemma 5.18. Lemma 5.42 excludes degree two, which would require $D \geq 14$. If $D \leq 9$, use Lemma 5.48. Suppose $D = 10$. Two ordinary facial pentagons are restored by Lemma 5.15; a unique one is handled by Lemma 5.47. Thus assume there are none.

The incidence calculation in Lemma 5.48 gives at least $4a - 2\Delta$ sole-heavy pentagonal incidences at degree-four vertices. Since $a - \Delta = 12 - h + e + Q > 0$ for $h \leq 10$, some such vertex has three sole-heavy faces and hence an adjacent pair. A facial hinge exists. We may assume no facial hinge has a separated hinged or bowtie partner, since Lemma 5.36 restores each such pair.

The nonnegative terms of (5.14) now force, for every facial hinge,

$$e = Q = \beta_+ = \nu_X = \sigma_X = \omega_X = 0.$$

Equation (5.15) gives no nonpentagonal heavy incidence and no multi-heavy pentagon meeting $X \cup B$. The centre of any hinge therefore has four single-heavy pentagons, so the set $\mathcal{C}$ in Lemma 5.51 is nonempty. Every hinge centred there satisfies all that lemma's hypotheses. Its conclusion $D = 6$ contradicts $D = 10$.

Every restorable deletion above removes at most seventeen vertices; minimum degree three and planar girth five give $|V(G)| \geq 20$. The base graph is nonempty and remains in the class, so induction and the cited restorations give the contradiction in each case. For $k \geq 5$ use Theorem 4.5; the frozen three-colouring of C_6 gives sharpness. □

Theorem 5.53. *For every nonempty planar graph G of girth at least five with $|\mathcal{H}_4(G)| \leq 7$, $\mathcal{R}_k(G)$ is Hamiltonian for every $k \geq 4$. This class has uniform Hamiltonian threshold four.*

Proof. The heavy-count class is hereditary. A minimum four-colour counterexample is connected with minimum degree at least two; Lemma 5.42 excludes degree two. If $D \leq 10$, Theorem 5.52 applies. Otherwise let o count ordinary facial pentagons. The incidence bound is

$$o \geq 12 - 3h + D + Q \geq 2.$$

Here $D \geq 11$. Restore two ordinary faces by Lemma 5.15. This deletes at most ten vertices, so the base is nonempty by $|V(G)| \geq 20$. It remains in the heavy-count class and has a Hamilton cycle by induction, contradicting restoration. Larger palettes follow from Theorem 4.5, and C_6 gives sharpness. □

Theorem 5.54. *For every nonempty planar graph G of girth at least five with at most thirty-three distinct 5-cycles, $\mathcal{R}_k(G)$ is Hamiltonian for every $k \geq 4$. This class has uniform Hamiltonian threshold four.*

Proof. A minimum four-colour counterexample in this hereditary class is connected with minimum degree at least two. The bound of thirty-four distinct 5-cycles in Lemma 5.42 excludes degree two. Pentagonal faces now give distinct 5-cycles, so

$$12 + 2D + Q = p \leq 33.$$

Hence $D \leq 10$, contrary to Theorem 5.52. For larger palettes use Theorem 4.5, and use C_6 for sharpness. □

Corollary 5.55. *Every nonempty two-degenerate planar graph G of girth at least five with $D(G) \leq 13$ has Hamiltonian $\mathcal{R}_k(G)$ for every $k \geq 4$. This class has uniform Hamiltonian threshold four.*

Proof. The class is hereditary. A minimum four-colour counterexample is connected with minimum degree at least two, and two-degeneracy forces a degree-two vertex in the graph itself. Lemma 5.42 then requires $D \geq 14$, contradicting the class bound. For $k \geq 5$, use Theorem 2.5 with degeneracy two. The graph C_6 belongs to the class and gives sharpness. □

6 The girth-five frontier

The threshold intervals in Table 1.1 leave two questions: does every $G \in \mathcal{P}_5$ have a Hamiltonian $\mathcal{R}_4(G)$, and is every such recolouring graph Hamilton-connected? The latter question also remains open for the four-colour subclasses of Theorem 1.3. Known connectivity [1] does not settle either question. The subclass theorems require any minimum Hamiltonian counterexample to have at least thirty-four distinct 5-cycles, $|\mathcal{H}_4(G)| \geq 8$, and $D(G) \geq 11$. Theorem 5.45 also requires a block with internal maximum degree at least four. If the counterexample contains a degree-two vertex, Lemma 5.42 gives at least thirty-four distinct 5-cycles and $D(G) \geq 14$; if that vertex is unique, Lemma 5.38 raises these bounds to forty-two and sixteen.

Proposition 6.1. *Let G be a minimum counterexample to four-colour Hamiltonicity in the full class $\mathcal{P}_5$. Suppose that $\delta(G) \geq 3$ and that G has at most thirty-five distinct 5-cycles. In the notation of Lemma 5.49, $D = 11$, $Q = 0$, and $p = 34$; every face has length five or six; $E = 0$; no degree-five vertex lies on a multi-heavy pentagonal face; every hexagon avoids heavy vertices; and every multi-heavy pentagonal face has exactly two heavy vertices, both of degree four. Consequently the heavy-degree multiset is one of*

$$\{4^{11}\}, \quad \{4^9, 5\}, \quad \{4^7, 5^2\}, \quad \{4^5, 5^3\}.$$

These are necessary conditions only; the proposition does not eliminate this regime.

Proof. Euler and Theorem 5.52 force $D = 11$ and $Q \leq 1$. The ordinary-face restorations, including Lemma 5.47, leave no ordinary face. The same incidence argument supplies a facial hinge, since $h \leq 11 < 12$. A separated partner would restore, so the equality identity has nonnegative terms. It forces $e = Q = \beta_+ = 0$ and

$$\nu_X + \sigma_X + \omega_X = 1 \tag{6.1}$$

for every facial hinge. Thus $p = 34$, and every face has length five or six.

In fact $E = 0$. If $E \geq 1$, (5.15) and (6.1) force $E = \omega_X = 1$ and $\nu_X = \sigma_X = 0$ for every hinge, with no near multi-heavy pentagon. A hinge centre has no nonpentagonal face: Lemma 5.50 permits only its special face, and that simple 5- or 6-cycle cannot contain the centre. Its edge to the shared light endpoint would be a chord producing a triangle or a 4-cycle. Hence a fully single-heavy centre exists. All such centres satisfy Lemma 5.51, which would give $D = 6$, a contradiction.

No degree-five vertex lies on a multi-heavy pentagonal face either. Such a vertex is outside every $X \cup B$, since the centre of X has degree four and $\beta_+ = 0$. Its incidence with a multi-heavy face reduces the number of far pentagons on which it can be sole heavy by at least one. Thus $\sigma_X \geq 1$ for every hinge, forcing $\sigma_X = 1$ and $\nu_X = \omega_X = 0$. Again every hinge centre is fully single-heavy and all the rigidity hypotheses hold, a contradiction. Consequently every hexagon avoids heavy vertices, and every multi-heavy facial pentagon has exactly two heavy vertices, both of degree four.

Since $\delta(G) \geq 3$, every vertex outside $\mathcal{H}_4(G)$ has degree three. Since $e = 0$, every heavy vertex has degree four or five. If a and b count the degree-four and degree-five vertices, then $a + 2b = 11$

and $a + b \geq 8$ by Theorem 5.53; these relations give the four displayed heavy-degree multisets. The number 34 counts pentagonal faces; the hypothesis counts all distinct 5-cycles. Degree-two cases with thirty-four or thirty-five distinct 5-cycles are not excluded here. □

7 Explicit constructions and finite verification

Appendices A–J describe the finite graphs, paths, cycles and auxiliary edge sets used in the local restoration lemmas. The accompanying programs enumerate the proper colourings of each local graph and check the specified paths and cycles, their symmetry classes, and the protected-edge conditions. Separate computations check the common edges of the chosen cycles at every admissible boundary transition in the original colour labels. Table 7.1 records their scope.

Scope of the finite verification. These computations establish the stated properties of the listed finite constructions, including the 23^2 product cases and the common-edge conditions under relative normalization. The existence of the underlying Hamilton cycles, the extensions to products of arbitrary order, component-graph connectivity along an arbitrary Hamilton path, spanning-tree splicing and structural induction are proved in the article. The finite verification supplies the local hypotheses of these arguments; it imposes no upper bound on the order of G. The two general girth-five threshold questions remain open.

References

[1] V. Bartier, N. Bousquet, C. Feghali, M. Heinrich, B. Moore, and T. Pierron, Recolouring planar graphs of girth at least five, *SIAM Journal on Discrete Mathematics* 37 (2023), no. 1, 332–350. doi: 10.1137/21M1463598.

[2] N. Bousquet and G. Perarnau, Fast recoloring of sparse graphs, *European Journal of Combinatorics* 52A (2016), 1–11. doi: 10.1016/j.ejc.2015.08.001.

[3] K. Choo and G. MacGillivray, Gray code numbers for graphs, *Ars Mathematica Contemporanea* 4 (2011), no. 1, 125–139. doi: 10.26493/1855-3974.196.0df.

[4] L. Cereceda, J. van den Heuvel, and M. Johnson, Connectedness of the graph of vertex-colourings, *Discrete Mathematics* 308 (2008), no. 5–6, 913–919. doi: 10.1016/j.disc.2007.07.028.

[5] H. Grötzsch, Zur Theorie der diskreten Gebilde, VII: Ein Dreifarbensatz für dreikreisfreie Netze auf der Kugel, *Wiss. Z. Martin-Luther-Univ. Halle-Wittenberg, Math.-Nat. Reihe* 8 (1958/1959), 109–120.

[6] H.-C. Hsu, T.-K. Li, J. J. M. Tan, and L.-H. Hsu, Fault Hamiltonicity and fault Hamiltonian connectivity of the arrangement graphs, *IEEE Transactions on Computers* 53 (2004), no. 1, 39–53. doi: 10.1109/TC.2004.1255789.

Table 5.1: The local components used in the protected-pentagon constructions of Appendices C–D. The labelled-boundary column gives the orbit size under rotations, reflections and colour permutations.

Boundary	Labelled	$\lvert F_t \rvert$	Component orders	Construction
Ordinary boundaries				
00000	4	30	$15 + 15$	two Hamilton cycles
00001	60	44	44	22 ordered blocks
00011	60	42	42	Hamilton cycle
00012	120	47	47	Hamilton cycle
00101	60	64	64	Hamilton cycle
00102	120	63	63	Hamilton cycle
00112	120	51	51	Hamilton path
00121	120	58	58	Hamilton cycle
00123	120	57	57	Hamilton cycle
01012	120	68	68	Hamilton cycle
01023	120	66	66	Hamilton cycle
Starred boundaries				
0000∗	20	54	54	Hamilton cycle
0001∗	120	60	60	Hamilton cycle
0010∗	120	78	78	Hamilton cycle
0011∗	60	62	62	Hamilton cycle
0012∗	240	71	71	Hamilton cycle
0101∗	60	88	88	Hamilton cycle
0102∗	240	85	85	Hamilton cycle
0110∗	60	74	74	Hamilton cycle
0112∗	120	74	74	Hamilton cycle
0120∗	120	83	83	Hamilton cycle
0123∗	120	82	82	Hamilton cycle

Table 7.1: Finite verification of the local constructions.

Appendix	Constructions and cases checked
Appendix A	Seven edge-covering cycle orbits and 53 endpoint paths; exact subset dynamic programming for 225 labelled list pairs and 3,225 boundary transitions.
Appendix B	Three normalized path cycles; 60 good boundaries and 504 directed transitions between adjacent good boundaries; safe edges for each allowed cycle pair and the required adjacencies at bad boundaries.
Appendices C–D	23 component types and 2,304 labelled pentagon boundaries; ordered path blocks and the exceptional two- and three-row cycles; auxiliary edges and protected edges in every row; all 313 transitions, partitioned as $47+236+30$; all 23^2 product constructions with alternating reserved edges; relative normalization in all $75{,}120 = 313 \cdot 240$ labelled overlapping cases.
Appendix E	Five order-four and fifteen order-five path cycles; 64 and 256 labelled boundaries and 576 and 3,072 directed transitions; at least three and five common chosen-cycle edges, respectively.
Appendix F	Seven degree-two families with 421 representatives and 96,036 states; 8,704 labelled boundaries and 158,976 directed transitions, including simultaneous changes at identified boundary positions; at least two common chosen-cycle edges.
Appendix G	Three pentagonal families with 781 representatives $(715 + 15 + 51)$, 786 component cycles $(720 + 15 + 51)$ and 101,121 states; 17,664 labelled boundaries and 486,912 directed transitions. Every overlapping component pair shares at least four chosen-cycle edges in the reservoir and at least two in the other configurations; every component overlaps a component at each adjacent boundary.
Appendix H	Five connected-cycle families with 771 representatives and 2,491,162 states; 75,008 labelled boundaries and 2,577,024 directed transitions; all permissible identifications of boundary positions and the corresponding chosen-cycle intersections.
Appendix I	558 bowtie fibres with 845,440 states and 1,116 cycles in products with K_2 and K_3; 2,251,776 directed transitions and their matching and protected-edge intersections; partitions into edges and triangles with at most two joining edges incident with each fibre vertex; agreement of the ordinary pentagon constructions with Appendices C–D.
Appendix J	410 hinged-pair fibres with 267,825 states and 1,230 cycles in products with K_2, K_3 and P_3; 446,976 directed transitions and their matching and protected-edge intersections; partitions into edges and triangles with at most two joining edges incident with each fibre vertex.

A Adjacent-pair endpoint constructions

For the article's fibre $F(A, B)$, write ab for (a, b) and use type $(|A|, |B|, |A \cap B|)$. Each endpoint word visits every state once; successive states differ in one coordinate. Table A.1 includes list-preserving colour permutations, allowable coordinate exchange and path reversal. Exact subset dynamic programming checks all feasible and excluded endpoint pairs.

Table A.1: Adjacent-pair endpoint constructions and their symmetry coverage.

Type	$\lvert V(F)\rvert$	Words	Feasible unordered pairs
331	8	6	28
332	7	7	19
333	6	1	6
342	10	16	45
343	9	8	36
443	13	11	78
444	12	4	66

A.1 Edge-covering cycles

The columns give the type, lists separated by a slash, and one cyclic word. Its list-preserving symmetry orbit, including allowable coordinate exchange, covers every edge.

```
331   012/034   03 04 14 10 20 24 23 13
332   012/013   23 03 01 21 20 10 13
333   012/012   10 20 21 01 02 12
342   012/0134  24 04 01 03 13 14 10 20 21 23
343   012/0123  12 10 20 21 01 02 03 23 13
443   0123/0124 32 12 10 20 21 01 02 04 24 14 34 31 30
444   0123/0123 23 21 20 10 12 02 32 30 31 01 03 13
```

The 53 endpoint words. The list pairs are given in the preceding cycle table.

For 331:

```
03 13 23 24 20 10 14 04
03 23 20 24 04 14 13 10
03 23 20 24 04 14 10 13
03 23 13 10 20 24 04 14
13 10 20 24 23 03 04 14
13 14 10 20 23 03 04 24
```

For 332:

```
01 21 23 20 10 13 03
01 21 20 23 03 13 10
01 03 23 21 20 10 13
01 21 20 10 13 03 23
03 01 21 23 20 10 13
03 01 21 23 13 10 20
03 01 21 20 10 13 23
```

For 333:

```
01 21 20 10 12 02
```

For 342:

```
01 21 24 20 23 13 10 14 04 03
01 21 23 20 24 14 04 03 13 10
01 21 24 23 20 10 14 04 03 13
01 21 23 24 14 04 03 13 10 20
01 04 14 24 23 03 13 10 20 21
01 21 24 20 10 13 14 04 03 23
03 23 24 14 13 10 20 21 01 04
03 23 24 20 21 01 04 14 10 13
03 13 10 20 24 23 21 01 04 14
03 23 24 21 01 04 14 13 10 20
03 13 23 24 20 10 14 04 01 21
03 01 21 20 24 04 14 10 13 23
03 23 13 14 10 20 21 01 04 24
20 24 23 13 10 14 04 03 01 21
20 10 13 14 04 24 21 01 03 23
23 21 20 10 14 13 03 01 04 24
```

For 343:

```
01 21 23 20 10 12 13 03 02
01 21 20 23 13 10 12 02 03
01 21 20 23 13 03 02 12 10
01 21 23 20 10 13 03 02 12
01 21 23 20 10 12 02 03 13
01 02 03 23 13 12 10 20 21
01 21 20 10 13 12 02 03 23
03 23 20 21 01 02 12 10 13
```

For 443:

```
01 31 34 24 21 20 30 32 12 10 14 04 02
01 21 31 34 32 30 20 24 14 10 12 02 04
01 21 31 34 32 30 20 24 14 04 02 12 10
01 31 32 30 34 24 21 20 10 14 04 02 12
01 31 32 30 34 24 21 20 10 12 02 04 14
01 21 20 30 32 31 34 14 10 12 02 04 24
01 21 31 32 30 20 24 14 10 12 02 04 34
04 24 34 32 31 30 20 21 01 02 12 10 14
04 34 14 24 20 21 01 31 32 02 12 10 30
04 34 32 30 20 21 24 14 10 12 02 01 31
04 24 20 30 32 31 21 01 02 12 10 14 34
```

For 444:

```
01 31 21 23 20 30 32 12 10 13 03 02
01 21 31 32 30 20 23 13 03 02 12 10
01 21 31 32 30 10 20 23 13 03 02 12
01 21 31 32 30 20 10 13 12 02 03 23
```

B Constructions for the $(2, 3, 2)$-path restoration

The fibres B_t, allowed colour images and path reversals, and special edge sets P_t are defined in the main text. The following explicit cycles supply the selected-layer identities used in the safe-common-edge lemma in the article.

B.1 The three normalized cycles

In the cyclic words below the coordinates are uvw. The rows for $001, 010, 012$ specify D_E, D_M, D_A, respectively.

001	120 123 323 313 213 210 230 130 132 232 212 312 310 320
010	101 121 321 323 123 103 303 302 301
	201 203 202 102 132 232 231 131
012	131 101 103 203 303 301 201 231
	230 130 120 123 323 320 321 121

(B.1)

B.2 Selected-layer identities and capacities

Order the end-, middle-, and all-distinct types as $E < M < A$. For distinct a, b, c, d, let $E(aab), M_0(aba), A(abc)$ be the images of D_E, D_M, D_A under $(0, 1, 2, 3) \mapsto (a, b, c, d)$, and let M_1 be the reversal of M_0. Unused-colour interchange fixes E, M_0, M_1; reversal followed by endpoint-colour interchange fixes A. Thus E, A each have one allowed edge set and M has the two stated sets. Up

to these symmetries, the good transitions to a type no smaller are

$$E(aab) : aac, abb, acb, bab, cab; \quad M(aba) : aca, cba; \quad A(abc) : abd, adc.$$

These exhaust one-coordinate changes; bad targets and smaller types are excluded, and unused colours and the two ends are identified by the stated symmetries.

In $E(aab)$, the middle-b layer is $2P_3$ and the middle-c, d squares are P_4. In $M_0(aba)$, the middle-a layer is P_8 with isolated bab, and the middle-c, d squares are P_4; reverse for M_1. In $A(abc)$, the middle-a, c layers are P_6 and the middle-d square is P_3 with isolated bdb. These are subgraphs of the respective selected cycles.

Each cycle has four middle-changing edges. For E, middle colour b joins c, d twice each, with last coordinate a twice and c, d once each. For M_0, middle colour a joins c, d twice each and every edge has first coordinate b; reverse for M_1. For A, middle colour d joins a, c twice each, with first coordinate in $\{b, c\}$ and last coordinate in $\{a, b\}$. These paths and counts are read directly from (B.1).

For common endpoint lists L, R and distinct allowed middle colours v, w, the layer H_v is $K_{|L\setminus\{v\}|}\square K_{|R\setminus\{v\}|}$, and there are $|L\setminus\{v, w\}|\,|R\setminus\{v, w\}|$ edges between middle colours v, w. Removing the special edges gives the safe universe size m; the original cyclic words give the retained counts x, y. The following seven families supply the capacities used in the proof:

$t \to s$	blocks	(m, x, y)
$aab \to aac$	H_*	$(4, 3, 3)$
$aab \to abb$	H_c	$(2, 2, 2)$
$aab \to acb$	H_*	$(2, 2, 2)$
$aab \to cab$	H_c, H_d	$(3, 2, 2), (1, 1, 1)$
$aba \to aca$	H_d	$(4, 3, 3)$
$abc \to abd$	H_*	$(6, 4, 4)$
$abc \to adc$	H_a, H_c	$(9, 5, 5)$ each

C Pentagon cycles and protected products

Five digits record the five colours in cyclic vertex order. In six-digit words the first digit is the path layer. The two constant-type component cycles are printed separately. Cycle words are cyclic; the 51-state and 47-state paths for 00112 are linear. An edge is represented by its two five-digit endpoint words. The ordered path blocks use the specified cycles and the two additional identities below; no eleven-type factor-word list is needed. The two identities below give the required product cycles explicitly.

Path for 00112 (51 words).

```
12303 12323 12023 13023 13020 12020 12320 32320 31320 21320
21323 21303 21203 21201 21301 21321 31321 32321 32301 31301
31201 31231 31230 21230 21231 21031 21030 31030 31031 32031
32021 31021 31020 32020 32030 12030 13030 13230 23230 23030
23020 21020 21021 21023 23023 23021 23031 23231 23201 23203
13203
```

Constant-boundary cycles (30 words).

```
12123 12323 12313 12312 32312 31312 31212 31232 31231 21231 23231 23131 23121 23123 13123
12132 13132 13232 13212 13213 23213 21213 21313 21323 21321 31321 32321 32121 32131 32132
```

Table C.1: Ordered pairs for 00001. Each pair $a_i b_i$ is a block; the connector to the next block is $a_i a_{i+1}$ when $\epsilon_i = 0$ and $b_i a_{i+1}$ when $\epsilon_i = 1$.

i	a_i	b_i	ϵ_i	i	a_i	b_i	ϵ_i
1	12132	32132	0	12	21310	31310	1
2	12130	32130	0	13	31312	32312	0
3	12120	32120	0	14	31212	31232	1
4	13120	23120	0	15	31230	21230	1
5	13123	23123	0	16	21210	31210	0
6	12123	12323	1	17	21213	23213	1
7	12313	12312	1	18	23210	13210	1
8	12310	32310	0	19	13212	13213	0
9	12320	32320	1	20	13232	13132	1
10	31320	21320	1	21	13130	23130	0
11	21323	21313	1	22	13230	23230	-

Cycle 00011 (42 words).

12102 13102 13103 23103 23123 13123 13120 23120 23130 13130
13230 23230 21230 31230 31232 31202 31302 32302 12302 12303
12103 12123 12323 12320 32320 31320 21320 21323 21303 21203
23203 13203 13202 13232 13132 12132 12130 12120 32120 32130
32132 32102

Cycle 00012 (47 words).

12103 12123 13123 13120 12120 32120 32130 12130 13130 13230
23230 21230 31230 31231 21231 23231 23201 23101 23131 23130
23120 23121 23123 23103 13103 13203 23203 21203 21201 31201
31301 32301 32101 32131 32121 32321 31321 21321 21301 21303
21323 21320 31320 32320 12320 12323 12303

Cycle 00101 (64 words).

12010 12020 32020 31020 21020 23020 13020 13010 23010 23210
13210 13212 13232 13032 13012 12012 32012 31012 31312 32312
12312 12310 32310 31310 21310 21320 31320 32320 12320 12323
12313 12013 12023 13023 23023 21023 21323 21313 21013 23013
13013 13213 23213 21213 21210 31210 31212 31232 31032 32032
12032 12030 32030 31030 31230 21230 23230 13230 13030 23030
21030 21010 31010 32010

Cycle 00102 (63 words).

12010 12030 32030 31030 31230 21230 23230 13230 13030 23030
21030 21031 23031 23231 21231 31231 31031 32031 32021 32321
31321 21321 21320 31320 32320 12320 12310 32310 31310 21310
21313 21323 21023 23023 13023 13020 13010 23010 21010 21020
23020 23021 21021 31021 31020 32020 12020 12023 12323 12313
12013 13013 23013 21013 21213 23213 13213 13210 23210 21210
31210 31010 32010

Cycle 00121 (58 words).

12010 13010 23010 23210 13210 13230 13030 12030 32030 31030
31032 32032 12032 13032 13232 13202 13212 13012 12012 32012
31012 31312 32312 12312 12310 32310 31310 21310 21210 21213
23213 13213 13203 23203 21203 21303 21313 21013 23013 13013
12013 12313 12303 12302 32302 31302 31202 31232 31212 31210
31230 21230 23230 23030 21030 21010 31010 32010

Cycle 00123 (57 words).

12010 12310 32310 31310 21310 21010 23010 13010 13210 13212
13202 13232 13032 13012 12012 32012 31012 31212 31232 31032
32032 12032 12030 32030 32031 31031 21031 21030 31030 31230
21230 21231 31231 31201 31202 31302 31312 32312 12312 12302
32302 32301 31301 21301 21201 23201 23231 23031 23030 13030
13230 23230 23210 21210 31210 31010 32010

Cycle 01012 (68 words).

10103 10203 13203 23203 20203 20303 10303 12303 12103 12123
10123 13123 23123 20123 20323 10323 12323 12320 32320 30320
10320 20320 20120 23120 13120 10120 30120 32120 12120 12130
32130 30130 10130 10230 30230 20230 23230 13230 13130 23130
20130 20131 23131 23231 20231 30231 30131 32131 32101 32301
30301 20301 20321 30321 32321 32121 30121 20121 23121 23101
23201 20201 30201 30101 20101 20103 23103 13103

Cycle 01023 (66 words).

10102 10302 30302 32302 12302 12312 10312 30312 32312 32310
12310 10310 30310 20310 20210 30210 10210 10212 30212 30202
10202 13202 13102 13132 13232 13212 13210 23210 23230 23130
23131 23231 23201 23101 20101 30101 30201 20201 20301 30301
32301 32101 32131 30131 30231 20231 20131 20130 20230 30230
10230 13230 13130 12130 32130 30130 10130 10132 10232 30232
30132 32132 12132 12102 32102 30102

C2 for 00112 (102 words).

012020 012030 032030 032031 032021 031021 021021 021031 031031
031231 031201 031301 032301 032321 032320 012320 012323 012303
112303 112323 112320 132320 132321 132301 131301 131201 131231
131230 031230 031030 021030 021230 121230 121030 131030 131020
121020 123020 123023 123021 123031 123030 123230 113230 113030
113020 113023 112023 012023 013023 013020 013030 013230 023230
023030 023031 023231 123231 121231 021231 021201 021301 021321
031321 131321 131021 131031 121031 121021 121023 121323 121303
021303 021203 121203 123203 113203 013203 023203 023201 123201
121201 121301 121321 121320 131320 031320 021320 021323 021023
023023 023021 023020 021020 031020 032020 132020 132021 132031
132030 112030 112020

Three-row insertion rule. Let C_2 be a Hamilton cycle of $A\square P_2$, with row digits $0, 1$. Suppose its second row contains three disjoint edges u_iv_i, and the states of A partition into three paths with respective endpoints u_i, v_i. Replace each cut edge $(1, u_i)(1, v_i)$ by its path in row 2 and the cross edges at both endpoints. These paths are internally disjoint and cover the new row, so they subdivide three edges of C_2 into one Hamilton cycle of $A\square P_3$. If D_A occurs in both old rows and in the new paths, and avoids the three cut edges, it survives in every row of the derived product cycle.

Common row edges 00112 (13 edges).

```
12023-13023 12320-32320 13020-13023 13030-13230
13230-23230 21020-23020 21020-31020 21030-21230
21030-31030 21301-21321 31201-31301 32030-32031
32301-32321
```

Generating the three-row product cycle for 00112. The following path P contains forty-seven distinct states and all thirteen edges above. Together with $12303, 12323, 13203, 23203$, its vertices are precisely the legal states of 00112.

```
21201 21203 21303 21301 21321 21320 21323 21023 23023 23021
23031 23030 23230 13230 13030 12030 32030 32031 32021 32020
31020 21020 23020 13020 13023 12023 12020 12320 32320 31320
31321 32321 32301 31301 31201 31231 31230 21230 21030 31030
31031 31021 21021 21031 21231 23231 23201
```

In row 1 of the printed C_2, cut

$$12303\text{–}12323, \qquad 13203\text{–}23203, \qquad 21201\text{–}23201.$$

The new row uses the first two edges as its two-point paths and P as the third path. All cut edges avoid D_A, which occurs in both rows of C_2 and in P. The insertion rule therefore gives the required three-row product cycle with all thirteen common edges.

D Explicit protected-pentagon constructions

Boundary digit 4 denotes the full list. Colour words use only 0,1,2,3. In six-digit cycle words the first digit is the path layer. The first two auxiliary edges for a Hamilton component join two or three path rows; any two remaining auxiliary edges supply block-chain reserves. Write P_A for the listed auxiliary edge set of a component A. Its protected set is $E(C_A) \setminus P_A$ for a component with a specified Hamilton cycle C_A, and $D_A \setminus P_A$ for an exceptional component, where D_A is its listed set of common row edges. The preceding appendix gives the ordinary cycles, the two constant-type 15-cycles and the product cycles for 00112. Each boundary transition is specified by its initial component (type:index), changed position and new colour, normalized final component, and a common protected edge. Positions start at zero. Two records per line are separated by semicolons.

PC2 for 00001 (88 words).

```
012120 012130 112130 112120 132120 132130 132132 112132 113132
113232 113230 113210 123210 123213 113213 113212 013212 013213
023213 021213 021210 023210 013210 013230 013232 013132 012132
032132 032130 032120 032320 032310 012310 012320 012323 012313
012312 032312 031312 031310 031210 031212 031232 031230 021230
023230 123230 121230 131230 131232 131212 131210 121210 121213
```

```
121313 121323 021323 021313 021310 121310 121320 021320 031320
131320 132320 112320 112310 132310 131310 131312 132312 112312
112313 112323 112123 012123 013123 023123 123123 113123 113120
123120 123130 113130 013130 023130 023120 013120
```

Common row edges 00001 (13 edges).

```
12132-13132 12132-32132 12310-32310 13120-23120
13123-23123 13210-23210 13230-13232 21210-21213
21230-23230 21230-31230 21313-21323 31212-31232
32130-32132
```

Generating the three-row product cycle for 00001. The following path P contains forty distinct states and all thirteen edges above. Together with $12312, 12313, 13212, 13213$, its vertices are precisely the legal states of 00001.

```
13210 23210 23213 21213 21210 21310 21313 21323 21320 31320
32320 32120 12120 13120 23120 23123 13123 12123 12323 12320
12310 32310 32312 31312 31310 31210 31212 31232 31230 21230
23230 23130 13130 12130 32130 32132 12132 13132 13232 13230
```

In row 1 of the printed C_2, cut

$$12312\text{–}12313, \qquad 13212\text{–}13213, \qquad 13210\text{–}13230.$$

The new row uses the first two edges and P. All cuts avoid D_A, which occurs in both rows of C_2 and in P. Apply the insertion rule in Appendix C to obtain the three-row product cycle, retaining every listed common edge.

Cycle for 00004 (54 words).

```
12120 12130 12132 13132 13130 23130 23230 23210 13210 13230
13232 13212 13213 23213 21213 21210 31210 31212 31232 31231
31230 21230 21231 23231 23131 23121 23123 23120 13120 13123
12123 12323 12313 12312 12310 32310 32312 31312 31310 21310
21313 21323 21321 21320 31320 31321 32321 32121 32131 32132
32130 32120 32320 12320
```

Cycle for 00014 (60 words).

```
12102 13102 13132 13232 13202 13203 13103 12103 12123 13123
13120 12120 32120 32320 12320 12323 12303 12302 32302 31302
31202 31232 31230 21230 23230 13230 13130 23130 23120 23121
23123 23103 23203 23201 23101 23131 23231 21231 31231 31201
21201 21203 21303 21323 21320 31320 31321 21321 21301 31301
32301 32321 32121 32101 32131 32130 12130 12132 32132 32102
```

Cycle for 00104 (78 words).

```
12010 12030 32030 31030 31230 21230 23230 13230 13030 23030
21030 21031 23031 23231 21231 31231 31031 32031 32021 32321
32320 12320 12323 12313 12013 12023 13023 23023 21023 21020
31020 32020 12020 13020 23020 23021 21021 31021 31321 31320
21320 21321 21323 21313 21310 31310 32310 12310 12312 32312
31312 31012 32012 12012 13012 13013 23013 21013 21213 23213
13213 13212 13232 13032 12032 32032 31032 31232 31212 31210
21210 23210 13210 13010 23010 21010 31010 32010
```

Cycle for 00114 (62 words).

12020 12320 32320 31320 21320 21323 21023 23023 23021 23031
21031 21021 21020 21030 21230 31230 31030 32030 12030 13030
13230 23230 23030 23020 13020 13023 12023 12323 12303 12302
32302 31302 31202 31232 31032 32032 12032 13032 13232 13202
13203 23203 21203 21303 21301 21321 31321 32321 32301 31301
31201 21201 23201 23231 21231 31231 31031 32031 32021 31021
31020 32020

Cycle for 00124 (71 words).

12010 12030 32030 32031 31031 31231 21231 21031 21030 31030
31230 21230 23230 13230 13030 23030 23031 23231 23201 23203
21203 21303 21301 21201 31201 31202 31302 31301 32301 32302
12302 12303 12313 12013 13013 13010 13210 13212 13012 12012
32012 31012 31312 32312 12312 12310 32310 31310 31210 31212
31232 31032 32032 12032 13032 13232 13202 13203 13213 23213
21213 21313 21013 23013 23010 23210 21210 21310 21010 31010
32010

Cycle for 01014 (88 words).

10102 10132 13132 13232 10232 30232 30132 32132 12132 12130
32130 30130 10130 10230 30230 20230 23230 13230 13130 23130
20130 20131 23131 23231 20231 30231 30131 32131 32101 32301
30301 20301 20101 30101 30201 20201 23201 23101 23121 20121
30121 32121 32321 30321 20321 20323 10323 12323 12320 32320
30320 10320 20320 20120 23120 13120 12120 32120 30120 10120
10123 12123 13123 23123 20123 20103 20203 20303 10303 10302
30302 32302 12302 12303 12103 10103 10203 13203 23203 23103
13103 13102 13202 10202 30202 30102 32102 12102

Cycle for 01024 (85 words).

10102 10202 30202 30212 10212 13212 13210 23210 20210 30210
10210 10213 10203 20203 23203 13203 13213 23213 20213 20313
10313 12313 12312 12310 32310 30310 20310 10310 10312 30312
32312 32302 30302 10302 10303 20303 20103 23103 13103 10103
12103 12303 12302 12102 32102 30102 30101 30201 20201 23201
23101 20101 20301 30301 32301 32101 32131 30131 30231 20231
23231 23131 20131 20130 23130 13130 13230 23230 20230 30230
10230 10130 30130 32130 12130 12132 32132 30132 30232 10232
10132 13132 13232 13202 13102

Cycle for 01104 (74 words).

10210 10310 20310 30310 32310 12310 12312 10312 30312 32312
32012 32032 32031 32021 32321 30321 20321 20320 10320 30320
32320 12320 12020 12010 32010 32020 32030 12030 12032 13032
13232 13212 13012 12012 12013 12313 10313 20313 20323 10323
12323 12023 13023 23023 23013 13013 13213 23213 20213 10213
10212 10232 10230 13230 23230 23030 13030 13020 23020 23021
23031 23231 20231 30231 30232 30212 30210 30230 20230 20210
23210 23010 13010 13210

Cycle for 01124 (74 words).

10202 10203 20203 23203 23201 20201 20301 20303 10303 12303
12302 32302 32301 30301 30201 30231 20231 20230 30230 10230
13230 23230 23231 23031 23030 13030 12030 32030 32031 32032
32012 32010 12010 12012 12032 13032 13232 10232 30232 30202
30212 10212 10213 10210 30210 20210 20310 30310 10310 12310
32310 32312 12312 12313 12013 13013 23013 23010 23210 13210
13010 13012 13212 13202 13203 13213 23213 20213 20313 10313
10312 30312 30302 10302

Cycle for 01204 (83 words).

10120 20120 23120 13120 13130 23130 20130 10130 30130 32130
12130 12132 32132 30132 10132 13132 13032 12032 32032 32012
12012 13012 13010 12010 32010 32020 12020 13020 23020 23010
23030 13030 12030 32030 32031 32131 30131 20131 23131 23031
23021 23121 20121 30121 32121 32021 32321 30321 20321 20320
10320 30320 32320 12320 12323 12023 13023 23023 23013 13013
12013 12313 12310 32310 32312 12312 10312 30312 30310 10310
20310 20313 10313 10323 20323 20123 23123 13123 10123 12123
12120 32120 30120

Cycle for 01234 (82 words).

10102 10302 30302 32302 12302 12303 10303 10323 20323 20303
20103 10103 12103 12123 13123 23123 20123 10123 10120 30120
32120 12120 13120 23120 20120 20320 10320 30320 32320 12320
12310 32310 30310 10310 20310 20313 10313 10312 30312 32312
12312 12313 12323 12023 13023 23023 23013 23010 13010 12010
12013 13013 13012 12012 32012 32010 32020 12020 13020 23020
23021 23121 20121 30121 32121 32021 32321 30321 20321 20301
30301 32301 32101 30101 20101 23101 23103 13103 13102 12102
32102 30102

Auxiliary edge set for 00000:0.

12123-13123 12312-12313 21231-31231 23121-23131 31212-31312

Auxiliary edge set for 00000:1.

12132-13132 13212-13213 21213-21313 21321-31321 32121-32131

Auxiliary edge set for 00001:0.

21210-21213 21230-31230

Auxiliary edge set for 00011:0.

12102-13102 13102-13103 13123-23123 13130-13230
13202-13203 21203-21303 31202-31302 32130-32132

Auxiliary edge set for 00012:0.

13130-13230 21301-21321 23120-23130 23121-23123
31201-31301 32101-32301 32121-32131 32121-32321

Auxiliary edge set for 00101:0.

12010-12020 12012-13012 12013-12023 12320-12323
13013-23013 13210-13212 21010-31010 23010-23210

Auxiliary edge set for 00102:0.

12013-13013 12313-12323 13013-23013 21013-21213
21023-21323 21031-23031 23020-23021 31031-31231

Auxiliary edge set for 00112:0.

21020-23020 21030-31030

Auxiliary edge set for 00121:0.

12010-13010 21010-21030 21010-31010 21210-21213
23010-23210 23030-23230 31012-31312 31030-32030

Auxiliary edge set for 00123:0.

13030-23030 13230-23230 21010-21310 21030-31030
21230-21231 31010-31210 31012-32012 31030-31230

Auxiliary edge set for 01012:0.
10103-13103 10123-13123 10130-30130 13130-13230
20101-30101 20230-30230 20301-20321 30131-30231

Auxiliary edge set for 01023:0.
12130-13130 20101-30101 20201-20301 20210-30210
20231-30231 23130-23131 30102-32102 30202-30212

Auxiliary edge set for 00004:0.
12312-12313 13130-13132 13212-13213 21210-21213
21321-21323 23121-23123 31231-31232 31310-31312

Auxiliary edge set for 00014:0.
12102-32102 12103-13103 12120-32120 12130-12132
13130-23130 21201-21203 23121-23123 32101-32121

Auxiliary edge set for 00104:0.
13012-13013 13210-23210 13212-13213 21020-21023
21320-21321 21321-21323 31320-31321 32320-32321

Auxiliary edge set for 00114:0.
12030-13030 21020-21021 21301-21321 21320-21323
23021-23023 31020-31021 31020-32020 32021-32031

Auxiliary edge set for 00124:0.
12310-32310 13210-13212 21030-31030 21201-31201
21213-21313 23010-23210 31010-32010 31301-31302

Auxiliary edge set for 01014:0.
10120-10123 10202-13202 10302-10303 12120-13120
12123-13123 13102-13103 20130-20131 20321-20323

Auxiliary edge set for 01024:0.
10103-13103 10132-13132 10302-10303 10312-30312
12130-12132 12312-12313 20103-20303 20213-20313

Auxiliary edge set for 01104:0.
10210-10310 12010-12020 12012-12013 12310-12312
13013-13213 20210-23210 23030-23230 32012-32312

Auxiliary edge set for 01124:0.
10202-10203 10210-10213 10230-13230 10310-30310
13010-13210 20210-20310 23230-23231 32310-32312

Auxiliary edge set for 01204:0.
10123-13123 12010-32010 12120-12123 12310-12313
12320-12323 23010-23020 30310-30312 32030-32031

Auxiliary edge set for 01234:0.
10102-30102 12102-32102 12123-13123 12302-32302
12310-12320 13012-13013 13102-13103 20101-30101

The thirty constant-source transitions. Write $K_i = E(C_i) \setminus O_i$ for the two constant components. Coordinate rotation fixes both components, since its order is five. Each auxiliary set of five edges O_i is a rotation orbit, so K_i is invariant under $\rho_r(x)_k = x_{k+r \pmod 5}$. The following edges belong to the unchanged protected set K_H of type 00001:

$$a = 31212 - 31232, \quad b = 12132 - 32132, \quad c = 21313 - 21323, \quad d = 13123 - 23123.$$

When boundary entry j changes from 0 to $q \in \{1, 2, 3\}$, its canonical normalizer is one of

$$N_{j,\epsilon}(x)_k = p_q(x_{j-4\epsilon+\epsilon k}), \qquad \epsilon \in \{+1, -1\},$$

where p_q is the first lexicographic permutation with $p_q(0) = 0$ and $p_q(q) = 1$. Indeed, the four repeated entries map to 0 and the unique changed entry maps to 1 at position 4.

The table gives $e_{i,q} \in K_i$ and its two normalized images; these memberships follow directly from the printed words and auxiliary edge sets.

i	q	$e_{i,q}$	$N_{0,+1}(e_{i,q})$	$N_{0,-1}(e_{i,q})$
0	1	$23121 - 23123$	a	b
0	2	$31231 - 31232$	c	d
0	3	$21231 - 23231$	d	c
1	1	$21213 - 23213$	b	a
1	2	$31321 - 32321$	d	c
1	3	$21321 - 21323$	c	d

At any position j, use $\rho_{-j}(e_{i,q}) \in K_i$. Its image is

$$N_{j,\epsilon}(\rho_{-j}(e_{i,q})) = N_{0,\epsilon}(e_{i,q}) \in K_H,$$

so it lies in both actual protected sets. These six families cover all thirty transitions.

Common-edge selections (236 transitions). Let K be either fixed protected set and B the opposite component, with auxiliary edge set P_B and a Hamilton cycle. At a common state x, put $d = \deg_B(x)$ and $g = |(K \cap \delta_B(x)) \setminus P_B|$. If $d - g \leq 1$, every Hamilton cycle in B takes two incident edges at x, at least one in $K \setminus P_B$. Only B needs a Hamilton cycle.

t	S_t
00001	13123, 21230, 23120, 32132
00004	12130, 13123, 32312
00011	12323, 21230, 21320
00012	12303, 13203, 13230, 31230
00014	12303, 31202, 31230
00101	12312, 13230, 13232, 21323, 31312
00102	13023, 13210, 13230, 23231, 31231, 32031, 32321
00104	12032, 12323, 13232, 23021, 23031
00112	12023, 13230, 31201
00114	12032, 21303
00121	12302, 12303, 21203, 21303
00123	12302, 12310, 23201, 32031
00124	12303, 23031, 32301
01012	10203, 13203, 23231, 32301
01014	12302, 23201, 23231, 32301
01023	12310, 13202, 20310, 23101
01024	23213, 32301, 32310
01104	20231, 30321, 32021
01124	12303, 23031, 32301
01204	10132, 13032, 23013, 30312, 32312
01234	23013, 23101

For each of the 236 nonconstant-source transitions not listed individually below, some $x \in S_t$ lies in both actual components and satisfies $d - g \leq 1$, with fixed source K against the restored target component, or restored target K against the source component. The opposite side has its displayed Hamilton cycle; the printed protected sets and auxiliary edge sets use the original transition's first normalizer. Its degree is

$$L_i(b, x) = \{0, 1, 2, 3\} \setminus \{x_{i-1}, x_{i+1}, b_i\}, \qquad d = \sum_i (|L_i(b, x)| - 1),$$

where b is its boundary in the original coordinates, omitting b_i when starred. This proves the stated common-edge identities in all 236 cases.

The 47 remaining individual nonconstant-source transitions.

```
00001:0 1 2 00102:0 21313-21323 ; 00001:0 1 3 00102:0 21313-21323
00001:0 2 1 00101:0 12310-32310 ; 00001:0 4 2 00001:0 13120-23120
00001:0 4 3 00001:0 12310-32310 ; 00011:0 0 2 00112:0 12130-12132
00011:0 0 3 00112:0 12120-12130 ; 00101:0 0 1 00101:0 21020-23020
00101:0 1 1 00101:0 12013-12313 ; 00101:0 2 0 00001:0 12310-32310
00101:0 3 2 00121:0 12010-32010 ; 00101:0 3 3 00121:0 12010-32010
00101:0 4 0 00001:0 12032-32032 ; 00102:0 0 1 01012:0 21013-23013
00112:0 0 1 00102:0 21030-21230 ; 00112:0 0 2 00112:0 12023-13023
00112:0 1 3 00123:0 12320-32320 ; 00112:0 2 2 00121:0 12023-13023
00112:0 2 3 00123:0 13020-13023 ; 00112:0 3 2 00112:0 13030-13230
00112:0 4 3 00112:0 12320-32320 ; 00121:0 2 2 00112:0 12010-32010
00121:0 3 0 00101:0 12010-32010 ; 00121:0 4 2 00112:0 12013-13013
```

```
00123:0 2 2 00112:0 12030-32030 ; 01012:0 0 1 00102:0 20103-23103
01012:0 1 2 01012:0 10120-13120 ; 00104:0 0 1 01014:0 21010-31010
00104:0 1 2 01204:0 13010-23010 ; 00104:0 1 3 01204:0 12010-32010
00124:0 2 3 00124:0 12010-32010 ; 01014:0 0 1 00104:0 20101-30101
01014:0 0 2 01024:0 10102-12102 ; 01014:0 0 3 01024:0 10103-10203
01014:0 1 2 01024:0 10120-30120 ; 01014:0 1 3 01024:0 10102-12102
01014:0 2 2 01024:0 10102-10132 ; 01014:0 2 3 01024:0 10102-10132
01024:0 0 3 01234:0 10103-12103 ; 01024:0 1 2 01014:0 10102-13102
01024:0 2 3 01234:0 10102-13102 ; 01204:0 1 0 00104:0 12012-13012
01204:0 1 3 01204:0 10120-30120 ; 01234:0 0 2 01024:0 10102-10302
01234:0 1 3 01024:0 10102-10302 ; 01234:0 2 0 01024:0 10103-12103
01234:0 3 1 01024:0 10102-10302
```

E Shared-hub path constructions

Let $v_0, \ldots, v_{r-1}$ be the restored path, where $r \in \{4, 5\}$. Its endpoints have the same outside neighbour w; the other outside neighbours of its internal vertices are distinct and different from w. In the boundary order $t = (a, b_1, \ldots, b_{r-2})$, the fibre consists of the proper path words with coordinate exclusions

$$(a, b_1, \ldots, b_{r-2}, a).$$

All colours in this appendix and its arrays lie in $\{0, 1, 2, 3\}$. The files `data/path4-family.json` and `data/path5-family.json` supply explicit cyclic arrays, one for each normalized boundary. Normalize colours in first-appearance order, then append unused colours in ascending order. Restore every cycle by the inverse palette map, without reflecting the path or choosing an additional relative map.

Table E.1: Complete shared-hub path coverage. Transitions are directed.

Order	Types	States	Boundaries	Transitions	Min. edges
4	5	160	64	576	3
5	15	1112	256	3072	5

Here "States" sums the lengths of the normalized arrays. The program reconstructs each legal fibre independently, then checks complete coverage, no repeated state, and a one-coordinate change at every cycle step, including the closing step.

The program expands to every labelled boundary and intersects the *chosen cycle edge sets* at every change at one boundary vertex. A change at w changes both endpoint exclusions simultaneously; it is one boundary transition, included among the 576 or 3072 directed cases. The endpoints of every shared edge are legal on both sides, so both matching outside-recolouring edges exist. The displayed minima exceed the two edges needed to select an outgoing splice after one incoming edge has been removed.

The program `verify_shared_hub_paths.py` checks these finite claims, including complete representative coverage, unique states, legal colour words, and the stated transition minima. Splicing along an arbitrary base Hamilton path and the article's structural induction remain analytic arguments; neither is inferred from finite verification.

F Degree-two configuration families

The seven files `data/degree-two/`*kind*`-family.json` give explicit cycles for the following fixed internal models. Paths have vertices $0, \dots, n-1$ in order; the star has centre 0. Write T_m for the cycle $0, \dots, m-1, 0$ with the additional vertex m and edge $0m$. A boundary is $t \in \{0,1,2,3\}^q$. At vertex v, exclude the colours t_i for $i \in I_v$. In the tuples below a single index denotes its singleton set. An empty set excludes no colour; a two-element set can exclude one or two colours, according to whether the corresponding excluded colours agree.

Kind	X	$(I_0, \dots, I_{n-1})$
`star3`	$K_{1,3}$	$(0,1,2,3)$
`path4heavy`	P_4	$(0,\{1,2\},3,0)$
`path5heavynear`	P_5	$(0,\{1,2\},3,4,0)$
`path5heavycenter`	P_5	$(0,1,\{2,3\},4,0)$
`path6`	P_6	$(0,1,2,3,4,0)$
`tailed5`	T_5	$(0,\varnothing,1,2,3,4)$
`tailed6`	T_6	$(0,\varnothing,1,2,3,4,5)$

The numbers of boundary positions q are respectively $4,4,5,5,5,5,6$. In every path configuration, position 0 records the common outside neighbour of both endpoints, so its move changes both endpoint exclusions at once.

Two boundary positions can represent one outside vertex only if every pair of their incident internal vertices has distance at least three; otherwise a triangle or a 4-cycle results. The complete additional pairs of positions that may be identified are

`path6` : $(1,4)$,
`tailed5` : $(1,4)$, $(2,4)$,
`tailed6` : $(0,2)$, $(1,4)$, $(1,5)$, $(2,5)$, $(3,5)$.

The first four models have none. Each compatibility graph is triangle-free, so no outside vertex occupies three boundary positions. Several disjoint pairs may be identified. A move changes one singleton boundary position or one listed pair, with equal old entries in the latter case.

Table F.1: Degree-two family coverage; transitions are directed.

Kind	Types	States	Boundaries	Transitions	Min. edges
star3	15	499	256	3072	4
path4heavy	15	376	256	3072	2
path5heavynear	51	2915	1024	15360	2
path5heavycenter	51	2921	1024	15360	2
path6	51	8593	1024	16128	9
tailed5	51	8535	1024	16896	12
tailed6	187	72197	4096	89088	19

The first computation enumerates all legal colourings of the seven fixed configurations and checks complete cycle coverage and the permissible identifications. A separate computation checks cycle legality and all boundary transitions in the original colour labels. Palette normalization is as in Appendix E. The counts in Table F.1 cover 421 representatives, 96,036 states, 8,704 boundaries and 158,976 directed transitions. At least two common chosen-cycle edges occur at every transition, as required for restoration. Splicing along a base path of arbitrary length and structural induction are proved in the article.

G Pentagonal reservoirs and full-list families

The three new arrays in `data/pentagonal-reservoirs/` have the following fixed models, using the exclusion notation of Appendix F. The cycle is $0, 1, 2, 3, 4, 0$; vertex 5, when present, is the additional degree-two vertex. The model R deletes an ordinary pentagon and a nonadjacent degree-two vertex; F deletes a pentagon with one degree-two vertex; P deletes an ordinary pentagon and an adjacent degree-two vertex.

Model	X	$(I_0, \ldots, I_{n-1})$	q
R	$C_5 \dot\cup \{5\}$	$(0, 1, 2, 3, 4, \{5, 6\})$	7
F	C_5	$(\varnothing, 0, 1, 2, 3)$	4
P	$C_5 + 05$	$(\varnothing, 0, 1, 2, 3, 4)$	5

The file for R is `reservoir-family.json`; the files for F and P are `pentagon-full-family.json` and `pentagon-pendant-family.json`, respectively. Their palette is $\{0, 1, 2, 3\}$, with first-appearance normalization as before. The two neighbours of vertex 5 in model R are distinct, but their colours may agree.

The complete pairs of positions that may be identified for R are (i, j) with $0 \le i < 5$ and $j \in \{5, 6\}$. They form $K_{5,2}$, with $1 + 10 + 20 = 31$ matchings. Model F has no identifications; model P has pairs $(1, 4)$ and $(2, 4)$, hence three matchings. The internal-distance test and absence of compatible triples are checked independently by both verification programs. Several disjoint pairs of identified positions introduce no additional move type: an actual recolouring changes one boundary position or one identified pair whose old entries are equal.

Table G.1: Pentagonal family coverage. Moves and component overlaps are directed.

Kind	Types	Cycles	States	Labels	Moves	Overlaps	Min. edges
reservoir	715	720	91410	16384	466944	470208	4
full	15	15	1105	256	3072	3072	11
pendant	51	51	8606	1024	16896	16896	8

In model R, a fibre has two components exactly when its first five boundary entries agree; otherwise it is connected. There are five such normalized disconnected boundaries. Each displayed cycle covers one *whole component*, not the disconnected fibre. The smallest component has order 30. Models F and P have connected fibres. The first computation enumerates all legal states and their components; an independent verification program reconstructs these states and components separately. Both check exact component coverage, cycle lengths, uniqueness and all closing steps.

At every labelled transition, *every overlapping component pair* has the stated common-cycle-edge margin. Every component on either side overlaps a component on the other side, including moves at a shared outside vertex. The independent computation transports explicit cycles into original colour labels and checks the corresponding selected-edge sets. The programs agree on the distributions of common-edge counts. In total the constructions contain 781 boundary types, 786 component cycles and 101,121 states, covering 17,664 labels and 486,912 directed moves.

Margin four supplies the reservoir tree-splicing reserve: along a base path with at most two components per layer, a component-graph node has degree at most four and loses at most three edges before its next attachment. For the connected families, margin two suffices along a base path. Unchanged boundaries retain the identical selected cycles, whose lengths are also checked against these degree bounds. The connected-layer argument, unbounded splicing and structural inductions are analytic proofs in the article. Only the three displayed families are used in these restorations.

H Linked pentagons and a three-contact path

The five compressed arrays in `data/extended-cycles/` are explicit Hamilton cycles for fixed local graphs. Their palette is $\{0,1,2,3\}$. A boundary position at a vertex excludes its boundary colour; vertices with no boundary position have full lists. The two shared-edge models have cycles $(0,1,4,3,2,0)$ and $(0,1,7,6,5,0)$, with the optional edge 36. Their exterior boundary orders are respectively $(2,3,4,5,6,7)$ and $(2,4,5,7)$. The three-contact path has vertices $0,\dots,6$ and exclusion pattern (a,b,c,a,d,e,a), so its retained hub is one boundary position with three contacts.

The linked models have cycles 012340 and 567895, with edge 05 and, in the two-link case, edge 27. Their boundary positions are the vertices without a joining edge, in increasing order. Thus they have eight and six boundary positions. The explicit files have the kind names in Table H.1, followed by `.json.gz`.

The shared-edge and path models use first-appearance palette normalization, appending unused colours increasingly. The one-link and two-link models also minimize over their eight and four declared graph automorphisms. Each automorphism is checked to preserve edges and contact sets; inverse vertex permutations and palette maps restore original coordinates. Both computations use the fixed graph and contact models, cover all canonical boundaries, and verify exact cyclic coverage.

Two boundary positions may represent the same outside vertex precisely when all pairs of their contacts have internal distance at least three. The complete extra pairs are

shared edge	$(0,4),(0,5),(1,3),(1,4),(1,5),(2,3),(2,4),$
shared edge with chord	$(0,3),(1,2),$
three-contact path	$(1,3),(1,4),(2,4).$

For the linked models they are all pairs between the two cycles' sets of boundary positions. The compatibility graphs have no triangle, hence no three distinct positions can represent the same outside vertex. Their numbers of complete identification matchings are respectively $22,4,5,209,34$. Several disjoint pairs of identified positions are allowed, while each actual move changes one boundary position or one equal pair simultaneously. The repeated hub boundary position changes three local exclusions at once.

Table H.1: Extended cycle families; transitions are directed recolouring moves.

Kind	Types	States	Labels	Transitions	Min. edges
sharededge	187	158054	4096	95232	56
sharededgechord	15	16958	256	3456	94
threecontactpath7	51	18843	1024	17664	10
onelinkpentagons	451	1917512	65536	2359296	219
twolinkpentagons	67	379795	4096	101376	390

In total there are 771 representatives with 2,491,162 states, covering 75,008 labelled boundaries and 2,577,024 directed transitions. The first computation checks legal states and all selected-cycle intersections. An independent computation reconstructs the fixed models and maps, restores colour words in the original coordinates, and reproduces every distribution of common-edge counts. Both check the displayed counts and all selected-cycle intersections. Only a margin of two is used in the analytic path-order splice.

I Bowtie product cycles and clique partitions

A bowtie has cycles 012340 and 056780, with vertex 0 of original degree four and the other vertices of original degree three. The chorded model adds edge 26. Their exterior orders of boundary positions are $(1, 2, 3, 4, 5, 6, 7, 8)$ and $(1, 3, 4, 5, 7, 8)$; all unlisted vertices have full lists. In `data/bowties/`, the two arrays of product cycles have suffix `-seeds.json.gz`, and the two tree arrays have suffix `-clique-trees.json.gz`; their prefixes are `bowseed` and `bowchord`. The file `ordinary-pentagon-seeds.json` extracts the ordinary product cycles from Appendices C–D.

Palette normalization is by first appearance, with unused colours appended increasingly, after minimizing over the eight bare or two chorded vertex permutations. The fixed maps preserve the graph and exterior boundary positions. Canonical boundaries and inverse maps are reconstructed by the verification programs before the product cycle checks.

For a bowtie fibre B_t, the product cycles are cycles of $B_t \square K_2$ and $B_t \square K_3$. They use the complete clique row adjacency, including moves between rows zero and two. Every edge of a listed matching M_t is retained in every row. The first two matching edges are distinct reserves; the rest form the protected set K_t. Every cycle is checked for complete coverage, uniqueness, legality and closure. An independent computation reconstructs the same finite graphs and restores original coordinates directly.

Table I.1: Bowtie product cycles and boundary transitions. The last columns give the minimum intersections of the matchings M_t and protected sets K_t.

Kind	Types	Fibre states	Product states	Transitions	M_t	K_t
bowseed	451	640608	3203040	2162688	9	9
bowchord	107	204832	1024160	89088	25	24

The 1,116 clique-product cycles contain 4,227,200 vertices in total. There are twelve and five permissible pairs of boundary positions, respectively; neither compatibility graph has a triangle. Both computations check all 65,536 and 4,096 labelled boundaries, and all 2,162,688 and 89,088 directed factor transitions. They reproduce the distributions of matching and protected-edge intersection sizes. These factor checks also handle outside vertices shared between distinct deleted factors; their combined transitions factorize analytically.

The clique arrays partition every bowtie fibre into edges and triangles. Every block edge is a legal one-vertex move. The listed connectors form a tree on the blocks, and at most two meet each fibre vertex. They cover 558 types and 845,440 states, with 422,601 blocks including 238 triangles, and 422,043 connectors. A three-vertex path does not meet the requirement that each three-vertex block be a triangle.

The ordinary-pentagon constructions have eleven boundary types, twelve components and 590 component states, with 24 two- and three-row product cycles. The program checks these cycles, auxiliary edge sets, protected sets, reserves and exceptional product cycles against the constructions in Appendices C–D. This preserves the protected-edge condition under relative normalization used in the article. The P_3 product cycles also give K_3 product cycles because $P_3 \subseteq K_3$; their protected sets need not be matchings.

The generalized protected-product lemma, its use over arbitrary base paths, and the completed-dual structural proof remain analytic.

J Hinged-pair product cycles and clique partitions

A hinged pair has cycles $(0,2,3,4,1,0)$ and $(0,1,7,6,5,0)$. Vertex 0 has original degree four and every other vertex has original degree three. The chorded model adds edge 36. The bare exterior boundary order is $(0,2,3,4,5,6,7)$, and the chorded order is $(0,2,4,5,7)$. Each boundary position excludes one outside colour; all unlisted vertices have full lists. In particular the heavy vertex has one outside exclusion.

The compressed files in `data/hinged-pairs/` are `cycles.json.gz` and `trees.json.gz`; they describe the fixed models `bare` and `chord`. Boundaries are seven- or five-digit strings over $\{0,1,2,3\}$. Matching words have eight digits; product cycle words prepend the row. Tree indices are zero-based in ascending lexicographic order of legal eight-digit words.

Normalize over the identity and arm exchange $(0,1,5,6,7,2,3,4)$, renaming colours by first appearance and appending unused colours increasingly. The least full normalization tuple fixes ties. Each map preserves the graph and contacts of the boundary positions; its inverse restores original coordinates. The complete compatible pairs of boundary positions are

$$\begin{aligned} \text{bare} \quad & (1,5),(1,6),(2,4),(2,5),(2,6),(3,4),(3,5), \\ \text{chorded} \quad & (1,4),(2,3). \end{aligned}$$

They are exactly the pairs at internal distance at least three. There is no compatible triple. The complete matchings of permissible identifications number 22 and 4; a move changes one boundary position or one equal pair simultaneously.

Every representative supplies a matching M and Hamilton cycles in its fibre product with K_2, K_3 and P_3. Each matching edge is retained in every row of all three product cycles. The first two edges are reserves, and the remaining edges form the protected set K. A K_3 product cycle may use row edges 02; a P_3 product cycle may only change between adjacent rows.

Table J.1: Hinged-pair product cycles and directed factor transitions. The last columns give the minimum intersections of the matchings M and protected sets K.

Kind	Types	Fibre states	Product states	Transitions	M	K
bare	379	241774	1934192	430080	6	6
chord	31	26051	208408	16896	21	20

The 410 representatives have 267,825 fibre states. Their 1,230 product cycles contain 2,142,600 vertices in total. Both computations enumerate the legal colourings and check each product cycle, its retained matching, and all 17,408 labelled boundaries and 446,976 directed transitions. The second computation specifies the graphs, contacts, symmetries and permissible identifications separately, and compares matching and protected-edge intersections in the original colour labels. The fixed bare boundary $(2,2,1,1,3,1,1)$ is also reconstructed: it has 526 states, 1,578 edges and three leaves, excluding a Hamilton path in that single fibre.

The first computation checks all 410 trees of clique blocks, with 133,811 blocks, including 203 triangles, and 133,401 connectors. Every block is an edge or a triangle. The joining edges form a tree on the blocks, and at most two are incident with each fibre vertex.

The proofs use the K_2/K_3 product cycles and trees of clique blocks. For an ordinary-pentagon partner, use its product cycles first and the hinged tree second, preserving the relative-normalization condition of Appendices C–D. The P_3 cycles are additional finite constructions.